Governance Concentration Beyond Token Allocation: A Protocol-Address-Corrected Cross-Sectional Audit of 52 DePIN and DeFi Protocols

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Abstract

Governance concentration in token systems is a post-distribution institutional-design outcome, not a launch-allocation or holder-list artifact; it cannot be read off initial tokenomics tables or naive holder lists. Across a 52-protocol cross-section spanning DePIN, DeFi, infrastructure, and social tokens, we compute Herfindahl-Hirschman Index (HHI) concentration after excluding protocol-controlled addresses (PCAs), and find that launch-design and protocol-financial covariates (insider, team, and investor allocation; protocol maturity; circulating float; valuation ratios) are each uninformative about steady-state concentration (insider allocation Pearson $r = 0.07$, $p = 0.68$, $N = 37$). Post-distribution and institutional-design factors are the larger correlates: protocols with more insider wallets among top holders are more concentrated (Spearman $\rho = 0.48$, $p = 0.003$, $N = 37$, surviving a non-insider HHI tautology check); DePIN governance is more concentrated than DeFi after correction (Mann-Whitney $p = 0.011$, Cohen's $d = 1.05$, robust in direction across leave-one-out, permutation, and a powered multivariate specification, with statistical significance conditional on the protocol-controlled-address and exchange-custody exclusion applied symmetrically across sectors, Section 4.6.2); and delegation amplifies voting power above token holdings in thirteen of eighteen protocols with sufficient governance data, with five design-driven exceptions (ENS, GMX, HNT, JUP, LPT). A subsidy-to-concentration association appears only through a single outlier (Pearson $r = 0.62$ including Livepeer, $r = 0.07$ excluding it).

The methodological contribution is a generalizable five-class PCA-exclusion typology that corrects systematic inflation in prior holder-list concentration studies (median inflation factor 2.3x, maximum approximately 18x); Gini and HHI capture distinct properties of the same holder set ($r = 0.58$) and inequality metrics cannot substitute for direct concentration measurement. All findings are descriptive associations from a single cross-section, not causal claims. The paper disciplines that descriptive status by specifying five forward predictions with explicit falsification thresholds and stated test windows, committing to pre-register their hypothesis specifications, statistical tests, and thresholds on the Open Science Framework before any panel-data or event-study extension. Together they reposition the question: token governance concentration is shaped less by launch-time allocation than by post-distribution institutional design (insider retention, architectural sector, and delegate-program structure), which is where measurement and governance-design attention should focus. The practical implication is a changed audit default: a serious decentralization audit should start from PCA-corrected current holder concentration and then add insider-retention, voting-HHI, and delegation or vote-escrow adjustment, rather than reasoning from launch allocation tables. Concentration of this kind bears on the legitimacy of decentralized governance, not only its efficiency: where a small set of holders or delegates commands decisive voting weight, the broad participation in rule modification that the commons self-governance ideal presumes is nominal rather than operative.

JEL Codes: C21, C81, D02, D71, D72, G23, G34, L22, O33

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1 Introduction

1.1 Motivation

Token-based governance is the default coordination mechanism for billions of dollars in decentralized physical infrastructure (DePIN) and decentralized finance (DeFi). Its votes are not symbolic: they can alter protocol rules, treasury use, emission schedules, fee switches, upgrade paths, and delegate programs. If voting weight is dispersed, token governance can approximate broad participant control. If voting weight concentrates, the same formal apparatus can place shared infrastructure under the discretion of a small holder or delegate class.

That makes the motivating question concrete: what shapes whether governance power concentrates or distributes after launch? Three candidate explanations dominate practitioner discourse: insider token allocation concentrates power; subsidy dependence (high token emissions relative to revenue) entrenches early holders; and younger protocols have not yet had time to decentralize. This paper tests all three and finds that none is sufficient as a standalone explanation of the cross-sectional variation observed across fifty-two protocols. The deeper problem is that all three explanations, and the decentralization claims practitioners build on them, are routinely read off launch allocation tables and raw holder lists: both are the wrong object for live governance power. Token allocation is a launch document. Governance concentration is a live institutional state. An audit that reasons from initial tokenomics is reasoning from audit metadata, not from governance evidence. The contrast tracks the distinction comparative political economy draws between de jure and de facto power (Acemoglu & Robinson, 2008): the formally specified allocation of authority is a different object from the power that groups actually exercise once a system is live.

Two empirical contrasts illustrate why the candidate explanations fail. Both ENS and DIMO let token holders delegate their voting power, yet the two protocols produce opposite governance concentration outcomes. ENS (the Ethereum Name Service) funds roughly 100 active community delegates, and the resulting voting-power distribution is less concentrated than the underlying token holdings (voting-to-holding amplification ratio 0.48x; ratios below 1.0 mean delegation disperses governance power relative to the holding baseline). DIMO, a vehicle-data DePIN protocol, has ten active voters across the most recent twelve months, and its voting-power distribution is 9.1 times more concentrated than its token holdings (amplification ratio 9.1x). Across the eighteen-protocol governance sample studied here, delegation amplifies concentration in thirteen of eighteen protocols (range 2.5x to 25.6x above the holding baseline), with five structural exceptions on the dispersing side (ENS 0.48x, GMX 0.87x, HNT 0.26x to 0.39x, JUP 0.12x, LPT 0.27x; Section 4.5). The cross-protocol variation is associated not with sector membership or governance venue but with institutional design choices at the delegate-program level: how many community delegates the protocol funds, how voting power flows from holders to delegates, and what incentives shape delegate participation (Section 5.1).

A parallel pattern emerges for subsidy economics: financial sustainability is uninformative as a concentration indicator. Protocols with near-identical subsidy ratios span nearly an order of magnitude in governance HHI, so financial sustainability is neither necessary nor sufficient for governance decentralization (Section 4.4). Governance concentration emerges as a secondary market outcome, shaped by post-distribution trading and accumulation dynamics that neither allocation design nor protocol financial characteristics explain in this cross-section. The allocation result is therefore best read as a practice correction rather than a bare null: it tells auditors that initial tokenomics is metadata about a launch, not evidence about who holds governance power today.

That concentration is a market outcome rather than a launch artifact matters normatively as well as descriptively. On a republican view of freedom as the absence of subjection to another’s arbitrary power, the relevant question is not whether concentrated holders have in fact interfered but whether they hold the standing capacity to alter the rules unilaterally; concentrated voting weight establishes exactly that capacity. The Neo-Brandeisian tradition associated with Khan (2017) and Wu (2018) treats such structural power over shared infrastructure as a harm worth scrutiny on political-economy terms, independent of whether any single decision has visibly harmed participants. Concentration measured after correction is therefore the object of scrutiny regardless of how the holders have so far chosen to use it.

The measurement instrument that makes every downstream result credible is the PCA correction: before any covariate can be tested, the holder list has to be cleaned of addresses that hold tokens but structurally cannot vote. We construct a 52-protocol governance concentration dataset spanning DePIN and DeFi, develop a Blockscout-verified (a blockchain contract verification tool) exclusion methodology to remove 133 protocol-controlled address exclusions (PCAs; addresses that appear on holder lists but structurally cannot vote, including staking contracts that hold tokens on behalf of stakers, exchange custodians that custody customer deposits, vesting locks that release tokens to insiders over time, and protocol treasuries) from HHI computation, and test every available cross-sectional covariate of concentration using both parametric and rank-based methods.

Four empirical contributions emerge, and they read as a sequence: naive holder lists are contaminated by addresses that cannot vote; once the holder list is corrected, launch allocation turns out to be uninformative about current concentration; and current institutional structure (sector, insider retention, and delegation design) is the more informative correlate. First, naive token holder lists include addresses that cannot vote: staking contracts, exchange custodians, vesting locks, and protocol treasuries. Across 38 protocols, 133 such address inclusions (125 unique addresses; five recur across multiple protocols, e.g., Binance 8 hot wallet across four tokens) inflated affected protocols' HHI in prior concentration measurements by up to 18x (RENDER, where Wormhole Token Bridge custody dominated naive top-1000 holdings; median inflation factor 2.3x across 32 protocols with complete pre-exclusion and post-exclusion HHI logged). The methodology generalizes via the five-class typology in Section 3.8, supporting uniform replication across ERC-20 and SPL token-governance datasets. Second, initial allocation design is uninformative about concentration while current insider wallet retention is associated with concentration (Spearman rho = 0.48, p = 0.003, N = 37; Section 4.4). Third, DePIN governance concentration is higher than DeFi after exclusion correction (Mann-Whitney p = 0.011, Cohen's d = 1.05), and the finding is robust to single-protocol exclusion (significant in all 30 of 30 leave-one-out iterations).

Fourth, delegation concentrates governance power in thirteen of eighteen protocols sampled, with amplification factors of 2.45x (Gnosis), 2.7x (Uniswap), 3.1x (Arbitrum, DRIFT), 3.6x (Optimism), 3.8x (WeatherXM), 4.4x (Aave), 5.55x (Synthetix), 6.3x (Lido), 9.1x (DIMO), 9.63x (Bittensor), 9.9x (Compound), and 25.65x (Polkadot). Five protocols (ENS at 0.48x; GMX at 0.87x; HNT at 0.26x-0.39x; JUP at 0.12x most-extreme dispersion outlier; LPT at 0.27x, the most pronounced DePIN governance disperser) show delegation-mediated dispersion, reflecting mature-delegate-program (ENS), methodology-sensitive crossover (GMX), Solana VSR lockup-weighted design (HNT), Solana delegated-vote with broad airdrop user base (JUP), and Livepeer orchestrator bonded-stake delegation across roughly 100 orchestrators (LPT) patterns where holders systematically delegate to a broader community-delegate set or vote-weight mechanism design distributes voting power below the holding baseline. Two further protocols (Uniswap, Optimism) would appear to show dispersion if holding HHI were measured without excluding addresses that cannot vote (burn wallets, foundation treasuries); under the consistent exclusion methodology applied throughout this paper, both protocols join the amplification pattern. Curve, Balancer, and Frax (vote-escrowed governance via veCRV, which is vote-escrowed CRV, veBAL, vote-escrowed BAL, and veFXS, vote-escrowed FXS) show extreme amplification (15x, 21x, and 11.4x) reflecting lock-duration weighting and form a distinct class discussed in Section 4.5.2. Institutional design within each delegate program is associated with whether delegation concentrates power and how strongly.

The sequence carries one practical conclusion: governance audits should measure live power across these four surfaces rather than infer it from inherited launch promises. The same corrected measurements make visible a further, method-enabled extension, a cross-protocol institutional concentration in which a small professional-delegate and exchange-custody class recurs across many protocols at once; this is previewed in Section 4.5.5 (Supplementary File S25) as a descriptive extension of the framework, not as a fifth equal contribution.

This audit framing responds to a stated field gap: recent scoping work by Alshater (2026) systematically maps the DePIN tokenomics design space, identifying the burn-and-mint equilibrium, governance-calibrated issuance, and provider staking as the dominant economic primitives across leading deployments. Alshater explicitly frames decentralization as a spectrum rather than a binary property, echoing analyses of decentralized governance for cyber-physical systems (Nabben et al., 2024), and identifies the lack of rigorous quantitative empirical work on these models as the field's most pressing research gap. The cross-sectional measurement reported here responds to that gap, applying Herfindahl-Hirschman Index analysis to fifty-two protocols spanning DePIN and adjacent categories. These results extend the governance concentration literature (Barbereau et al., 2023; Feichtinger et al., 2023; Cong et al., 2025) through a methodology that corrects systematic measurement artifacts in prior holder-list studies and through cross-sectional evidence on allocation, sector, subsidy, and delegation covariates of concentration.

1.2 Companion Theory and Normative Bridge

Governance concentration is not only a statistical pattern; it bears on institutional legitimacy when a small set of holders or delegates can set rules affecting many participants. A companion theoretical paper develops the Polanyian interpretation (Polanyi, 1944): governance rights packaged as transferable tokens function as a candidate fourth fictitious commodity, and concentration patterns are consistent with commodification of constitutive political authority rather than transient implementation failure (Zukowski, 2026e). This empirical paper does not develop that framework at length. It supplies the cross-sectional measurement base Zukowski (2026e) cites for Test 2 and documents where delegate-program design, not sector or launch allocation, shapes voting-power outcomes.

A second companion paper develops the normative standard these patterns imply (Zukowski, 2026f). It argues that legitimate decentralized governance requires non-domination, namely that those subject to a protocol’s rules can reliably check arbitrary power over them, operationalized through Ostrom’s design principles for enduring self-governance and constrained by allocative justice. This empirical paper does not adjudicate that standard; it supplies the concentration measurements against which the standard is applied. Read together, the theoretical companion accounts for why concentration is the structural equilibrium the design produces rather than a transient failure to finish decentralizing, and the normative companion specifies the conditions under which a given concentration distribution is or is not legitimate.

1.3 Research Questions

Three empirical research questions guide the analysis:

RQ1: After PCA exclusion, which cross-sectional protocol features are associated with post-distribution holder concentration (HHI)?

RQ2: How does delegated voting power deviate from post-exclusion token holdings, and is amplification associated with sector or governance venue?

RQ3: Which measurement and design implications follow for governance audits that rely on launch-time allocation or naive holder lists?

1.4 Scope, Assumptions, and Limitations

1.4.1 Scope

The empirical analysis focuses on tokenized systems between 2017 and 2025 across DeFi, infrastructure protocols, DAOs, and DePIN networks. The core analytic target is **governance**

concentration measurement and its associations with protocol characteristics, not short-term market performance.

1.4.2 Assumptions

- **Holder lists require classification.** Addresses that custody, burn, lock, bridge, or treasury-hold tokens should not be treated as independent governance holders.
- **Snapshots measure states, not trajectories.** The March 2026 cross-section supports association and comparison, not causal claims about deconcentration over time.
- **Voting power can diverge from holdings.** Delegation and vote-escrow mechanics require separate voting-HHI measurement where data are available.

1.4.3 Limitations (preview)

Three classes of limitation deserve early notice. Measurement scope: concentration is measured from holder lists and governance voting surfaces after PCA exclusion; informal influence, off-chain coordination, and proposal-specific quorum rules are not fully captured. Causal inference scope: a March 2026 cross-section supports descriptive association, not causal identification. Data limitations: voting-HHI is available for eighteen protocols, PCA classification is weaker on some Solana- and Substrate-native tokens, and the sample is still modest for fully specified multivariate models. The statistical significance, though not the direction, of the sector contrast is additionally load-bearing on the exchange-custody exclusion, developed in Sections 4.6.2 and 5.7. These limitations are developed in Sections 5.3 and 5.7 with the relevant robustness analyses.

1.5 Paper Outline

Section 2 reviews related work. Section 3 specifies data, PCA exclusion methodology, and statistical tests. Section 4 presents results. Section 5 discusses design implications and limitations. Section 6 concludes. Supplementary materials provide replication scripts, robustness analyses, and the canonical statistics ledger (Supplementary File S0).

2 Related Work

Sections 2.1 through 2.7 synthesize related literature on token governance, institutional design, DePIN, and concentration measurement. Section 3 specifies the empirical pipeline, PCA exclusion methodology, and measurement instruments used in Sections 4 and 5. Readers focused on the empirical contribution may read Section 3 first.

2.1 Overview

Three measurement gaps in the token-governance literature motivate this paper. First, prior holder-list HHI computations have not systematically excluded protocol-controlled addresses (staking contracts, treasuries, bridge custody, exchange wallets), inflating concentration estimates by a median 2.3x and up to 18x in this paper’s sample. Second, the cross-sectional relationship between launch-time token allocation and post-distribution concentration has not been tested with sample size sufficient for null findings to be informative. Third, the gap between holding HHI and delegated voting HHI has not been measured across sectors using comparable governance data surfaces (Tally, Snapshot, Solana on-chain governance).

The literature spans voting-power concentration in major DAOs (Fritsch et al., 2024), insider trading by proposal managers (Cong et al., 2025), the “decentralization illusion” (Aramonte et al., 2021), and structural concentration patterns across categories (Barbereau et al., 2023; Feichtinger et al., 2023). The subsections below review the strands relevant to each gap: tokenomics as mechanism and institutional design (§2.2-2.3), DAO governance concentration and delegation (§2.4), DePIN tokenomics (§2.5), constitutional and polycentric framings (§2.6), and synthesis (§2.7).

2.2 Tokenomics and Cryptoeconomic Design

2.2.1 Tokenomics as a Mechanism Design Problem

Tokenomics can be understood as a form of mechanism design (Hurwicz, 1973; Myerson, 1981) and, more broadly, as an exercise in institutional invention (Hurwicz, 1987): the “reverse engineering” of game theory in which a designer specifies rules and incentives to induce desired behavior among strategic agents (users, validators, operators, token holders). Token-governance protocols are explicitly invented institutions, designed from whitepaper plus code rather than evolved from social practice, which sharpens the design question Hurwicz 1987 poses for invented institutions generally: any implementable mechanism must satisfy incentive compatibility (participants find truthful participation in their interest) and informational feasibility (the mechanism cannot require information no participant possesses). The mechanism design literature provides foundational formal tools for reasoning about incentive compatibility, equilibrium selection, and adverse selection. These tools remain highly relevant when tokenized systems must function under manipulation pressure and incomplete information. The Nobel Prize’s summary of mechanism design theory emphasizes incentive-compatible direct mechanisms and the revelation principle as foundational concepts, offering a bridge between classical economics and modern crypto-economic system design.

Token systems, however, often introduce additional constraints absent from classical mechanism design (Roughgarden, 2021) settings: open entry, pseudonymity, composability, and adversarial execution environments. These constraints place a premium on robust incentive design, fraud resistance, and governance structures that can adapt without becoming arbitrary or captured.

2.2.2 Transaction Fee Mechanisms and “Sinks” as Economic Policy

Some of the clearest illustrations of tokenomics-as-policy appear in fee mechanism design. Ethereum’s EIP-1559 (Buterin et al., 2019; an Ethereum Improvement Proposal is a community-authored specification document defining a protocol change for Ethereum, analogous to a constitutional amendment for the system’s economic rules), for instance, introduced a base fee adjusted by network demand and burned rather than paid to miners/validators, creating a direct “sink” (a mechanism that permanently removes tokens from circulation, reducing supply and tying the token’s value to system usage) linked to usage and shaping the economic properties of the asset. Tim Roughgarden’s work on transaction fee mechanism design treats EIP-1559 not as marketing but as mechanism engineering: explicitly analyzing goals and trade-offs, including alternative designs such as smoothing and revenue-forwarding.

Subsequent research (Roughgarden, 2020) highlights that even well-designed sinks can introduce new strategic behaviors (e.g., manipulation or minority attacks on base fee dynamics), underscoring that tokenomics is not "set-and-forget" but a governance problem requiring transparency and monitoring.

These studies motivate a broader design intuition central to this paper: token mechanics that tie usage to scarcity (through burns, locks, buybacks, or fee routing) represent a form of fiscal policy embedded in code (cf. Werner et al., 2022).

The leading formal model of token adoption (Cong, Li, & Wang, 2021) shows that equilibrium token price is determined by aggregating heterogeneous users' transactional demand rather than by initial supply distribution, implying that post-distribution dynamics dominate design-time allocation choices. This theoretical result anticipates the allocation design null documented in Section 4.4.

2.3 Incentives, Liquidity Mining, and the "Mercenary Capital" Debate

A major wave of Web3 growth (notably in DeFi) was catalyzed by token distribution mechanisms: initial coin offerings (Howell, Niessner, & Yermack, 2020), liquidity mining (issuing newly-minted tokens to users who deposit assets into a protocol, as a recruitment incentive), staking rewards (Schär, 2021; issuing tokens to participants who lock existing tokens to secure the protocol), and user rebates, designed to bootstrap two-sided markets. A key controversy is whether these incentives generate durable adoption or merely transient participation driven by extrinsic rewards.

Empirical and conceptual analyses increasingly emphasize the risk of "mercenary capital" (Schär, 2021): incentives attract liquidity or usage temporarily, but participation exits when emissions decline, leaving weak organic demand. While the quality of evidence varies across sources, the consistent theoretical point is that token incentives can create short-run growth while undermining long-run sustainability unless paired with credible sinks, retention design, and governance discipline. This paper integrates that debate into a more general claim: **incentives are political economy**, not simply growth marketing, because they distribute resources, create winners and losers, and can entrench power.

Academic work on liquidity mining (Schär, 2021; Roughgarden, 2021) explicitly treats it as an innovative mechanism that changes investor behavior and protocol dynamics, suggesting a need for systematic evaluation rather than anecdotal interpretation.

2.4 DAO Governance: Concentration, Delegation, and the "Decentralization Illusion"

The legitimacy of tokenized governance depends not only on formal voting rules but also on power distribution and participation structure (Aramonte et al., 2021; Nadler & Schär, 2020). Decentralized Autonomous Organizations (DAOs; on-chain organizations whose governance rules and treasury controls are encoded in smart contracts and exercised via token-weighted voting) and similar token-governed protocols face a central challenge: token-based governance can produce concentrated influence (Barbereau et al., 2023) including "whales" (very large individual holders), delegates (parties who receive voting power from many other holders), or insiders, leading to outcomes that are formally decentralized but substantively centralized. The political-theory foundations of decentralized autonomous organizations have been developed across two complementary lines of work: an early political-theory line (DuPont, 2014, “The Politics of Cryptography”; Atzori, 2017) treating DAOs as new constitutional sites, and a more recent commons-governance line (Schneider, 2024, “Governable Spaces”) that engages Ostromian polycentricity directly and provides a synthesis between DAO design and commons-management scholarship. This paper uses that literature to motivate measurement of who can actually exercise governance power. The separation of ownership from control has been a central concern of corporate-governance theory since Berle and Means (1932): nominal ownership shares do not by themselves determine who directs an organization (Shleifer & Vishny, 1997). Delegated token governance reproduces that separation and the principal-agent costs it creates (Jensen & Meckling, 1976; Tirole, 2001, on managerial accountability and minority protection), so holder lists and voting power must be measured separately rather than assumed to coincide.

The Bank for International Settlements (BIS) argues (Aramonte et al., 2021) that DeFi contains a “decentralization illusion,” emphasizing that protocols marketed as decentralized frequently concentrate decision-making power among a small number of token holders, developers, or validators. This critique is significant for this paper because it links governance concentration to systemic risks and questions the legitimacy of “decentralized” claims. Walch (2019) sharpens the point from the legal side: the label “decentralized” can function as a veil that obscures identifiable sites of concentrated power and the responsibility attaching to them, so a corrected concentration measurement is what replaces the label with evidence.

Gersbach, Schneider, and Shahkar (2024) analyze vote delegation and find it can meaningfully change outcomes under highly unbalanced voting rights but has negligible impact under balanced systems, suggesting delegation is not a universal solution to legitimacy problems. Weidener et al. (2025) provide a systematic scoping review of delegated voting in DAOs, synthesizing that voting power concentration in DAOs typically exceeds token-holding concentration because delegation chains amplify the effective influence of large delegates beyond their direct token holdings, reinforcing that delegation can either mitigate or entrench concentration depending on design. Hall and Miyazaki (2024) analyze liquid democracy across 18 DAOs, finding that only 17% of tokens are delegated yet the top five delegates routinely capture over 50% of effectively voted weight, with high-SSPI delegates more likely to signal early and cascade alignment; the 2.5x to 25.6x delegation amplification ratios documented in Section 4.5 are consistent with this clumping pattern. A loose analog appears in corporate governance, where a small number of professional proxy advisors shape voting outcomes across many firms: Malenko and Shen (2016) estimate by regression discontinuity that a negative recommendation from a single advisor can move say-on-pay voting support by roughly twenty-five percentage points. The parallel is imperfect, but it suggests treating delegate programs as instruments that require concentration measurement of their own rather than as a presumed source of dispersion. Esposito, Tse, and Goh (2025) evaluate quadratic voting and delegated governance against Ostrom’s common-pool resource (CPR) framework, documenting that tokenization alone is insufficient for inclusive governance.

Most directly relevant, Cong, Rabetti, Wang, and Yan (2025) document that blockvoters control 76.2 percent of voting power across the DAOs they study, while proposal managers engage in insider trading yielding an average market-adjusted return of 9.5 percent. They argue that the combination of concentrated voting power and insider trading is value-destroying in the long run, particularly during crisis periods when governance becomes central to organizational survival. These stylized facts on persistent concentration in DAOs and governance token trading dynamics reinforce the need for institutional safeguards rather than assuming "token voting" equals democracy. Panel evidence from DAO-quarter data documents that effective governance control concentrates as proposal volume scales, consistent with monitoring costs outpacing broad-based participation (Tchuente, 2025); the cross-sectional allocation null documented here (Pearson $r = 0.07$, $p = 0.68$, $N = 37$) is compatible with this scale-driven concentration mechanism operating independently of initial token distribution.

Calzada (2025) corroborates these patterns ethnographically across seven Web3 ecosystems, documenting whale dominance and voter turnout below 10% in major DAOs.

The governance concentration patterns documented in Section 3 are consistent with Williamson's (1999) "probity hazard": token holders can comply with smart contract rules while extracting value that contradicts the protocol's institutional mission, a risk that intensifies as governance power concentrates in fewer addresses.

Voting power inequality in Compound, Uniswap, and ENS exceeds wealth inequality in real-world economies, with voting power more unequally distributed than income in any country (Fritsch, Müller, & Wattenhofer, 2024), a pattern consistent with the delegation amplification documented in Section 4.5; notably, these concentrated delegates rarely overturn community-preferred outcomes, suggesting that concentration creates domination risk rather than routine domination in the sense the republican tradition makes precise: on Pettit's (1997) account of freedom as non-domination, the standing capacity for arbitrary interference constitutes domination even when that power sits unexercised, so the dependent position of other participants is a structural concern and not only a matter of observed behavior. Across 10,639 proposals in 151 DAOs, three or fewer entities exert effective control over most governance decisions (Appel & Grennan, 2023), consistent with the insider dominance this paper documents at the token-holder level.

Together, these literatures (see also Feichtinger et al., 2023; Rikken et al., 2023) motivate two design imperatives advanced in later chapters: (i) governance systems require **contestability and due process** beyond raw voting, a concern sharpened by work asking whether on-chain governance mechanisms operate as genuine accountability protocols or only nominal ones (Nabben & De Filippi, 2024), and (ii) legitimacy depends on measurable concentration constraints and multi-constituency representation.

Han, Lee, and Li (2023) formalize the governance consequences of ownership concentration, showing that whale dominance reduces platform growth but that staking mechanisms and token illiquidity can mitigate it. The finding that ownership concentration is structurally harmful, not merely descriptive, motivates the normative framework developed here: if concentration damages governance quality, the political-philosophical traditions that specify what good governance requires become directly relevant to token system design. Han, Lee, and Li (2024) provide a comprehensive review of DAO governance mechanisms and the emerging agency problem between large and small token holders.

2.4.1 Cooperatives as the Nearest Institutional Ancestor

Among established institutional forms, the cooperative bears the closest structural resemblance to the tokenized protocol. Both organize collective production without a single owner who captures surplus; both distribute governance rights to participants rather than to external shareholders; both face the challenge of sustaining voluntary participation under conditions where exit is available. The cooperative literature (Hansmann, 1996; Birchall, 2011) identifies these properties as the defining features of member-owned enterprise, distinguishing cooperatives from investor-owned firms. The International Cooperative Alliance’s seven cooperative principles (ICA, 1995), including voluntary membership, democratic member control, and concern for community, map onto legitimacy constraints derived from political philosophy.

The clearest contemporary illustration is SporkDAO, the Colorado Limited Cooperative Association behind the ETHDenver conference. SporkDAO tokenizes patronage rather than selling ownership: members earn $SPORK by contributing to the ETHDenver ecosystem, stake it to activate membership, and cast staked balances in quadratically weighted Snapshot elections for a nine-member Board of Stewards that carries the association’s legal and financial responsibility. Elections run on an annual cycle tied to the conference, which serves as the cooperative’s members’ meeting, and the membership is broad, numbering tens of thousands of token holders. The case shows why the cooperative is the nearest institutional ancestor for tokenized protocols, and also why the analogy is incomplete. Legal cooperative form, contribution-based token issuance, and annual member meetings do not by themselves settle the distribution of live governance power; operational custody, board responsibility, staking design, and concentrated multisignature and treasury balances still have to be measured directly. Tokenized protocols do not merely resemble cooperatives; in cases like SporkDAO they can become cooperatives in law. But tokenization reopens the cooperative legitimacy problem at the level of live token custody, staking, and delegated voting power, which is what the empirical results in Section 3 set out to measure.

Five structural differences separate protocols from cooperatives. First, cooperatives require application and capital contribution; protocols permit permissionless entry under pseudonymity. Second, cooperative ownership is purchased; well-designed protocol ownership is earned through contribution (staking, hardware deployment, service delivery), a property Scholz (2016) identifies as the platform-cooperativism premise. The governance concentration data presented in Section 3 indicate that most protocols have not collapsed the capital-labor boundary in practice: holding HHI values span from cooperative-grade distribution (Anyone Protocol 0.013) to single-party control. Third, cooperatives enforce through bylaws, elected boards, and courts; protocols enforce through code, gaining credible neutrality at the cost of contextual judgment unless appeal mechanisms are designed in. Fourth, cooperatives operate within one jurisdiction’s cooperative law; protocols operate across jurisdictions simultaneously, making Ostromian polycentricity (developed in Section 2.6) a structural necessity rather than a design preference. Fifth, cooperative fiscal discipline operates through dues and patronage dividends; protocol fiscal discipline operates through emissions and burns, with the Spend-to-Reward ratio the protocol equivalent of cooperative fiscal sustainability.

These distinctions define the contribution space: protocols inherit the cooperative’s legitimacy challenge (governing collective production without a centralized principal) while adding permissionless entry, programmable enforcement, and borderless jurisdiction as additional design constraints. The polycentric framework in Section 2.6 takes up this expanded design space; the empirical results in Section 3 indicate how far current protocols remain from satisfying it.

2.5 DePIN and Tokenized Physical Infrastructure: Demand Coupling and Service Verification

DePIN expand tokenized coordination into the physical world (Bane & Gala, 2025, 2026): hardware deployment, service provision, and maintenance. DePIN intensifies the institutional nature of tokenomics because it coordinates capital formation (hardware investment), labor (operations), and service quality (SLA performance; SLA stands for service-level agreement, a contractually-specified threshold for network uptime, latency, or coverage that operators must meet to earn rewards). The result is a system that resembles public infrastructure governance more than application monetization.

Recent taxonomies formalize DePIN as a distinct coordination category with three architectural layers spanning distributed ledger, cryptoeconomic, and physical infrastructure dimensions (Ballandies et al., 2023), while Alshater (2026) synthesizes the token flywheel mechanism through which burn-and-mint equilibria (a DePIN-specific design pattern in which service users burn tokens to pay for service while operators receive newly-minted tokens as reward; the balance between burns and mints determines whether token supply contracts or expands with demand) bootstrap supply-side hardware deployment.

A key design pattern in DePIN is dual-asset economics (Voshmgir, 2020) separating speculative governance/stake tokens from stable-value usage credits. Helium provides a prominent example: network usage is paid in Data Credits, a non-transferable utility token used for service payments and onboarding actions.
This dual-asset structure motivates a general principle developed later: stable-value usage pricing reduces user friction, while governance/stake tokens enable long-term alignment and security. Separate simulation work stress-tests DePIN mechanism designs under adversarial conditions using cadCAD (a Python-based simulation framework), showing that mechanism-level choices (emission model, slashing parameters where “slashing” means automated penalty mechanisms that confiscate a validator’s staked tokens for malicious or negligent behavior) produce macro-level institutional outcomes that the normative framework developed here can evaluate.

2.6 Constitutional Political Economy and Polycentric Governance

2.6.1 Constitutions as Rule-Selection Under Constraints

Constitutional political economy (Buchanan & Tullock, 1962) treats political rules as objects of analysis and choice rather than fixed background conditions. Buchanan and Tullock frame constitutions as foundational constraints shaping later policy choices and collective decisions; this resonates with tokenized systems where “hard parameters” (e.g., inflation bounds, timelocks, emergency powers) are embedded in code. Buchanan and Tullock explicitly describe their work as analyzing the political organization of a society of free persons, using economic reasoning to examine constitutional constraints.

This literature (Buchanan & Tullock, 1962; extended in Brennan & Buchanan, 1985, on rule-following and the asymmetry between rules and the discretionary decisions they constrain; North, 1990, on the role of institutions as humanly-devised constraints structuring economic exchange) informs later chapters' distinction between a protocol's "constitutional layer" (hard-to-change constraints) and "policy layer" (adjustable parameters).

2.6.2 Ostrom and Polycentricity

Ostrom's (1990) work provides one of the strongest empirical foundations for governance of commons and public service industries, building on and overturning Hardin's (1968) "tragedy of the commons" by documenting sustained commons-management institutions that avoid both privatization and centralized state control. Her concept of polycentric governance emphasizes multiple overlapping decision centers, local experimentation, and rule diversity. Ostrom's AER article explicitly characterizes polycentric systems as alternatives to the binary "market vs state" framing, offering an analytical framework for complex economic systems with multi-level governance. Rozas et al. (2021) situate blockchain's institutional affordances against Ostrom's design principles directly, arguing the technology can support commons self-governance without collapsing into either techno-determinism or the view that centralization is necessary. Van Vulpen and Jansen (2023) map Ostrom's eight design principles to DAO governance structures, providing a standardized evaluation framework for collective action sustainability. Li and Chen (2024) conceptualize DAOs as collective stewards of digital commons, adapting Ostrom's design principles to emphasize equitable distribution of decision-making authority; the governance concentration data in Section 3 test whether operating protocols achieve the distributional outcomes this framing requires. De Filippi et al. (2024) analyze overlapping decision-making structures in blockchain systems through Ostrom's polycentricity framework, identifying structure, process, and outcome dimensions that map onto the governance concentration, delegation, and participation metrics measured here. This paper draws on this literature to motivate why governance concentration is worth measuring, not to score legitimacy directly; the optional legitimacy-scoring instruments are retained in the supplements.

Polycentricity (Ostrom, 1990) is particularly salient for DePIN, where service conditions differ across geography and where local knowledge is crucial for operational decisions. The paper later proposes polycentric sub-DAOs/councils as governance primitives for range-based or region-based tuning.

2.7 Synthesis and Research Gap

Across economics, computer science, political philosophy, and institutional analysis, a shared conclusion emerges: tokenized systems create constitutions and policies in code (Werner et al., 2022). Yet existing literature rarely measures who can exercise governance power after token distribution, PCA correction, and delegation. This paper addresses the gap empirically; §3 specifies the data and methods, §4 reports the results, and the four contributions stated in §1.1 are developed across them.

3 Data and Methods

3.1 Empirical Approach

The empirical pipeline has three parts. First, the paper constructs a 52-protocol holder-concentration cross-section and computes post-exclusion HHI, Gini, and top-holder shares. Second, it applies a protocol-controlled-address (PCA) exclusion typology before computing concentration, so staking contracts, burn addresses, treasuries, vesting contracts, bridge custody, and exchange custody do not count as independent governance holders. Third, it computes voting-HHI for protocols with sufficient delegation or voting data and compares voting concentration with post-exclusion holding concentration.

The analysis is descriptive. It tests whether observed protocol characteristics are associated with concentration in a March 2026 snapshot. It does not claim causal identification. Longitudinal panel and event-study designs are specified as future work.

Sample flow (March 2026 snapshot). The full cross-section includes N = 52 protocols with post-exclusion holding HHI (Table 3). The allocation battery uses N = 37 protocols with initial insider allocation data. The headline DePIN-vs-DeFi contrast uses 15 DePIN and 15 DeFi protocols. Subsidy analysis uses N = 23 protocols with non-zero subsidy under either Token Terminal or on-chain metrics, and N = 22 when Livepeer is excluded. Voting-HHI comparison uses N = 18 protocols with sufficient governance data (Table 6). In market-capitalization terms, the sampled DeFi and DePIN protocols aggregate to approximately $20.5 billion and $2.7 billion respectively at the snapshot, on the order of one-third of the DeFi category and one-fifth to one-quarter of the DePIN category against contemporaneous CoinGecko category totals (DeFi approximately $56 billion, DePIN approximately $9.4 billion in early 2026); the DePIN share is sensitive to category-boundary differences across aggregators (for example, whether Bittensor is classed as DePIN or, as here, as an infrastructure layer-1).

3.2 Units of Analysis and Sampling Strategy

Primary unit: **protocol-token system** (including its governance institutions). Secondary unit: **governance episodes/events** (emission changes, fee switches, unlocks, enforcement controversies, governance reforms).

Sampling principles. Protocols were selected to (i) span multiple categories (L1/L2, DeFi, infrastructure/oracles, DePIN, social), (ii) include “dead” or stagnated projects to reduce survivorship bias, and (iii) stratify by chain and launch epoch to control for regime effects.

Sample-selection criteria. Protocols were chosen to maximize variation across three dimensions: sector (15 DeFi, 15 DePIN, 8 L1/L2 infrastructure protocols where L1 denotes a base-layer blockchain like Ethereum and L2 denotes a scaling-layer chain that settles to L1, 2 failed or stagnated protocols included to avoid studying only survivors), maturity (2017-2024 launches), and governance model (token-weighted, delegate, futarchy where futarchy means governance through prediction-market betting on the outcomes of proposed policies). The expanded 52-protocol sample was constructed by adding protocols from the Dune Analytics ecosystem (Dune is a community blockchain-data platform where analysts publish SQL queries that aggregate on-chain transactions into human-readable tables) that met three criteria: (i) publicly traded governance token, (ii) holder data queryable via Dune or Helius DAS API (a Solana-blockchain data service that provides holder-balance queries via the Digital Asset Standard), and (iii) at least 50 non-zero-balance holders after protocol-controlled address exclusion. MetaDAO is included as the sole futarchy-governance protocol in the sample. Solana-native tokens (META, DRIFT, GRASS, W) required a separate data collection pipeline (Helius DAS API) because the primary Ethereum-focused indexer undercounted Solana holders by two orders of magnitude; details are in Supplementary File S5.

Holder-list cutoff. The top-1,000 holder cutoff follows established convention in the token-governance concentration literature (Sai et al., 2021 use comparable cutoffs across blockchain governance studies). The cutoff captures the holder population materially relevant to governance decisions; long-tail holders below this threshold collectively hold less than five percent of supply across all sample protocols. Robustness across cutoffs is supported by the Top-1%, Top-5%, and Top-10% columns in Table 3: rank-ordering across protocols is preserved at all three cutoffs (Spearman rho > 0.95 between any two columns), indicating that absolute HHI values may shift modestly with deeper-tail inclusion but cross-protocol comparison remains robust to the cutoff choice. The canonical methodology pulls top-(1000 + number of PCAs) holders by balance and applies PCA exclusion to yield 1000 post-exclusion non-PCA holders; Table 3 N column reflects this pre-exclusion sample size (e.g., Optimism N = 1007 reflects top-1007 pre-exclusion sample with 7 PCAs excluded to yield 1000 non-PCA holders for HHI computation). For protocols with fewer than 1,000 total unique non-zero holders (e.g., Token Engineering Commons with N = 742), the actual holder count is reported as the ceiling rather than the 1,000+PCAs convention.

Gini as ledger-architecture artifact, not governance signal. Inequality (Gini) is near-universal across all sectors (range 0.52 to 0.99), while concentration (HHI) varies by an order of magnitude (0.005 to 0.199 after excluding protocol-controlled addresses). Prior work applying the Gini coefficient to eight major cryptocurrencies found inequality levels comparable to real-world economies (Sai, Buckley, & Le Gear, 2021); because permissionless address creation produces millions of near-zero-balance wallets that inflate Gini toward 1.0 regardless of governance structure, the near-universal inequality this sample documents is a measurement artifact of ledger architecture rather than a governance signal, reinforcing the case for HHI as the primary concentration metric.

Reading Table 3. Top-1% and Top-10% columns report literal cumulative shares: Top-1% is the supply share of the single largest holder; Top-10% is the cumulative share of the top 10 holders, both computed as percentages of the post-exclusion top-1,000-holder sample total. The Top-5% column (in the canonical CSV; not displayed in this table) reports the top 5 holders cumulative share under the same convention. For protocols with fewer than 1,000 unique non-zero holders after exclusion, the same literal-top-N convention applies to the actual N. For protocols with more than 1,000 unique non-zero holders post-exclusion, the top-1,000 holders are sampled and the literal top-1 and top-10 shares within that top-1,000 sample are reported. The HHI and Top-N values in Table 3 reflect the canonical top-1,000 pre-exclusion computation; strict application of the (1,000 + PCAs) pre-exclusion convention reported in the N column would yield HHI shifts below 0.0001 and Top-N shifts below 0.1 percentage points, immaterial for the cross-protocol comparisons reported in Section 4.

3.3 Data Sources (Replicable, Multi-Disciplinary)

Six source streams inform the analysis. (1) On-chain holder and balance data: Dune Analytics (EVM chains) and the Helius DAS API (Solana SPL tokens), with the Sim API used for Solana balance and token-count verification. (2) Entity and address labels for protocol-controlled-address classification: Etherscan public name tags and the hildobby exchange-address compilation, Blockscout verified-contract labels, the Nansen Address Labels API, and the Polkawatch operator registry for Polkadot; on chains with sparse label coverage a transaction-pattern behavioral-signature method (Supplementary File S21) serves as a recovery layer, and centralized-exchange custody is confirmed against entity labels rather than token-count heuristics alone. (3) Governance and voting data: Tally (on-chain delegate voting), Snapshot (off-chain vote-weighted governance), and Realms (Solana SPL Governance), with Subscan (Polkadot conviction-weighted referenda) and Taostats (Bittensor validator stake) for the chain-native protocols. (4) Analytic aggregators: Token Terminal, DeFiLlama, and CoinGecko, with Blockworks dashboards supplying Helium and GEODNET financial data and Messari used to cross-validate Livepeer fee estimates. (5) Protocol documentation: whitepapers, governance forums, and tokenomics and parameter documentation. (6) Public academic and policy sources. A systematic coverage gap between DeFi and DePIN protocols (Token Terminal covers approximately 100% of DeFi but only about 25% of DePIN) constrains the cross-sector revenue regression. The protocols Token Terminal does not cover, the on-chain and aggregator fallbacks applied, and the resulting partial-data subsets are documented in Supplementary File S6.

Snapshot date discipline for governance-power data: the Table 6 voting HHI values are computed from a March 2026 snapshot of Tally and Snapshot governance data, consistent with the March 2026 holder-list snapshot used for Table 3 holding HHI. Voting-power distributions can drift materially over multi-month windows: replication runs using fresh Tally and Snapshot data should expect comparable but not identical voting HHI values, and rank ordering across protocols is the more durable inferential property than absolute magnitude. Section 4.6 reports a PCA-symmetric voting-HHI sensitivity check.

Voting-HHI sampling methodology. The Table 6 voting-HHI column uses comprehensive sampling on both Tally and Snapshot sides: Tally pulls top-1,000 delegates per protocol via paginated GraphQL queries (Aave pulled at all- delegates depth N = 150,911 as a methodology-floor sensitivity anchor); Snapshot pulls ALL proposals in the 12-month rolling window 2025-05-22 to 2026-05-22 and aggregates unique voters across all proposals weighted by maximum voting power per voter. For high-volume protocols (ARB, COMP, LDO), this comprehensive sampling yields large unique-voter counts (ARB N = 5,466; LDO N = 333; COMP N = 138). Low-volume protocols (DIMO and WeatherXM with 4 or fewer proposals total) experience no methodology-induced N expansion; DIMO N = 10 is genuine across the protocol's entire 12-month proposal history. Solana- native protocols (JUP, HNT, DRIFT) use on-chain governance program parsing because Tally is not deployed for Solana protocols (JUP via Dune `jupiter_solana.govern_call_setvote` per-signer max-weight aggregation; HNT and DRIFT via Helius RPC parsing of SPL Governance VoteRecordV2 accounts).

The asymmetric capping reflects what each platform's N represents: Tally caps at top-1,000 to truncate the deadweight long tail of registered-but-inactive delegates (validated as immaterial to HHI by the Aave all-delegates sensitivity anchor at N = 150,911); Snapshot and Solana on-chain

N are uncapped because every entry is by construction an active participant within the window, so no analogous truncation is warranted.

Protocol	Source	N	Notes
DIMO	Snapshot	10	all 4 proposals in window
Lido	Snapshot	333	all 41 proposals in window
WeatherXM	Snapshot	73	3 proposals in window
Compound	Snapshot	138	all 24 proposals in window
Aave	Tally	150,911	all-delegates pull (anchor)
Uniswap	Tally	1,000	top-1,000 delegate ceiling
Arbitrum	Snapshot	5,466	all 39 proposals in window
Optimism	Tally	1,000	top-1,000 delegate ceiling
GMX	Tally	1,000	top-1,000 delegate ceiling
ENS	Tally	1,000	top-1,000 delegate ceiling
Jupiter (JUP)	Solana on-chain	128,476	per-signer max-weight aggregation
Helium (HNT)	Solana on-chain	1,835	VSR position-state via Helius RPC
Drift (DRIFT)	Solana on-chain	641	VoteRecordV2 parsing via Helius RPC

Table 1. *Per-protocol active-voter sample size and methodology source for the Table 6 voting-HHI computation. Source-specific N reflects either Tally delegate-registry depth (top-1,000 ceiling; Aave at all-delegates anchor), Snapshot voter-pool union across the 12-month proposal window, or Solana on-chain governance program parsing.*

3.4 Codebook and Variables

Insider taxonomy: the analysis distinguishes two related but distinct constructs. Insider allocation (insider_pct) is the regression covariate measured at token generation event (TGE; the moment a protocol first mints and distributes its token to a defined set of recipients), defined as team_pct + investor_pct from primary tokenomics sources. Insider count fraction in top-10 holders is a descriptive mechanism variable measured post-distribution from holder snapshots, defined as the post-exclusion fraction of wallets in the top-10 classified as insider via an evidence-based procedure. Classification combines third-party on-chain entity labels (Nansen Address Labels), Blockscout verified-contract labels and Dune exchange tags, vesting-contract bytecode matching, and, for multi-signature wallets (Safe and GnosisSafe treasuries where several authorized parties must each sign a transaction to authorize fund movement), deployer-and-signer tracing. The governing rule for multi-signature wallets is team-confirmation: a Safe counts as insider only when its deployer or signing set resolves to the protocol team, founder, or a named investor, rather than treating any multisig as insider by default. This rule corrects a systematic over-count present in keyword-only classification, in which independent large holders' Safes were attributed to the team: of seventeen high-share bare Safes traced across the cross-section, six resolve to protocol teams and eleven to independent holders, with three deployer traces documented as worked exemplars in Supplementary File S22 (the insider classification of record and the seventeen-Safe verdict table) within the replication package (Section 5.9).

Protocol-controlled addresses (PCAs) are excluded from holding HHI computation entirely at the top of the pipeline (133 address exclusions across 38 protocols documented in the exclusions log; 125 unique addresses with five recurring across multiple protocols) and are not counted as insiders for the purposes of the count fraction. Solana-native protocols were re-classified against the same entity-label and deployer-tracing standard: the ambiguous large holders of the Solana-native tokens resolve to exchange, custody, and retail addresses rather than hidden insiders, so the conservative not-insider coding holds without insider undercounting. The earlier acknowledged labeling gap (io.net classifying as zero of ten insiders for want of Blockscout-style contract labels at the scale available on Ethereum and EVM (Ethereum Virtual Machine; the execution environment for Ethereum-compatible smart contracts) Layer 2 networks) is therefore a coverage rather than a substantive limitation; residual lower-attribution-coverage considerations are discussed in Section 5.7.

We code each protocol across standardized dimensions via a machine-readable table capturing metadata, supply and issuance rules, distribution, value routing and sinks, staking and lockup parameters, governance architecture, decentralization metrics, incentive windows, and data/identity posture. Full field definitions and the codebook schema are documented in Supplementary File S1. Optional governance-scoring instruments preserved during the broader project (Supplementary Files S2 and S3) are not used in the main-text regressions, tables, or headline findings.

3.5 Outcome Variables

Three primary metrics are used. The Herfindahl-Hirschman Index (HHI), ranging from 0 (dispersed) to 1 (monopoly), measures overall token concentration; values above 0.25 indicate high concentration by antitrust standards. Section 4.3 notes no protocol exceeds this threshold. The Gini coefficient captures distributional inequality. Top-N ratios (Top-1, Top-5, Top-10 share) complement HHI by identifying specific whale addresses.

Full metric definitions, computation procedures, and Dune query specifications are provided in Supplementary File S1.

3.6 Testability and Empirical Pipeline

The measurement claims are testable in principle through cross-sectional governance analysis (this paper), longitudinal panel models, event studies around governance decisions, and qualitative process tracing. This paper reports the cross-sectional governance concentration analysis. The full quantitative pipeline, including panel regressions, event-study specifications, and robustness protocols, is documented in supplementary material (Supplementary File S5) for replication and extension by future researchers.

3.7 Ethical and Practical Considerations

All data are drawn from public sources: on-chain transactions, published governance proposals, and open protocol documentation. No private or proprietary data are used. Governance token distribution data are computed from Dune Analytics multi-chain queries. The author served as Senior Investment Analyst at Borderless Capital, a cryptocurrency-focused investment firm, through February 2026 and retains no decision-making role, carried interest, or position-dependent compensation with the firm. The author may hold small personal positions in some of the protocols analyzed; positions were established outside the research period and are not position-dependent on Borderless Capital. A complete disclosure is provided in the Conflict of Interest statement accompanying this submission.

3.8 Generalizable Protocol-Controlled-Address (PCA) Exclusion Methodology

The protocol-controlled-address (PCA) exclusion methodology applied across Sections 3.2 through 3.4 generalizes beyond this paper's specific protocol set. Any cross-protocol governance HHI analysis on ERC-20 (the Ethereum standard for fungible tokens) or SPL (Solana Program Library; Solana's analog for fungible tokens) token holders should apply the five PCA classes documented below before computing concentration metrics. The methodology contribution is not just the specific exclusions applied to this paper's 52-protocol sample (the exclusions log contains the address-by-address documentation), but the generalizable typology that future replication and extension work can apply uniformly. The logic has a precedent in corporate-ownership analysis, which warns against treating the registered holder as the effective unit of control: ownership recorded at a single address may sit with custodians, nominees, or aggregation vehicles rather than an independent controller (La Porta, Lopez-de-Silanes, & Shleifer, 1999). PCA exclusion applies that control-identification logic to token holder lists.

Class 1: Burn-destination addresses. Canonical burn addresses (0x000…000 and 0x000…000dead on EVM chains; chain-specific patterns; protocol-specific burn destinations confirmed via official documentation, Blockscout/Etherscan labels, or Alchemy/Arkham labels) should be excluded from governance HHI computation. These addresses hold tokens that are inert by construction and cannot participate in governance under any condition; including them in the holding distribution inflates the holding denominator and understates effective concentration. The Uniswap 0x000…000dead exclusion (102.46M UNI, 11.27 percent of supply at measurement) is the canonical case in this paper; the Hyperliquid burn-pattern addresses (0x222…222 and 0xfefe…fefe) were caught by a confirmed-pattern methodology. The universal burn-rule applied here is generalizable to any ERC-20 or SPL token governance analysis.

Class 2: Foundation and treasury custody. Protocol-controlled foundation and treasury custody addresses should be excluded. These addresses hold tokens that are governance-relevant in principle (the foundation may delegate or vote with them) but are structurally distinct from independent holder concentration: the foundation position represents institutional rather than community ownership. Specific cases in this paper: Optimism Foundation GnosisSafe (0x2a82ae142…; 32 percent of pre-exclusion top-1000 OP); Arbitrum

FixedDelegateErc20Wallet (0xf3fc178157…; 30 percent of pre-exclusion top-1000 ARB); WeatherXM treasury (74.8 percent of pre-exclusion top-1000 WXM); DIMO Foundation (52.3 percent of pre-exclusion top-1000 DIMO). The methodology requires per-protocol identification typically via Blockscout verified-contract labels, GnosisSafeProxyFactory creation records, or foundation disclosures.

Class 3: Staking-aggregation contracts. Staking contracts that aggregate underlying staker holdings into a single custody address should be excluded from holding HHI computation, with the caveat that underlying stakers retain pass-through voting power. The stkAAVE staking contract (0x4da27a545c0c…; 21.9 percent of post-PCA-exclusion top-1000 AAVE) is the canonical case; AAVE stakers vote through stkAAVE rather than the contract itself making governance decisions. The sGMX RewardTracker plays an analogous role for GMX. The exclusion is methodologically warranted at the holding-HHI computation step because the contract is not an independent governance participant; the exclusion at the voting-HHI computation step (Section 4.6 PCA-symmetric robustness check) is methodologically distinct: stkAAVE votes as a single delegate per Tally's data model, even though the underlying voters are dispersed. The paper treats the holding-side exclusion as canonical and reports the voting-side exclusion as a sensitivity check.

Class 4: Bridge custody and migration addresses. Bridge custody addresses (Wormhole token bridge for cross-chain wrapped tokens; the GRT BridgeEscrow for L1-to-Arbitrum custody; LayerZero canonical bridges) hold tokens in transit or as cross-chain accounting positions; these are not governance-participating holders. Migration contracts (the SYRUP Migrator for MPL-to-SYRUP migration; the Polygon StakeManagerProxy for MATIC-to-POL transition) hold tokens during migration windows and should be excluded.

Class 5: Centralized exchange custody. Centralized exchange (CEX) hot and cold wallets that custody customer deposits should be excluded from governance HHI computation. Customer-deposited tokens held in CEX custody are not governance-participating holdings: the exchange holds custody of tokens belonging to many individual customers, and exchange operational practice is to abstain from governance voting on customer-held tokens. Specific cases in this paper: Binance hot wallets across OP, ENS, GRT, AAVE, COMP, UNI, ARB, and other protocols (the canonical Binance address 0xf977…41acec is the most common case); Bithumb 162 (0x54d1…22cbf9) and Upbit 59 (0x377b…190873) on AXL; analogous patterns on other protocols. CEX address identification relies on public name tags (Etherscan, Arkham, hildobby compilations) and behavioral signals (high transfer volume, many counterparties, no governance participation history).

Class 3 versus Class 5 disambiguation via transaction-pattern signatures. Class 3 (staking-aggregation contracts) and Class 5 (centralized exchange custody) share an underlying behavioral signal of many small transfers, but the directionality and turnover characteristics differentiate them at the transaction-pattern layer. Class 5 CEX hot wallets exhibit a *deposit-collection signature*: many small inbound transfers from independent counterparty addresses funneling to a single outbound consolidation, rapid turnover (minutes to hours; auto-rotation cadence diagnostic in canonical Binance and Coinbase patterns), and low-balance steady state with lifetime-cumulative-inflow to current-balance ratio commonly exceeding 10x. Class 3 staking-aggregation contracts exhibit the opposite directionality: a *batch-payout signature* of many small outbound transfers from a single protocol-controlled inbound source (the staking-rewards distribution mechanism), slow rotation matching the protocol's reward-distribution cadence, and high-balance steady state accumulating reward pools between distribution events (lifetime-inflow to current-balance ratio typically below 5x). The distinction is architecture-agnostic and applies uniformly to EVM staking contracts (stkAAVE; Lido stETH; Rocket Pool minipools), Solana SPL stake-pool accounts (Marinade msol; Jito jitoSOL), and Substrate-native nominator-validator-stake patterns (Polkawatch-attributed institutional staking-providers; protocol-pallet reward distributions). The Polkadot fifth-protocol extension established the canonical worked examples (Supplementary File S21): rank-2 `13Z7KjGn...` plus rank-5 `12ouvKS...` on AssetHub Polkadot classified as LIKELY-CEX via deposit-collection signature (Scenario C-narrow post-PCA-exclusion HHI = 0.0043); ranks 10/11/12 (`163egH5d-...` distributed staking-infrastructure beneficiaries) preserved as legitimate Class 3 via batch-payout signature. The methodology operates as the recovery layer when explicit entity attribution (Etherscan public name tags, Nansen Address Labels API, Polkawatch operator_id registry) is unavailable, particularly at lower-attribution-coverage chains (Section 5.7 third limitation); misclassification of a Class 3 staking-aggregation address as Class 5 CEX, or vice versa, materially affects exclusion magnitude and downstream HHI computation. S21 documents the full four-property signature definitions, diagnostic decision matrix, cross-architecture application notes (EVM + Solana + Substrate), and audit framework for systematic application across the full N=52 cross-section.

Across the five classes, this paper applies 133 PCA exclusions across 38 protocols (the supplementary exclusions log dataset). The five classes are unevenly populated within the cross-section: on the N = 30 DePIN-vs-DeFi sector contrast (125 exclusions across 36 protocols), Class 2 (foundation and treasury custody) accounts for 76 of the 125 exclusions (61 percent), Class 5 (CEX custody) for 19 (15 percent), Class 3 (staking-aggregation contracts) for 12 (10 percent), Class 1 (burn destinations) for 10 (8 percent), and Class 4 (bridge custody and migration) for 8 (6 percent). The Class 2 dominance has a practical implication for replication: future researchers operationalizing this methodology can prioritize Class 2 identification (label-based foundation-multisig detection plus on-chain treasury-contract bytecode matching) as the highest-yield exclusion class, with the remaining four classes contributing the residual 39 percent. The generalizable methodological contribution: any future cross-protocol governance HHI analysis should apply the five-class PCA exclusion uniformly before reporting holding concentration metrics. The specific addresses depend on the protocol sample; the methodology is sample-independent.

4 Results

4.1 Scope and Claims Boundary

The cross-section covers 52 protocols across four sectors (DePIN, DeFi, infrastructure, social). Table 2 delimits the claims boundary: all findings are descriptive within the observed scope. The governance data represent a March 2026 snapshot, not a time series. Causal claims about the relationship between institutional design and governance outcomes require the longitudinal evidence beyond the scope of this paper. All governance token distributions were queried from Dune Analytics multi-chain queries. Concentration metrics are point-in-time snapshots;

trajectory analysis requires the longitudinal extension specified in the Future Research discussion (Section 5.8).

Claim	Evidence Grade	Pointer
PCA exclusion corrects systematic HHI inflation	Methodological	Section 3.8
Allocation null with insider-retention contrast	Descriptive	Section 4.4; Table 4
DePIN concentration exceeds DeFi after correction	Descriptive	Section 4.3; Table 5
Delegation often amplifies voting above holdings	Descriptive	Section 4.5; Table 6
Subsidy association is Livepeer-driven only	Descriptive (null)	Section 4.6; S8

Table 2. *Claims boundary: what this paper does and does not claim.*

4.2 Cross-Protocol Concentration

Governance concentration spans two orders of magnitude across the 52-protocol sample (Table 3): HHI ranges from 0.005 (Hyperliquid) to 0.199 (Livepeer), with a median near 0.04. DePIN protocols cluster in the upper half of this distribution.

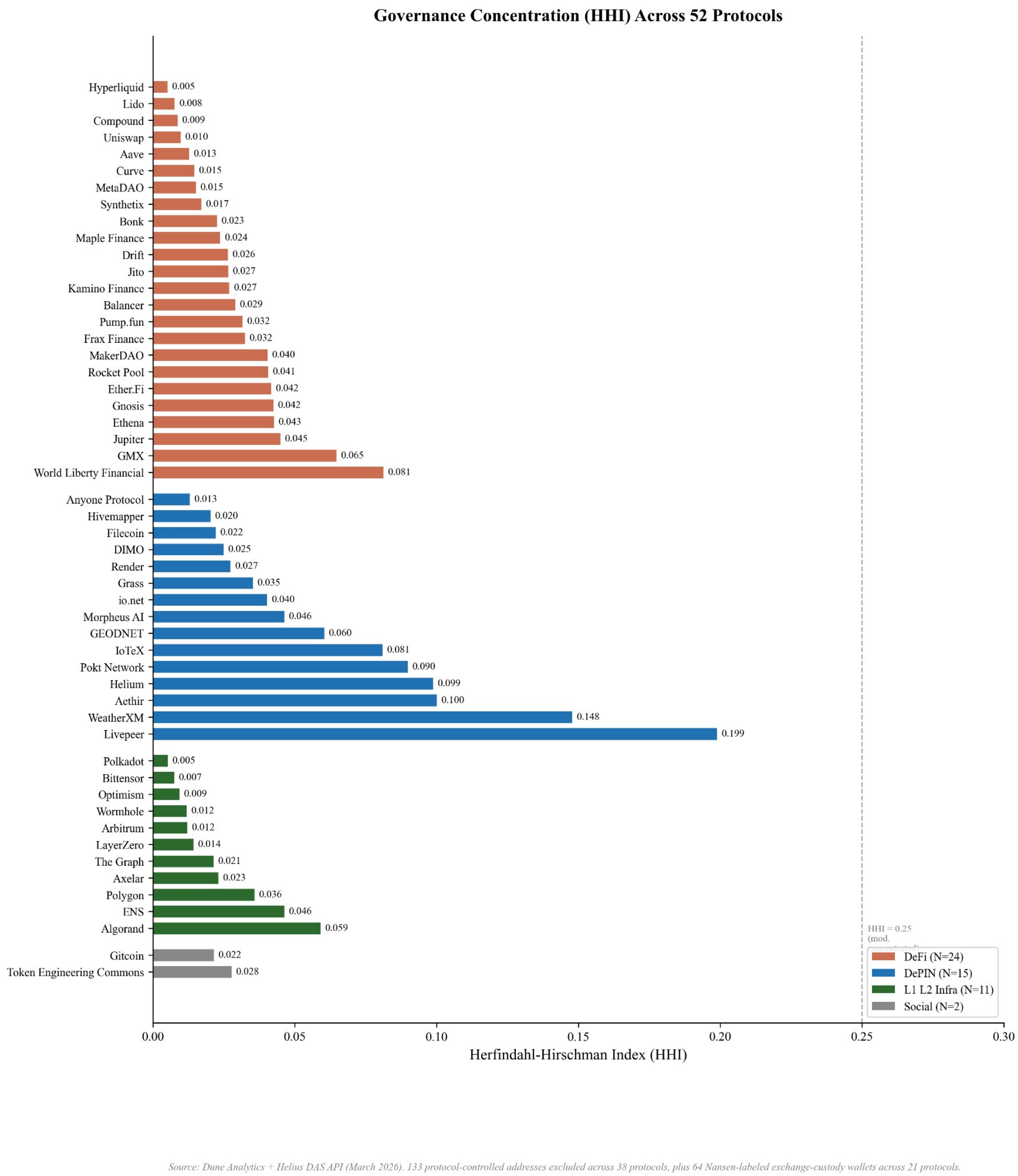


Figure 1. *Holding-based HHI across 52 governance tokens (March 2026 snapshot; top-1,000 holders; 133 protocol-controlled addresses excluded across 38 protocols per Section 3.8 five-class typology, plus 64 Nansen-labeled centralized-exchange deposit wallets excluded across 21 protocols per the 2026 exchange-custody completion audit, Supplementary File S13). Bars colored by category (DeFi orange; DePIN blue; L1/L2 Infrastructure green; Social grey); sorted by category then HHI ascending. Dashed vertical line at 0.25 marks the antitrust concentration threshold. Highest HHIs cluster in DePIN (Livepeer 0.199; WeatherXM 0.148; Aethir 0.100; Helium 0.099); DeFi and L1/L2 protocols sit predominantly below 0.05.*

A measurement caveat applies to interpretation: the 0.25 HHI antitrust threshold is borrowed from market-concentration analysis (DOJ Horizontal Merger Guidelines) and is reported here as a benchmark for descriptive interpretation only. A market-concentration threshold is not equivalent to a democratic-legitimacy threshold; a protocol may have HHI below 0.25 while remaining politically dominated through quorum rules (the minimum fraction of voting power required for a proposal to pass; a high quorum may concentrate decisive influence in whoever holds enough tokens to meet it), delegation patterns, or proposal-rights concentration. The HHI captures one dimension of concentration (post-exclusion holding distribution) and is not a sufficient condition for distributed governance.

Livepeer is the clearest structural outlier; excluding it leaves a distribution in which DePIN still exhibits higher concentration than DeFi but without a dominant leverage point.

Concentration metrics are computed from Dune Analytics multi-chain queries and the Helius DAS API for Solana SPL tokens (March 2026 snapshot, top-1000 holders per protocol), after applying 133 protocol-controlled address exclusions across 38 protocols (staking contracts, exchange custodians, vesting locks, protocol treasuries; see the exclusions log). Solana-native tokens use getTokenAccounts with page-based pagination to correct systematic undercounting in Dune's indexed tables. Replication scripts and input CSVs are listed in Supplementary File S5.

Three additional DePIN protocols are included to improve statistical power for the subsidy analysis: Aethir (ATH, Ethereum, GPU compute, $156M annualized revenue (Token Terminal, April 2026)), Hivemapper (HONEY, Solana, mapping), and io.net (IO, Solana, GPU compute, $21M revenue). Governance concentration was computed using the same methodology (Dune for ERC-20 tokens, Helius for Solana SPL tokens, top-1000 holders, exchange-labeled and protocol-controlled addresses excluded).

Protocol	Token	Category	HHI	Gini	Top-1	Top-10	N
Hivemapper	HONEY	DePIN	0.020	0.91	5.0%	32.7%	1000
Morpheus AI	MOR	DePIN	0.0464	0.88	12.6%	55.4%	1003
Render	RENDER	DePIN	0.0273	0.88	10.3%	43.3%	1001
Grass	GRASS	DePIN	0.0352	0.76	16.0%	38.2%	1000
DIMO	DIMO	DePIN	0.0249	0.85	5.9%	39.3%	1002
Anyone Protocol	ANYONE	DePIN	0.0130	0.71	4.9%	30.2%	1003
Filecoin	FIL	DePIN	0.0221	0.74	9.3%	35.6%	1001
Helium	HNT	DePIN	0.0988	0.98	22.1%	75.3%	1001
IoTeX	IOTX	DePIN	0.0810	0.96	12.3%	80.9%	1008
io.net	IO	DePIN	0.0402	0.94	12.3%	54.1%	1000

Protocol	Token	Category	HHI	Gini	Top-1	Top-10	N
GEODNET	GEOD	DePIN	0.0605	0.91	16.2%	62.3%	1004
WeatherXM	WXM	DePIN	0.1479	0.89	35.0%	77.9%	1001
Pokt Network	POKT	DePIN	0.0899	0.94	25.7%	66.7%	1001
Aethir	ATH	DePIN	0.1001	0.94	25.4%	60.1%	1008
Livepeer	LPT	DePIN	0.1989	0.94	42.9%	73.7%	1002
Hyperliquid	HYPE	DeFi	0.0052	0.65	2.4%	15.5%	1000
Lido	LDO	DeFi	0.0077	0.77	3.2%	18.3%	1009
Synthetix	SNX	DeFi	0.0171	0.90	8.4%	36.5%	1000
Bonk	BONK	DeFi	0.0226	0.86	7.9%	40.8%	995
Maple Finance	MPL_SYRUP	DeFi	0.0237	0.88	9.2%	39.0%	1002
Compound	COMP	DeFi	0.0086	0.84	3.4%	19.4%	1005
Jito	JTO	DeFi	0.0266	0.91	9.9%	41.2%	997
Balancer	BAL	DeFi	0.0290	0.87	13.2%	35.2%	1011
Kamino Finance	KMNO	DeFi	0.0269	0.93	8.0%	44.2%	999
Frax Finance	FXS	DeFi	0.0324	0.91	11.1%	50.3%	1000
Rocket Pool	RPL	DeFi	0.0406	0.87	12.1%	52.6%	1002
Pump.fun	PUMP	DeFi	0.0316	0.88	9.7%	47.9%	994
MakerDAO	MKR	DeFi	0.0404	0.88	13.9%	50.6%	1002
MetaDAO †	META	DeFi	0.0152	0.88	6.2%	28.6%	999
Gnosis	GNO	DeFi	0.0425	0.91	13.8%	51.7%	1000
Ethena	ENA	DeFi	0.0427	0.89 1	5.8%	47.4%	998
GMX	GMX	DeFi	0.0647	0.92	20.1%	59.3%	1004

Protocol	Token	Category	HHI	Gini	Top-1	Top-10	N
Aave	AAVE	DeFi	0.0128	0.83	4.4%	27.5%	1003
Ether.Fi	ETHFI	DeFi	0.0417	0.93	15.7%	47.8%	1002
Uniswap	UNI	DeFi	0.010	0.89	4.2%	22.9%	1002
Drift	DRIFT	DeFi	0.0265	0.92	10.8%	37.5%	1001
Jupiter	JUP	DeFi	0.0450	0.98	12.0%	58.3%	1001
Curve	CRV	DeFi	0.0146	0.79	6.7%	31.5%	1010
World Liberty	WLFI	DeFi	0.0812	0.94	25.9%	54.0%	994
Bittensor	TAO	L1/L2/Infra	0.0075	0.68	3.6%	25.9%	1000
Polkadot	DOT	L1/L2/Infra	0.0052	0.68	3.1%	21.0%	1000
Axelar	AXL	L1/L2/Infra	0.0231	0.85	8.7%	38.6%	1004
LayerZero	ZRO	L1/L2/Infra	0.0143	0.89	4.0%	29.9%	1002
Wormhole	W	L1/L2/Infra	0.0119	0.86	7.3%	18.5%	1001
Polygon	POL	L1/L2/Infra	0.0358	0.92	12.7%	49.4%	1002
Arbitrum	ARB	L1/L2/Infra	0.0122	0.73	8.9%	22.4%	1007
The Graph	GRT	L1/L2/Infra	0.0214	0.92	10.7%	32.9%	1004
Optimism	OP	L1/L2/Infra	0.0094	0.77	5.8%	21.1%	1007
ENS	ENS	L1/L2/Infra	0.0463	0.86	19.6%	38.6%	1004
Algorand	ALGO	L1/L2/Infra	0.0591	0.87	19.9%	54.6%	1000
Token Engineering Commons ‡	TEC	Social	0.0278	0.91	9.2%	44.2%	742
Gitcoin ‡	GTC	Social	0.0215	0.89	7.1%	36.8%	996

Table 3. *Governance concentration cross-section (March 2026 snapshot; Dune Analytics + Helius DAS API; top-(1,000 + number of PCAs) holders per token pre-exclusion, yielding 1,000 post-exclusion non-PCA holders for HHI computation; 133 protocol-controlled-address exclusions across 38 protocols per Section 3.8 five-class typology, plus a 2026 exchange-custody*

completion audit that excluded an additional 64 Nansen-entity-labeled centralized-exchange deposit wallets across 21 protocols, reflected in the HHIs reported here, per Supplementary File S13; N column reflects pre-exclusion sample size). Grouped by category (within-category rows are not strictly HHI-ordered). HHI values measure raw token-holding concentration, not voting power; vote-escrowed concentration (Curve veCRV, Balancer veBAL) is expected to be higher due to lock-duration multipliers (Section 4.5.2). Hyperliquid is excluded from the subsidy regression because 97% of trading fees route to an Assistance Fund that permanently burns HYPE tokens (sink-mechanism in framework). Aethir HHI (0.100) reflects May 2026 Dune top-1008 pull with eight PCA exclusions plus three Nansen-labeled exchange-custody exclusions; Hivemapper + io.net rows from Helius DAS API. Six tokens (WLFI, ENA, PUMP, JTO, BONK, KMNO) report raw top-1000 token-holder HHI (holder measurement) rather than governance-token HHI; they appear in this distribution but are excluded from the governance sector contrast in Section 4.3.

Table 3 notes:

† = futarchy governance (MetaDAO).

‡ (in protocol name column) = survivorship-bias control (TEC, Gitcoin).

Methodology notes: HHI for Curve reflects raw CRV holdings; veCRV-weighted voting concentration is approximately 0.26 computed from cumulative CRV deposit volume across the top-50 historical veCRV lockers (Convex Finance, an aggregator that pools user-deposited CRV into a single locked veCRV position and votes as a meta-governance layer on behalf of depositors, holds 48% of the top-50 deposit-weighted share; Yearn Finance, a yield-aggregation protocol that runs a similar meta-governance position, holds 14%; Stake DAO holds 9%; the remaining top-50 lockers each hold less than 4%); the methodology is discussed in Section 4.5.2. HHI for Uniswap reflects post-burn-rule exclusion of 0x000...dead (102.46M UNI, 11.27% of supply). Top-1% and Top-10% columns for AAVE, UNI, ARB, GRT, and OP report post-exclusion shares, applying the same protocol-controlled-address exclusion methodology used for HHI computation.

HHI ranges from 0.005 to 0.199 across the full sample. Among DeFi governance tokens, HHI ranges from 0.005 (Hyperliquid, after excluding the Assistance Fund and burn addresses) to 0.065 (GMX). Curve's raw CRV holding HHI of 0.014 reflects the post-Convex-exclusion distribution; the veCRV-weighted voting concentration is substantially higher (approximately 0.26 computed from cumulative deposit volume across the top-50 veCRV lockers; Section 4.5.2), and is discussed in Section 4.5.2. DePIN protocols span a similar range: 0.020 (Hivemapper) to 0.199 (Livepeer); Morpheus AI registers 0.046 after PCA exclusion (per Table 3 row 0.0464).

Several protocols exhibit much lower concentration in the post- exclusion top-1,000 sample than in the raw holder list. Applying the holder-list correction for Solana tokens (Helius DAS API enumeration versus the systematically-undercounting Dune indexer) and the PCA exclusion methodology (38 protocols with documented exclusions; Section 3.8), the DePIN-DeFi concentration gap is computed on a holder sample materially different from the naive top-1,000 alternative. For DRIFT, the raw holder distribution yields HHI 0.419 against the post-exclusion value of 0.026 (Helius enumerates 27,888 distinct holders versus 104 retrieved from the Dune indexer). Wormhole (W) moves from 0.126 (raw) to 0.012 (post-exclusion); GRASS moves from 0.132 to 0.035. The exclusion methodology (Section 3.8; the address-by-address exclusions log) is particularly consequential for protocols with large foundation treasuries or exchange custodian addresses in the top-10 holders.

Several protocols show that low governance concentration is achievable across sectors. Hyperliquid (HHI 0.005), Lido (0.008), Compound (0.009), and Uniswap (0.010) in DeFi; Optimism (0.009), Arbitrum (0.012), Wormhole (0.012), and LayerZero (0.014) in L1/L2 infrastructure; and Hivemapper (0.018) in DePIN all achieve HHI below 0.02. These counterexamples indicate that low governance concentration is achievable across multiple sectors, constraining explanations that treat high concentration as structurally inevitable in any single sector. The fourth sector, social token protocols, is too small in the sample ($N = 2$) to support independent sector-level claims.

Three measurement choices affect these results and merit attention as scope conditions. First, Top-N cutoffs at 1,000 holders may undercount long-tail holders; increasing the cutoff to 10,000 addresses would change absolute HHI values but is unlikely to change the rank ordering. Second, the snapshot methodology captures a single point in time; several protocols (notably DIMO) have announced token distribution programs that may reduce concentration over subsequent months. Third, the analysis uses raw token holdings; for protocols with delegation (Compound, Uniswap), delegated voting power may differ from raw holdings. We use delegated power where available, but not all protocols instrument delegation.

4.3 DePIN vs. DeFi Contrast

DePIN protocols have approximately 2.6 times the governance concentration of DeFi protocols (mean HHI 0.067 vs. 0.026, a ratio of 2.6; Mann-Whitney $p = 0.011$, Cohen's $d = 1.05$, $N = 15/15$). The gap holds at the conventional 0.05 threshold in Models 1 and 2 (Table 5) after adjusting for protocol age and valuation, and falls just above the threshold in Model 3 ($p = 0.078$) after adding the initial insider allocation control. Prior review work documents that DePIN sectors vary in tokenomic architecture (Alshater, 2026): storage networks prioritize staking, wireless coverage emphasizes service verification, compute networks focus on billing. The concentration gap between DePIN and DeFi is not implied by this sectoral variation; it is a separate finding about who holds governance tokens.

The sector difference is robust to sample composition: the result holds under leave-one-out, permutation, and bootstrap resampling (full battery in Section 4.6). One candidate mechanism: in capital-intensive networks, scale economies produce concentrated market shares (Arnosti & Weinberg, 2022). Fleet operators deploying hundreds of devices achieve lower per-unit costs than solo participants, consistent with the concentration gap observed here.

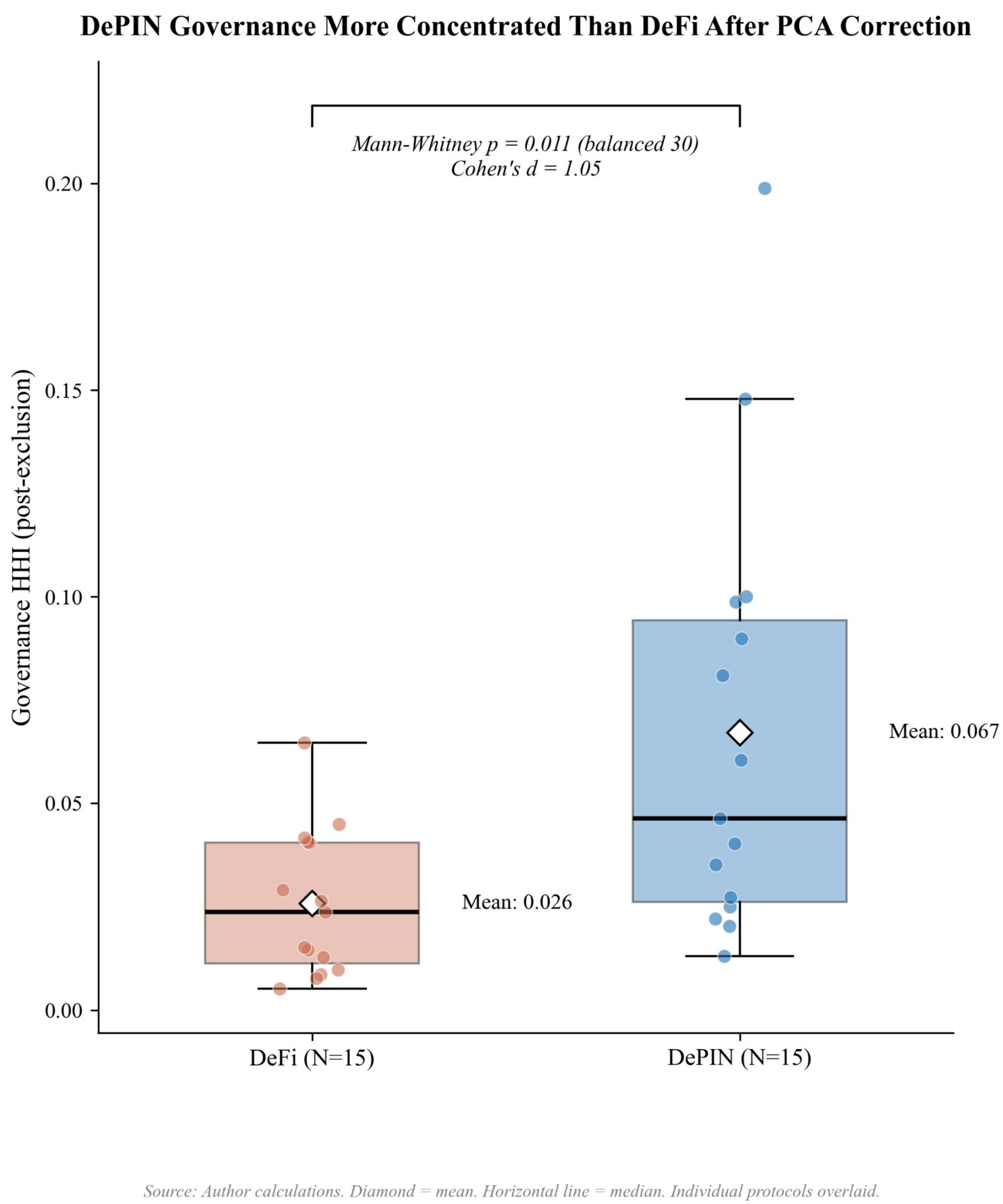


Figure 2. *Governance concentration (HHI) by sector. DePIN mean (0.067) exceeds DeFi mean (0.026), with distributions overlapping substantially: 9 of 15 DePIN protocols fall within the DeFi range. The Mann-Whitney test (p = 0.011, d = 1.05) is robust to sample composition (significant in all 30 leave-one-out iterations). Reading the figure: box plots show the concentration distribution within each sector; the DePIN-DeFi gap is visible in median position despite overlap in tails. The contrast uses a balanced sample (15 DePIN and 15 DeFi) to equalize the two groups for the nonparametric test; the sector effect does not depend on this balancing, holding on the full unbalanced cross-section (15 DePIN versus all 24 DeFi: Mann-Whitney p = 0.017, Cohen's d = 1.05) and generalizing in the three-class contrast (Figure 2b, N = 44) and the multivariate specification (Table 5, Model 4, N = 50).*

Beyond the balanced sector contrast, a multi-factor model addresses what concentrates governance across the full cross-section (Table 5, Model 4). On the covariate-complete sample ($N = 50$), architectural sector is the robust correlate: in the powered specification the DePIN coefficient is positive and statistically significant (log-HHI, $p = 0.0107$; same-signed and significant at $p = 0.019$ in untransformed HHI) and does not attenuate when revenue intensity and protocol maturity are added. At twelve and a half observations per predictor, this parsimonious specification clears the conventional observations-per-predictor floor that the prior, more heavily controlled specification did not. Revenue intensity, protocol maturity, and initial insider allocation are each uninformative, extending the launch-design null sweep to economic-activity intensity. Insider retention, the significant top-holder measure reported in Section 4.4, is reliably measurable only on the established-protocol cohort, where it survives the non-insider-HHI tautology check (Spearman rho = 0.54); it is not carried in the powered model because in the newer cohort insider holdings concentrate through addresses the exclusion methodology removes (foundation multisigs, exchange custody, and staking contracts), so the post-exclusion top-holder retention measure reads near zero by construction there. Whether that reflects a structural shift in how insiders hold rather than a weaker underlying mechanism is a question the cross-section cannot resolve, and is left to future work. The specification reported here substitutes protocol maturity for insider retention so that it reproduces over the full covariate-complete sample, and serves as the reproducible robustness anchor. With the insider-retention vector now re-derived and persisted under the evidence-traced classification of record (Section 3.4), the alternative specification that enters insider retention in place of maturity also reproduces a significant DePIN sector coefficient, with the insider-retention regressor itself not significant at the full sample (the cohort difference noted above); the sector result holds across six independent insider-classification schemes (Section 4.6.4). Both specifications are documented in the replication package described in Section 5.9.

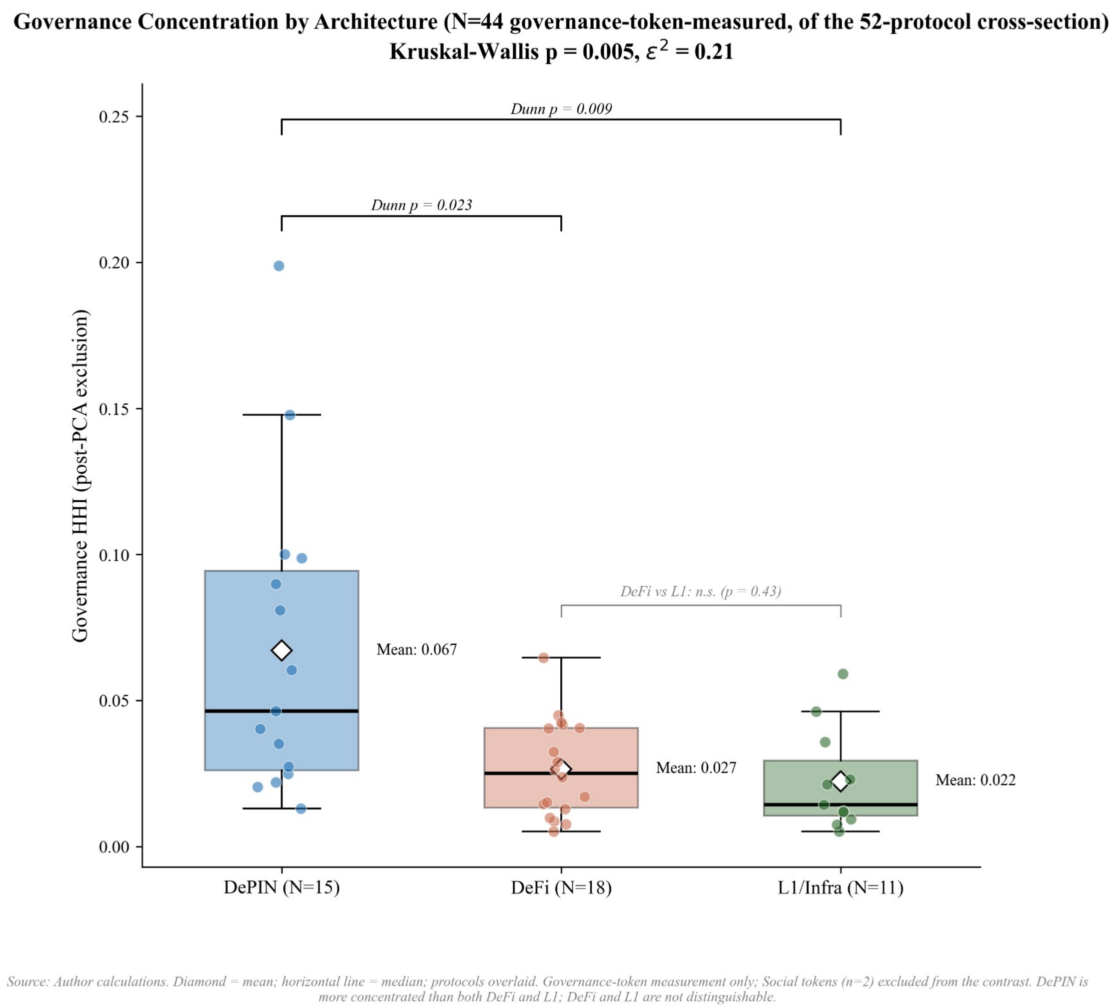


Figure 2b. *Post-exclusion HHI by architectural sector across the cross-section (DePIN, DeFi, and Layer-1/infrastructure). The distribution is descriptive; the inferential test of architecture as a concentration correlate, controlling for revenue intensity and protocol maturity, is the regression in Table 5, Model 4. Diamonds mark means; horizontal lines mark medians; individual protocols overlaid; governance-token measurement, social tokens excluded.*

Revenue dynamics reinforce this structural contrast: during 2025's market downturn, Helium's onchain revenue grew over 800% while leading DeFi protocol fees declined 27-70% over the same period (Bane & Gala, 2026; Blockworks, 2026b), suggesting that physical infrastructure revenue streams exhibit countercyclical properties absent in purely financial protocols.

This concentration gap is not explained by protocol maturity: Compound (2018 launch, HHI 0.009) and MakerDAO (2017, HHI 0.040) are older than DIMO (2022, HHI 0.025) and WeatherXM (2024, HHI 0.148), yet maintain far lower concentration. One candidate explanation is distribution design (Table 3): DePIN token allocations frequently reserve large shares for founding teams and early investors, while DeFi protocols have had longer periods of community distribution through liquidity mining, delegation, and governance participation incentives. However, the regression analysis below finds no significant correlation between insider allocation share and HHI (Section 4.4), suggesting that the distribution design channel operates through qualitative mechanisms not captured by the allocation percentage alone. The Anyone Protocol counterexample supports this interpretation: its low concentration (HHI 0.013) is consistent with governance mechanism choices (delegation incentives, activity-based caps, distribution schedule structure) that most DePIN protocols have not adopted.

This governance mechanism interpretation aligns with practitioner evidence from points-based token distribution programs (Sawinyh, 2025). Mature DeFi protocols distributed tokens through multi-season liquidity mining programs, safety module staking, and delegation incentives that implement sub-linear scaling (diminishing returns per marginal dollar deposited), activity-based caps, and time-weighted multipliers; these mechanisms structurally compress holding concentration over years. DePIN protocols that allocated tokens through conventional venture rounds, reserving 30 to 50 percent for founding teams and early investors with cliff-based vesting, bypassed these distribution-broadening mechanisms entirely. The governance concentration patterns in Table 3 may reflect distribution mechanism design operating through the sector channel: Table 5 shows DePIN membership is positively associated with concentration at $p < 0.05$ after adjusting for protocol age, valuation, and insider allocation. However, the regression null on insider allocation (Section 4.4) cautions against treating initial token allocation as a complete measure of distribution design: compression mechanisms (sub-linear scaling, activity-based caps, delegation incentives) operate through ongoing protocol governance, not through the initial percentage split alone.

4.3.1 Sector and chain-architecture decomposition

The sector contrast documented above (Cohen's d = 1.05) is robust to chain-architecture decomposition. When the cross-section is split by chain ecosystem, the ordering in which DePIN protocols carry higher governance concentration than DeFi protocols holds within the EVM cohort and is directionally consistent, though negligible in magnitude, within the Solana cohort. Chain architecture therefore enters this analysis as an additional observational axis, not as a substitute for the sector finding. The decomposition draws on an extended comparative cohort assembled for robustness, with both sector and chain architecture coded for each protocol; because this cohort differs from the 30-protocol primary sample, the cell-level magnitudes are not directly comparable to the headline effect size, and the question of interest is whether the sector ordering survives the split rather than the size of the gap.

Cell	Cross-section N	Mean HHI	Median HHI	Panel N	Net-decline share
EVM DeFi	12	0.025	0.019	11	9 of 11 (82%)
EVM DePIN	8	0.084	0.071	8	5 of 8 (63%)
Solana DeFi	7	0.028	0.027	7	4 of 7 (57%)
Solana DePIN	5	0.044	0.035	4	1 of 4 (25%)

Table 3.5. *Sector-by-chain-architecture decomposition of governance concentration (extended comparative cohort). The cross-section columns report the March 2026 holding HHI (cell mean*

and median); the net-decline column reports the share of protocols in each cell whose HHI declines over the post-distribution monthly panel. Cross-section cell sizes exceed panel sizes where a protocol lacks sufficient monthly history. EVM DeFi includes Hyperliquid (HHI 0.005), the lowest-concentration protocol in the cell; the cell mean sits modestly above its median because of the higher-concentration protocols at the top of the cell (GMX, Ether.Fi, MakerDAO), not because of any single high outlier. Cell-level sector effect sizes: EVM cohort Cliff's delta -0.71, Cohen's d -1.41; Solana cohort Cliff's delta -0.43, Cohen's d -0.79. The Cross-section N and Panel N columns both count protocols per cell.

Within the EVM cohort the DeFi-versus-DePIN contrast is large and in the expected direction, with DeFi less concentrated than DePIN (Cliff's delta = -0.71, a rank-based non-parametric effect-size measure that is robust to outliers; Cohen's d = -1.41). Within the Solana cohort the sector difference is smaller but directionally consistent (Cliff's delta = -0.43, Cohen's d = -0.79, DeFi less concentrated than DePIN as in the EVM cohort); the Solana DeFi cell is small (seven protocols), so this within-Solana comparison is reported as descriptive context rather than as an independent test of the sector effect.

The longitudinal column summarizes net post-distribution trajectories for the protocols with sufficient monthly panel coverage. Net HHI decline is the majority pattern in every cell, and a larger share of EVM protocols decline (DeFi 9 of 11; DePIN 5 of 8) than Solana protocols (DeFi 4 of 7; DePIN 1 of 4). The difference in DeFi deconcentration rates between the two architectures is not statistically significant in this panel (Fisher exact p = 0.33). These trajectory comparisons are descriptive: the present cross-section cannot support causal or trajectory claims (Section 5.7), and the full panel analysis is specified as future work (Section 5.8).

A matched-sample design within the DeFi sector cannot adjudicate the chain-architecture difference at present. A strict three-to-five-year maturity window leaves three DeFi protocols (Maple Finance and GMX on EVM; MetaDAO on Solana); loosening to a two-to-six-year window yields thirteen, with a persistent ten-to-three EVM-to-Solana imbalance. Neither window supports chain-architecture-level inference within sector. The cohort also exhibits chain-architecture-age collinearity, with the Solana DeFi sub-cohort averaging roughly 2.3 years of maturity against roughly 6.0 years for EVM DeFi, which prevents a clean attribution of trajectory direction to chain ecosystem rather than distribution-phase age. The decomposition is therefore reported at the descriptive layer only; a multivariate specification with a chain-architecture indicator is deferred to the age-balanced extended cohort described in Section 5.8 (falsifiable prediction 5).

4.4 Allocation Design and Post-Distribution Dynamics

Initial token allocation design is not associated with governance concentration. The percentage of total supply allocated to team and investors at launch shows no significant association with post-distribution HHI (Pearson r = 0.07, p = 0.68, N = 37; Spearman rho = 0.05, p = 0.76). At N = 37 this test has roughly 80 percent power to detect a population correlation of about |r| = 0.45 (two-sided, alpha = 0.05), so the null rules out a large allocation-concentration association but is not powered to exclude small or moderate effects. The allocation-design null is therefore an upper bound on effect size rather than evidence of exact independence, reported as a bounded effect rather than as proof of no effect (Wasserstein & Lazar, 2016). This null persists across alternative specifications: token unlock progress, float-adjusted concentration, and the interaction of allocation with protocol maturity all produce coefficients indistinguishable from zero (Table 5, Models 1 through 3; Table 4 summarizes the bivariate battery). Statistically Validated Network analysis across DeFi protocols documents that governance concentration shifts dynamically in correlation with total value locked and secondary market valuations (Eisermann et al., 2025), consistent with the post-distribution market dynamics that render initial allocation design uninformative in the cross-section.

Methodological note on AAVE: the stkAAVE staking contract is excluded from the holding Herfindahl-Hirschman Index per the protocol-controlled-address rule, but stakers retain pass-through voting power, so the reported AAVE holding HHI of 0.013 understates effective insider control. Attributing the staked balance back to the underlying stakers resolves what the wholesale exclusion hides: stkAAVE holds 21.67 percent of AAVE supply, of which roughly 4.4 to 6.3 percent of total supply is insider AAVE (Aave team multi-signature wallets plus the founder) invisible to the top-1000 holder snapshot. At the address level this attribution leaves the holding HHI essentially unchanged, and if anything marginally lower: the recovered insider stake re-enters as several distinct staker addresses rather than as a single block, so it enlarges the post-exclusion holder denominator more than it concentrates the squared-share sum, which lowers the HHI slightly rather than raising it. The substantive result is therefore the recovered insider share, not a movement in the holding HHI. This reinforces the reading that the holding measure understates insider control: the concentration the holding HHI does not register surfaces instead in the insider share and in delegated voting power, the latter captured by the companion Tally-sourced AAVE delegation HHI of 0.058 in Table 6.

Longitudinal panel evidence consistent with the cross-sectional null. A companion quarterly HHI panel for 14 governance tokens over 8 quarters post-token-generation event (Supplementary File S7) shows that 11 of 14 protocols exhibit monotonic governance HHI decay over the 24-month observation window, with the three exceptions reflecting mechanism-specific lock-in: CRV (vote-escrow), COMP (reservoir-drip emissions), and ENS (claim-window distribution). The longitudinal pattern reinforces the cross-sectional finding that initial allocation is not associated with steady-state concentration: distributions tend to deconcentrate over time across heterogeneous allocation designs, indicating that the relevant institutional-design margin is post-distribution architecture (delegation programs, lock-in mechanisms, distribution-channel diversity) rather than launch-time allocation. The panel design supports the cross-section's descriptive claim without resolving causal identification (which requires the panel-and-event-study extension specified in Section 5.8).

Covariate	**Test**	**Statistic**	**p**	**N**	**Status**
Insider allocation (%)	Pearson r	0.07	0.68	37	Null
Team allocation (%)	Pearson r	-0.08	0.62	37	Null
Investor allocation (%)	Pearson r	0.14	0.39	37	Null

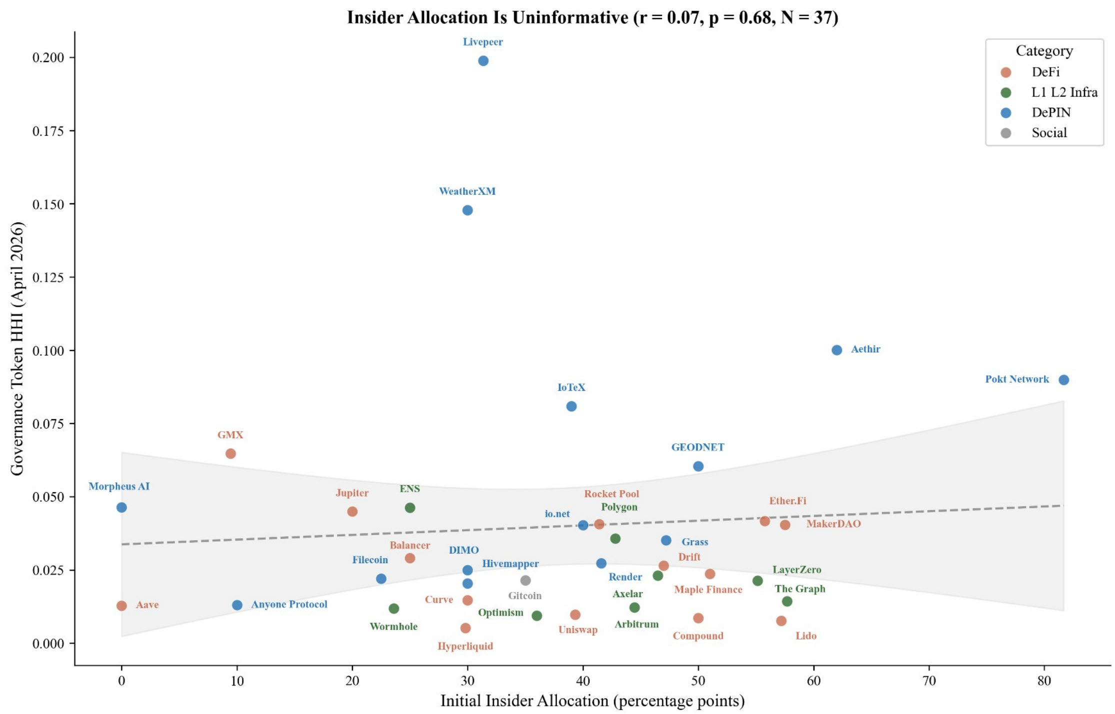
Covariate	Test	Statistic	p	N	Status
Protocol maturity (years)	Pearson r	0.02	0.92	38	Null
Circ/total supply ratio	Pearson r	0.03	0.86	31	Null
MCap/FDV ratio	Pearson r	0.21	0.25	33	Null
Exclusion-adjusted float	Pearson r	−0.003	0.99	30	Null
Insider count fraction (top-10)	Spearman ρ	0.48	0.003	37	Significant*
Non-insider HHI, V1 (count)	Spearman ρ	0.54	0.001	34	Tautology-checked
Non-insider HHI, V2 (balance)	Spearman ρ	0.33	0.060	34	Does not survive
Subsidy ratio (cross-sector)	Pearson r	0.62	0.002	23	Fragile (LPT-driven)
Subsidy ratio (TT expanded)	Pearson r	0.12	0.61	20	Null (robustness check)
Subsidy ratio (excl. Livepeer)	Pearson r	0.07	0.76	22	Null
DePIN sector indicato	r Mann-Whitney	U=174	0.011	30	Robust (30/30 LOO)
HHI-Gini	Pearson r	0.58	<0.001	44	Significant

Table 4. *Bivariate associations between protocol characteristics and governance concentration (HHI). Allocation, maturity, and float specifications are null. Insider wallet presence is significant but the balance-based measure does not survive a tautology check (Section 4.4). Subsidy is significant in levels but driven entirely by Livepeer (Section 4.6). * LOO-robust (37/37). MCap = market capitalization; FDV = fully diluted valuation. N is the number of protocols in each test's covariate-complete subsample (varying with per-covariate data availability).*

To understand why allocation is uninformative, we classify each protocol's top-10 post-exclusion holders using a three-tier methodology: Tier 1 identifies addresses via Blockscout labels and Dune exchange tags; Tier 2 inspects contract bytecode (SafeProxy, GnosisSafe, vesting contracts) via Blockscout's implementation field; Tier 3 applies Etherscan named-address matching. Of 390 addresses classified across 39 tokens, 119 (30.5%) are identifiable as insider wallets (team, foundation, or investor multisigs), 67 (17.2%) are exchange addresses, 78 (20.0%) are protocol contracts, and 126 (32.3%) remain unclassified. The 78 protocol contracts identified here differ from the 133 protocol-controlled address exclusions applied to HHI computation. The 78 figure counts protocol-contract addresses across the top-10 holders of 39 tokens. The 133 figure counts excluded address inclusions across the top-1,000 holders of the 38 protocols with documented PCA exclusions (125 unique addresses; five addresses recur across multiple protocols). The samples and windows differ; the two numbers are not interchangeable.

Protocols whose insider wallets retain top-holder positions tend to exhibit higher governance concentration (Spearman rho = 0.48, p = 0.003, N = 37), robust across all 37 leave-one-out iterations (rho range: 0.45 to 0.55). However, this association has a mechanical component: insider wallets with large balances contribute directly to HHI through the squared-share calculation. To verify that the relationship is not a tautological artifact, we recomputed concentration after excluding insider addresses and renormalizing remaining shares. The count-based measure survives: non-insider HHI remains significantly correlated with the fraction of top-10 holders that are insiders (Spearman rho = 0.54, p = 0.001, N = 34; N drops from 37 to 34 because the three zero-insider protocols, Hyperliquid, Optimism, and Arbitrum, have no insiders to exclude, so the tautology check is undefined for them), indicating that insider-heavy protocols have concentrated governance ecosystems, not merely concentrated insider positions. The balance-based measure loses significance (rho = 0.33, p = 0.060), consistent with it capturing the mechanical contribution of large insider stakes rather than a structural relationship. The informative finding is not the insider correlation per se but its contrast with allocation design: what matters is not how tokens were designed to be distributed, but whether the recipients retained or sold their positions.

The divergence between allocation design (null) and current insider presence (descriptive mechanism) carries a design implication. Governance concentration is not fixed at launch; it emerges from post-distribution dynamics. The allocation null (Section 4.4) is also consistent with Michels' (1911) observation that organizational complexity inevitably concentrates power in a specialized leadership class: early protocols' commitment to governance minimization, the assumption that minimal on-chain rules would produce emergent decentralization is inconsistent with the concentration patterns the ideology sought to prevent. Protocols whose insiders retain their positions exhibit more concentrated governance outcomes, regardless of how generously the initial allocation was designed. The count-based tautology check (rho = 0.54 on non-insider HHI) suggests this is not merely arithmetic: insider-heavy protocols attract or retain concentrated non-insider holders as well, consistent with a pattern where concentrated insider presence attracts or coexists with concentrated non-insider holders, rather than insider balances alone inflating the metric.



Source: Author calculations from Dune Analytics and Token Terminal (March 2026).

Figure 3. *Initial insider allocation (%) vs. post-exclusion holding HHI for 37 regression-ready protocols. The null relationship (r = 0.07, p = 0.68) is the paper's lead finding: initial allocation design is not associated with post-distribution governance concentration. Reading the figure: the absence of a downward slope is the finding; if initial insider allocation drove concentration we would expect a positive slope, and if exclusion overcorrected we would expect a negative slope.*

Hyperliquid provides the strongest available test of the allocation design hypothesis. The protocol accepted zero venture capital, distributed 31% of total supply via a single community airdrop (the largest in crypto history by dollar value), and operates a deflationary model where 97% of trading fees purchase and burn the native token; it has no ongoing emissions. Five protocol-controlled addresses were excluded from the HHI computation: the Assistance Fund system address (confirmed in Hyperliquid documentation), the burn address, the community rewards treasury (241 million tokens, matching the published allocation), the Foundation treasury (60 million tokens, exactly 6% of total supply), and the staking manager contract (confirmed via Hypurrscan). The resulting HHI of 0.005 is the lowest in the DeFi subsample (Lido at 0.008 is the second-lowest after the post-PCA-audit exclusions, and Compound at 0.009 the third-lowest). Despite the most community-favorable distribution architecture in the sample, this single observation cannot support a causal identification claim on its own. The cross-sectional null (Pearson r = 0.07, p = 0.68, N = 37) shows that protocols across the full range of allocation designs achieve comparable concentration levels, consistent with the finding that allocation design is not associated with post-distribution concentration.

The governance concentration patterns documented across the sample (Section 4.2, Table 3; HHI range 0.005 to 0.199) are descriptive, but they matter for governance audits because protocols often cite token distribution as evidence of decentralization. The corrected data show that current holder composition, insider retention, and delegation mechanics are more informative audit targets than launch allocation percentages alone.

The null pattern extends beyond allocation to financial sustainability more broadly. Three protocols with similar subsidy ratios (Aethir at 0.36x, Aave at 0.37x, io.net at 0.40x) exhibit governance HHI values of 0.100, 0.013, and 0.040 respectively. All three are revenue-dominant; their concentration spans much of the 52-protocol sample range. Financial sustainability is neither necessary nor sufficient for governance decentralization.

Table 5 reports multivariate OLS regressions of HHI on protocol characteristics. The DePIN coefficient is positive and statistically significant in Models 1 and 2 (Model 1: 0.045, $p = 0.012$, $N = 38$; Model 2: 0.043, $p = 0.045$, $N = 36$) and just above the conventional 0.05 threshold in Model 3 (0.043, $p = 0.078$, $N = 35$). No other covariate achieves $p < 0.10$. In particular, the initial insider allocation percentage used as a Table 5 covariate is not significant, consistent with the bivariate null in Table 4. This is distinct from the current insider-retention measure reported in Section 4.4 (Spearman rho = 0.48), which counts how many of a protocol's top-10 holders are insiders rather than measuring the share of supply allocated to insiders at launch. The adjusted R-squared ranges from 0.15 to 0.19, indicating that sector membership accounts for more cross-sectional variation than protocol age, valuation, or allocation structure. Models 1 through 3 carry between 35 and 38 observations across five to seven predictors, the observations-per-predictor range that limits how reliably they isolate independent effects.

Model 4 addresses this directly. On the covariate-complete sample of 50 governance-token protocols, it regresses log HHI (log-transformed to address the right-skew of the raw measure) on sector, revenue intensity, and protocol maturity: four predictors at 12.5 observations each, comfortably above the conventional floor. The DePIN coefficient remains positive and significant (+0.73, $p = 0.0107$, HC3 robust standard errors) and is essentially unchanged from the sector-only coefficient (+0.74), so the architecture association is not mediated by revenue intensity or maturity; both controls are individually uninformative ($p = 0.89$ and $p = 0.60$), extending the null sweep to economic-activity intensity. In untransformed HHI the DePIN coefficient is the same sign and also significant (+0.036, $p = 0.019$), so the powered specification holds under both the raw and the skew-correcting log measures rather than depending on the transform. The pipeline specification (Supplementary File S5) documents the replication infrastructure.

	Model 1	**Model 2**	**Model 3**
DePIN (vs DeFi)	0.045**	0.043**	0.043*
	(0.020)	(0.025)	(0.029)
L1/L2/Infra (vs DeFi)	-0.005	-0.002	-0.004
	(0.011)	(0.014)	(0.016)
Social (vs DeFi)	-0.010	-0.020	-0.019
	(0.079)	(0.064)	(0.146)

	Model 1	Model 2	Model 3
Protocol age (years)	n/a	0.002	0.001
		(0.005)	(0.006)
Log FDV	n/a	-0.003	-0.003
		(0.004)	(0.005)
Initial insider allocation (%)	n/a	n/a	0.000
			(0.001)
Constant	0.032***	0.076	0.063
	(0.007)	(0.089)	(0.099)
N	38	36	35
Adjusted R^2	0.173	0.190	0.151

Table 5. *OLS regression of HHI on protocol characteristics. Model 1 includes sector dummies only (DePIN, L1/L2/Infra, Social; DeFi omitted). Model 2 adds protocol age (years since token genesis) and log FDV (log fully diluted valuation). Model 3 adds the initial insider allocation percentage (team plus investor share at launch); cells marked n/a indicate a covariate not entered in that model. Successive models lose observations as covariate availability declines (N = 38 to 36 to 35 protocols). Robust standard errors (HC3; a heteroskedasticity-consistent estimator that produces valid standard-error estimates even when error variance is not constant across observations, the typical condition in cross-sectional cross-protocol data) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The powered Model 4 specification (covariate-complete sample, N = 50) is reported in the panel below.*

Predictor (DV = log HHI)	Coefficient	Robust SE	p
DePIN (vs DeFi)	+0.733	0.275	0.011
L1/L2/Infra (vs DeFi)	-0.353	0.307	0.256
Log revenue-intensity	-0.010	0.077	0.894
Protocol maturity (years)	+0.026	0.050	0.601
Constant	-3.866	0.304	< 0.001

N = 50 protocols; observations per predictor = 12.5; adjusted R-squared = 0.18. The sector-only baseline DePIN coefficient is +0.74, so the controlled estimate is essentially unattenuated.

Table 5, Model 4 (powered specification). *OLS regression of log HHI on sector, revenue intensity, and protocol maturity over the covariate-complete sample of 50 governance-token protocols (HC3 robust standard errors). The dependent variable is log-transformed to address the right-skew of raw HHI; in untransformed HHI the DePIN coefficient is the same sign and also significant (+0.036, p = 0.019).*

A bootstrap robustness check supports the sector coefficient is reliably positive across resampling under the full Model 3 specification. With 5,000 bootstrap resamples of the N = 35 sample, the DePIN coefficient bootstrap distribution centers at +0.054 with a 95% percentile interval of [+0.001, +0.091], strictly above zero. The bootstrap-positive fraction is 98.0%. The Model 3 attenuation relative to Models 1 and 2 is attributable to partial collinearity between the DePIN sector indicator and the insider-allocation control: in the cross-section, DePIN protocols cluster on higher insider allocation than DeFi protocols, so a fraction of the sector coefficient's signal is absorbed by the insider variable when both appear in the same model. This is consistent with the cross-sectional allocation null (Table 4, Pearson r = 0.07 for insider allocation alone) rather than indicating that the sector effect is fragile to covariate inclusion. The N = 35 multivariate sample also limits the precision with which the sector coefficient can be statistically isolated from correlated controls; the bootstrap robustness check above is the more informative robustness statement under small-sample multivariate conditions. Model 4 corroborates this reading directly: with the collinear insider-allocation control replaced by revenue intensity and maturity and the sample enlarged to 50, the DePIN coefficient is both unattenuated and significant.

4.5 Voting Power vs Holding Concentration

Delegation usually amplifies governance concentration above holding levels in the protocols with comparable voting data. Table 6 compares post-exclusion holding HHI with voting-HHI for eighteen protocols. Thirteen protocols amplify, with ratios from 2.5x (Gnosis) to 25.6x (Polkadot) and a mean of approximately 6.8x (median 4.4x). Five protocols disperse voting power below the holding baseline: ENS (0.48x), GMX (0.87x), HNT (0.26x to 0.39x), JUP (0.12x), and Livepeer (0.27x, the most pronounced DePIN governance disperser). The holding-based HHI reported in Table 3 measures token distribution, not governance power; delegated voting data are therefore necessary for protocols where votes flow through delegates, governance platforms, or lockup-weighted voting systems. To quantify this gap, the analysis uses Tally, Snapshot, and Solana on-chain governance data from the 12 months preceding the voting snapshot. The expanded sample also amplifies under nominally linear weighting (Gnosis, 2.45x) and under quadratic election weighting designed to disperse (Synthetix, 5.55x), consistent with a small, self-selected active electorate more concentrated than the broad holder base; Supplement S15 documents this participation reading together with the top-100-voter sampling convention that bears on its strength.

Amplification varies by sector composition rather than by sector membership. Within DeFi, the range spans 2.5x (Gnosis) to 9.9x (Compound); DePIN spans 3.8x (WeatherXM) to 9.1x (DIMO); infrastructure spans 3.1x (Arbitrum) to 25.6x (Polkadot). The five dispersion exceptions are discussed individually in §4.5.1 (ENS), §4.5.1.1 (GMX), §4.5.4 (HNT and JUP under within-Solana heterogeneity), and the Livepeer orchestrator bonded-stake result later in this section (LPT at 0.27x). Sector membership is not associated with how strongly delegation concentrates power; delegate-program design is the more plausible measurement surface. DIMO's voting HHI (0.228) is 9.1 times its holding HHI (0.025), though this ratio is based on only 10 Snapshot voters and should be interpreted with caution, and WeatherXM's (0.556) is 3.8 times its holding HHI (0.148). A small number of delegates, likely foundation-affiliated or early-investor wallets, control governance outcomes despite relatively distributed token holdings.

Protocol	Holding HHI	Voting HHI	Ratio	Source	N
DIMO	0.025	0.228	9.1x	Snapshot	10
Lido	0.008	0.050	6.3x	Snapshot	333
WeatherXM	0.148	0.556	3.8x	Snapshot	73
Compound	0.009	0.089	9.9x	Snapshot†	138
Aave	0.013	0.058	4.4x	Tally§	150,911
Uniswap	0.010	0.027	2.7x	Tally	1,000
Arbitrum	0.012	0.038	3.1x	Snapshot†	5,466
Optimism	0.009	0.033	3.6x	Tally	1,000
DRIFT	0.026	0.083	3.1x	Solana VSR	641¶
GMX	0.065	0.057	0.87x	Tally‡	1,000
HNT	0.099	0.026-0.039	0.26x-0.39x	Solana VSR	1,835¶
ENS	0.046	0.022	0.48x	Tally‡	1,000
JUP	0.045	0.0055	0.12x	Solana del-vote	128,476¶
Gnosis	0.042	0.104	2.5x	Snapshot	481
Synthetix	0.017	0.095	5.6x	Snapshot	30
Bittensor	0.007	0.072	9.6x	Taostats	75
Polkadot	0.005	0.133	25.6x	Subscan	239
Livepeer	0.199	0.054	0.27x	Livepeer orch‖	100

Table 6. *Holding versus voting concentration for protocols with sufficient governance activity (12-month lookback). Voting HHI from Tally delegate voting power (on-chain) or Snapshot vote-weighted power (off-chain) or Solana on-chain governance (VSR lockup-weighted or delegated-*

vote per protocol design); N is unique-voter count (Snapshot rows; 12-month rolling window), top-1000 delegate sample (Tally rows), all-delegates count (Aave Tally all-delegates pull), or unique-voter count from Solana on-chain VoteRecord parsing (Solana rows). Ratio above 1.0 indicates delegation concentrates governance above token-holdings; below 1.0 indicates delegation distributes. Holding HHIs are post-exclusion per Table 3. Thirteen of eighteen sampled protocols amplify; five structural exceptions show dispersion (ENS 0.48x and GMX 0.87x per Sections 4.5.1 + 4.5.1.1; HNT 0.26x-0.39x sensitivity-range per Section 4.5.4 VSR-lockup-weighted design; JUP 0.12x per Section 4.5.4 delegated-vote + broad-airdrop-user-base, the most-extreme dispersion outlier in the cross-section; and LPT 0.27x, the most pronounced DePIN governance disperser). The Livepeer voting HHI (‖) is computed over orchestrators by total bonded stake (self plus delegated) from the Livepeer protocol subgraph, as-of Arbitrum block 468186375 / round 4216, with N the active-orchestrator count; this governance-layer dispersion sits on a different axis from the Livepeer subsidy-outlier finding (Section 4.4), and LPT's holding concentration remains the cross-section maximum (0.199, Table 3), so the DePIN-concentration headline is unaffected. Curve veCRV, Balancer veBAL, and Frax veFXS (15x, 21x, and 11.4x respectively) form a distinct ve-token class discussed separately in Section 4.5.2. Gnosis, Synthetix, Bittensor, and Polkadot are the N=52 additions (Supplement S15): Gnosis Snapshot linear; Synthetix Snapshot council-election weighted-debt (small-N, 30 voters); Bittensor validator-stake share via Taostats; Polkadot conviction-weighted referendum vote via Subscan, pooled over 20 decided referenda.

† Compound and Arbitrum voting HHI sourced from Snapshot (n = 138 and n = 5,466 unique voters respectively in the 12-month rolling-window measurement period) where Snapshot's active-voter pool exceeded Tally's top-100-delegate sampling. AAVE, Uniswap, Optimism, GMX, and ENS use Tally (EVM-side delegate registry); DIMO, WeatherXM, and Lido use Snapshot (Tally is not deployed for these EVM protocols). JUP, HNT, and DRIFT use Solana on-chain governance program parsing (JUP via Dune `jupiter_solana.govern_call_setvote` decoded namespace with per-signer max-weight aggregation; HNT and DRIFT via Helius RPC parsing of the respective SPL Governance program VoteRecordV2 accounts) because Tally is not deployed for Solana protocols and on-chain governance state is the apples-to-apples comparable surface to Tally's EVM delegate registries.

‡ GMX and ENS values from paginated Tally top-100 delegate pulls (May 2026). Snapshot date drift between the March 2026 baseline (used for the eight protocols with March 2026 voting data) and May 2026 (used for GMX and ENS) means rank-ordering across protocols is the durable inferential property, with absolute magnitudes subject to delegate-pool changes over multi-month windows.

§ Aave voting HHI 0.058 from Tally all-delegates pull (N = 150,911); the all-delegates depth establishes ARB at maximum 437,453-delegate depth as the deepest-N anchor for the methodology-floor sensitivity analysis. Four deeper-N sweeps (AAVE + COMP + UNI + ARB) produce sub-0.005 HHI shifts vs top-1000 baseline, confirming the top-1000 methodology floor.

¶ Solana voting HHI values from on-chain governance program parsing via Helius RPC and Dune decoded namespace: JUP via `jupiter_solana.govern_call_setvote` (per-signer max(weight) aggregation; delegated-vote design; each token equals one vote); HNT and DRIFT via Voter Stake Registry (VSR) lockup-weighted vote design where vote weight is locked-amount multiplied by lockup-duration multiplier (HNT VSR saturates at four-year lockup; DRIFT VSR uses analogous lockup multiplier; both classified as VSR-lockup-weighted). HNT presented as sensitivity range under VSR position-state reconstruction with 14.8 percent closed-position rate bracketing the bounds. The within-Solana sample (N=3) is discussed in Section 4.5.4 within-Solana governance heterogeneity subsection; JUP at 0.12x ratio is the most-extreme dispersion outlier in the cross-section.

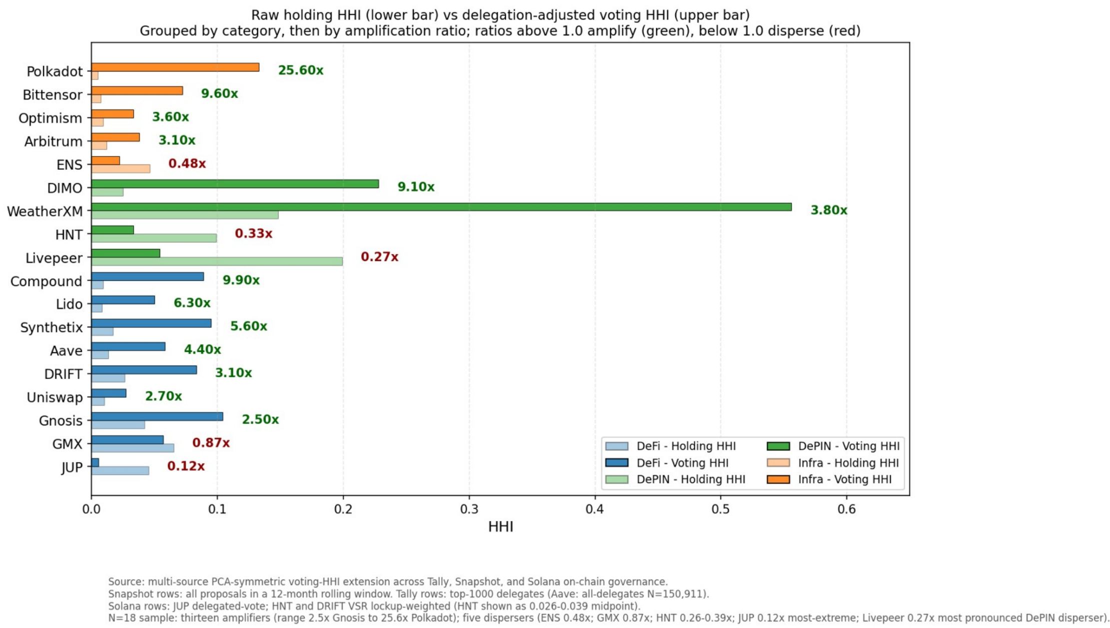


Figure 4. *Raw holding HHI (lower bar) vs. delegation-adjusted voting HHI (upper bar) for the eighteen protocols with sufficient governance activity, grouped by category (DeFi, DePIN, infrastructure), then by amplification ratio within category. Reading the figure: the ratio label at the right of each pair gives voting HHI divided by holding HHI; ratios above 1.0 indicate delegation amplifies concentration (the upper bar is longer than the lower bar; green annotation), ratios below 1.0 indicate delegation distributes it (red annotation). Thirteen of eighteen protocols amplify, with ratios ranging from 2.5x (Gnosis) to 25.6x (Polkadot); five protocols show dispersion: ENS (0.48x), GMX (0.87x), HNT (0.26x to 0.39x), JUP (0.12x, the most-extreme dispersion outlier), and Livepeer (0.27x, the most pronounced DePIN governance disperser). Per-protocol values and source notes are in Table 6 and Supplement S15. Snapshot rows aggregate unique voters across all proposals in a 12-month rolling window; Tally rows reflect top-1000 delegate samples per protocol.*

Seven DeFi protocols sampled show varying delegation outcomes: JUP (0.12x, most-extreme dispersion outlier per Section 4.5.4 Solana delegated-vote with broad-airdrop-user-base mechanism), GMX (0.87x, disperses; methodology-sensitive crossover from amplification 1.2x at top-100 sample to dispersion at top-1000 sample; Section 4.5.1.1), DRIFT (3.1x, amplifier; Solana VSR-lockup-weighted per Section 4.5.4), Uniswap (2.7x), Aave (4.4x), Lido (6.3x), and Compound (9.9x; the highest among the protocols drawn in this figure). The Uniswap result is worth examining in detail. Earlier analyses, using a holding HHI of 0.032, produced an amplification ratio of 0.84x and were cited as evidence that Uniswap's mature delegate ecosystem can disperse governance power. That earlier HHI included the canonical UNI burn address (0x000…000dead), which holds 102.5 million UNI (11.3 percent of supply) inert and cannot participate in governance under any condition: it has no delegate, no votes, and no path to ever cast one. After excluding this burn address, the post-exclusion holding HHI is 0.010, and the amplification ratio is 2.8x. Uniswap's delegate program concentrates voting power above its post-exclusion holding baseline, in line with every other protocol sampled. The pre-exclusion result was driven by including a wallet that structurally cannot vote in the holding denominator.

Both infrastructure protocols sampled show delegation amplification when holding HHIs are computed with the same exclusions applied to Table 3. Optimism's voting HHI (0.033) is 3.6 times its post-exclusion holding HHI (0.009). Arbitrum's voting HHI (0.038) is 3.1 times its holding HHI (0.012). Both protocols have invested in delegate-program infrastructure (published delegate platforms, recurring delegate compensation, active recruitment of long-form policy positions), and both produce amplification ratios in the 3.1x to 3.6x range, which sits below the cross-protocol mean. Delegate-program investment does not differentiate the two protocols on amplification magnitude. The institutional layers built around the delegate programs differ substantially and may differentiate outcomes that HHI cannot capture, as discussed in the next paragraph.

Optimism's Token House shows 3.6x delegation amplification, below the cross-protocol mean but consistent with the predominant direction of effect (amplification in thirteen of eighteen protocols). The Citizens' House moderates Optimism's governance in practice, though it does not enter the token-weighted HHI calculation. The Citizens' House is a separate governance body weighted by reputation rather than tokens, with formal veto power over economic parameters proposed by the Token House and a veto threshold that lowers when multiple stakeholder groups align (Optimism, 2024). Because reputation weights are not token weights, the Citizens' House provides accountability that the amplification ratio alone cannot capture.

Lido's 6.3x amplification catalyzed a parallel institutional response. The "Dual Governance" redesign grants stETH holders formal veto power over LDO governance proposals, with a programmatic "rage quit" exit mechanism as the ultimate accountability lever. The reform precedes the holding-HHI recomputation reported here and was justified at the time by the amplification ratio measured from a curated active-governance holder subset (N = 189; minimum balance threshold approximately 414K LDO). Under the universal top-1000 holder methodology applied in this manuscript with the full protocol-controlled-address audit (Section 3.8; Bybit and Binance custody addresses excluded as Class 5 CEX custody), Lido's amplification ratio (6.3x) is substantial, and the direction (concentration above stake) is consistent with the predominant-amplification pattern documented across thirteen of eighteen protocols.

Both responses share the same shape. Token-weighted delegation concentrates voting power above stake holdings across the sample, and the institutional reforms operate by adding an accountability layer outside the token-weighted system rather than redesigning the delegation mechanism itself. Both protocols with named accountability reforms (Optimism, Lido) are protocols where token-weighted delegation shows positive amplification, consistent with reading these reforms as responses to the predominant-amplification pattern that this section documents.

The delegation-adjusted analysis reported in this section extends recent empirical work by Fritsch, Müller, and Wattenhofer (2024), who apply Nakamoto coefficients and Gini measures to voting power in Compound, Uniswap, and ENS governance systems and document extreme concentration in delegate-level voting power. The cross-protocol comparison reported here spans eighteen protocols across DeFi (Compound, Aave, Uniswap, GMX, Lido, Synthetix, Gnosis), DePIN (DIMO, WeatherXM, Livepeer), and infrastructure (Optimism, Arbitrum, ENS, Bittensor, Polkadot). Amplification factors range from 2.5x (Gnosis) to 25.6x (Polkadot) across the thirteen amplifying protocols; ENS at 0.48x; GMX at 0.87x; HNT at 0.26x-0.39x; JUP at 0.12x; and LPT at 0.27x are the five structural exceptions (ENS: mature delegate program where holders systematically delegate to a broad community-delegate set; GMX: methodology-sensitive crossover per Section 4.5.1.1; HNT: Solana VSR lockup-weighted design per Section 4.5.4; JUP: Solana delegated-vote with broad airdrop user base per Section 4.5.4, the most-extreme dispersion outlier in the cross-section; LPT: Livepeer orchestrator bonded-stake delegation across roughly 100 orchestrators, the most pronounced DePIN governance disperser, with holdings concentration unchanged at the cross-section maximum). Within the sample, the 2.5x to 25.6x amplification range is distributed across sectors with no clear concentration in any one; sector membership does is not associated with how strongly delegation concentrates power.

The pattern is consistent with the broader claim of this paper: design choices within each delegate program (delegate selection criteria, compensation structure, voting threshold rules) are associated with the magnitude of governance concentration through delegation, while the direction of the effect (concentration above holdings) is the same across sector and governance venue. Whether a protocol concentrates governance power through delegation does not depend on whether it is DeFi, DePIN, or infrastructure. How much it concentrates is shaped by design choices made at the level of the delegate program itself. The amplification ratio carries welfare consequences for DePIN that scale with operator exit costs: hardware deployed for one network is a co-specialized asset that cannot be redeployed to a competing protocol the way liquidity can move between DeFi pools, a form of asset specificity that raises exit costs (Williamson, 1979).

A power-index extension across the N = 16 cross-source voting-HHI sample (Tally + Snapshot + Solana proper-weight) refines the binary amplify-versus-disperse framing into a three-class amplification typology. Class 1 (structural-majority): top-1 voter holds an absolute majority of weight, producing Shapley-Shubik HHI mechanically at 1.000 (WXM only example in current sample; +30.85 percentage-point SS-share amplification over nominal share). Class 2 (coordinated-amplification): top-1 voter dominant with disparate second-tier weight distribution; positive but sub-majority amplification (AAVE +4.80 pp, GMX +2.93 pp, COMP-snap +2.25 pp, LDO +2.20 pp, HNT +5.45 pp, DRIFT +8.52 pp). Class 3 (track-share): top-1 SS-share tracks nominal share within 1 percentage point (UNI, COMP-tally, ARB, OP, ENS, DIMO, JUP). Pearson r between SS-HHI and Voting-HHI is 0.9756 across the N = 16 sample, with rank-stability holding across all six quorum thresholds from 0.33 through 0.75; the cross-sectional rank ordering of governance concentration is preserved whether measured by HHI or by Shapley-Shubik pivotal power. Per-class quorum-sensitivity profile: Class 1 protocols are highly quorum-sensitive (WXM SS-HHI transitions from 1.000 to 0.609 at the q = 0.74 to 0.75 boundary), Class 3 protocols are quorum-stable, and Class 2 protocols sit in between. The Class 1 versus Class 2 distinction matters for design implications because the underlying institutional dynamics differ: Class 1 reflects a single-actor absolute-veto position; Class 2 reflects coordination capacity across a dominant top-tier voter and a fragmented secondary tier where the dominant voter's coalition-building costs scale with the second-tier fragmentation. Full per-protocol SS + Banzhaf values, quorum-variation table, and 3-class assignments are in Supplementary File S14.

4.5.1 ENS as institutional-design counterexample

ENS is one of five protocols in the cross-section where delegation empirically disperses voting power below the holding-HHI baseline (ratio 0.48x; voting HHI 0.022 vs holding HHI 0.046, the latter computed after the ENS DAO Wallet and Treasury PCA exclusions codified in Section 3.8). The dispersion case is informative for the framework's institutional-design thesis because it provides an existence-proof that delegation outcomes are design-tractable rather than mechanism-inevitable. The ENS delegate program funds approximately 100 active delegates (including the ENS DAO ecosystem grants for delegate work), and the top-100 delegate voting-power distribution is structurally broader than the holding distribution: top-1 post-exclusion holding share is 16.8% while top-1 voting-power share is 12.1% (fireeyesdao.eth, a community-elected delegate). Holders systematically delegate to a broader community-delegate set rather than self-delegating or delegating to whales.

The ENS + GMX + HNT + JUP dispersion findings contrast with the thirteen protocols where delegation amplifies: in those protocols, the top-delegate identity is typically either the protocol foundation itself, a staking-aggregation contract that aggregates underlying staker voting power, or an institutional delegate firm (see Section 4.5.3). The structural pattern across the eighteen-protocol sample is consistent with the framework's thesis that institutional design at the delegate-program level governs the direction of delegation's effect on voting concentration. The generalizable design recommendation: delegate-program architectures that incentivize broad community-delegate distribution (recurring delegate compensation; public delegate platforms; active recruitment; operational support for non-whale delegates) can produce delegation-mediated dispersion of voting power, contrary to the default-amplification pattern documented across most token-governance protocols.

4.5.1.1 GMX as a structural-exception counterexample

GMX is another of the five protocols in the cross-section where delegation empirically disperses voting power below the holding-HHI baseline (ratio 0.87x; voting HHI 0.057 vs holding HHI 0.065, the latter computed after the universal burn-rule exclusion plus GMX-specific Binance-custody exclusions codified in Section 3.8). The dispersion case is methodology-sensitive: at a top-100 delegate sample GMX appears to amplify at 1.2x, while at the canonical top-1000 sample the broader delegate distribution surfaces voting-power-share tails less concentrated than the holding-share tail at the same sampling depth. The crossover from amplification (1.2x at N = 100) to dispersion (0.87x at N = 1000) indicates that GMX top-100 amplification readings are sampling-depth artifacts rather than stable properties of the underlying delegate-program design; the top-1000 sample better captures the actual voting-power distribution.

The finding strengthens the framework's institutional-design thesis: GMX's delegate program is observably broader than its holder concentration when measured at appropriate sampling depth, consistent with the mature-delegate-program pattern documented at ENS (Section 4.5.1) at a different concentration level. The five structural-exception cases (ENS at 0.48x; GMX at 0.87x; HNT at 0.26x-0.39x; JUP at 0.12x; LPT at 0.27x) together preserve the dispersion pattern as informative counterexamples to the predominant-amplification thesis rather than refuting the broader argument that token-weighted delegation predominantly concentrates voting power above holdings.

4.5.2 Secondary extension: vote-escrowed governance

Three protocols using vote-escrowed governance (Curve veCRV; Balancer veBAL; Frax veFXS) form a distinct extreme-amplification class that operates through structurally different mechanisms than the delegate-based amplification documented in the Table 6 sample. Curve's veCRV system locks CRV for up to four years; lock-duration multipliers convert locked CRV into proportional veCRV voting power, with longer locks weighted more heavily. Convex (the dominant meta-governance protocol for Curve) aggregates user-deposited CRV into a single locked position that votes as a meta-governance layer; Convex holds approximately 48 percent of the top-50 historical veCRV deposit-weighted share per the direct cumulative-deposit computation (Section 4.5.2; supplementary file veCRV_voting_concentration). The resulting veCRV voting concentration (HHI 0.26 from cumulative-deposit aggregation across top-50 lockers) is approximately 15 times the raw post-Convex- exclusion CRV holding HHI of 0.014.

Balancer's veBAL system exhibits a structurally identical pattern: Snapshot voting on Balancer's governance space (12-month lookback; 36 active voters) yields voting HHI 0.626 against holding HHI 0.030, a 21x amplification ratio. The top-1 active voter holds 78 percent of total voting power; the top-10 voters cumulatively hold 99.85 percent. The Aura Finance protocol (Balancer's Convex-equivalent meta-governance aggregator) plays the same role for veBAL that Convex plays for veCRV.

Frax's veFXS system shows the same vote-escrow amplification at approximately 11.4x. This figure is the Snapshot veFXS-weighted-vote concentration (per-voter maximum veFXS-weighted vote across the frax.eth governance space, per Supplement S15); unlike the veCRV and veBAL figures it is measured at the Snapshot voting layer rather than re-derived from on-chain veFXS-balance aggregation, so it is reported as the lower-magnitude member of this class.

The veCRV and veBAL cases are not included in the Table 6 row set because the underlying methodology differs: Table 6 measures HHI of delegate voting power within token-governance delegate programs; the ve-token cases measure HHI of lock-duration-weighted CRV/BAL deposits, with meta-governance aggregation. The two mechanism classes produce concentration through different paths but converge on the same direction (amplification) with much larger magnitudes (11.4x to 21x versus the 2.7x to 9.9x range observed in delegate-program-based amplification). For practitioners designing token governance, the ve-token class is a separate design choice with substantially different concentration implications than standard delegate-program governance.

4.5.3 Foundation and aggregation-contract top-delegate pattern

Three of the eighteen Table 6 protocols (plus the ve-token class protocols in Section 4.5.2) have their largest single voting block held by a protocol-controlled aggregation entity rather than an independent community delegate: Compound Foundation (21.5% of total delegated voting power in May 2026 Tally pull); Aave stkAAVE staking contract (21.9%; aggregates underlying staker voting power as a single delegate); Curve Convex (48% of top-50 deposit-weighted veCRV share per Section 4.5.2). The three Solana-native protocols added at the earlier Table 6 expansion (JUP, HNT, DRIFT; Section 4.5.4) do not exhibit the foundation-or-aggregation-entity top-delegate pattern at the Solana SPL Governance member-set layer; among the four N=52 additions, Polkadot's largest effective voter is an independent delegate collective (ChaosDAO

OpenGov) rather than a protocol-controlled aggregation entity; Section 4.5.4.1 documents the Drift Foundation proposal-authorship-versus-vote-weight decoupling pattern that further differentiates Solana governance topology from the EVM precedents described here.

The aggregation-entity pattern raises practical measurement concerns relative to independent-delegate-dominated protocols (Uniswap top-1 is an investor delegate; Optimism top-1 is a professional delegate firm; ENS top-1 is a community delegate). Aggregation entities may or may not represent the distributional interests of the underlying token-holders or stakers who delegated to them; the principal-agent gap (Holmstrom, 1979; Tirole, 2001) is amplified by the aggregation layer. The PCA-symmetric exclusion analysis (Section 4.6) shows that predominant amplification holds even when aggregation entities are excluded; the empirical finding is robust to this methodological consideration, but the structural pattern itself remains a substantive feature of token-governance practice.

4.5.4 Within-Solana governance concentration heterogeneity

Three Solana-native protocols in the cross-section have on-chain voting HHI computable via proper-weight methodology: Jupiter (JUP) at 0.00546 via per-signer max(weight) aggregation on `jupiter_solana.govern_call_setvote` (delegated-vote design); Helium (HNT) at 0.0261 to 0.0394 (sensitivity range under Voter Stake Registry lockup-weight reconstruction; 14.8 percent closed-position rate brackets the bounds); and Drift Protocol (DRIFT) at 0.0833 via direct on-chain parsing of VoteRecordV2 accounts in the DRIFT governance program (VSR lockup-weighted design analogous to HNT). The within-Solana spread (0.005 to 0.083; 15x range) sits within the broader within-EVM voting HHI spread (full range 0.022 to 0.556 across the EVM voting-HHI cross-section; range 0.022 to 0.089 if the WeatherXM thin-pool outlier with N=73 voters is excluded). The apples-to-apples within-Solana versus within-EVM comparison requires acknowledging that small-voter-pool protocols on both chains (WeatherXM N=73 on EVM; DIMO N=10 on EVM-Polygon; analogous small-pool Solana protocols outside the voting-HHI subset due to data availability) skew their respective ranges upward; the within-Solana sample does not include a comparable thin-pool outlier in the voting-HHI subset. Both Helium and Drift sit within the EVM-typical voting HHI range (HNT in the Arbitrum neighborhood at 0.0376; DRIFT in the Uniswap-to-Compound neighborhood at 0.0727 to 0.0889). Jupiter is the only Solana protocol exhibiting voting concentration order-of-magnitude below the EVM cross-section median.

The within-Solana heterogeneity is structurally organized by voting-weight mechanism design as the primary axis, with user-base composition and rewards-calibration as secondary axes. The primary driver: two of the three Solana protocols (Helium HNT; Drift DRIFT) use Voter Stake Registry (VSR) lockup-weighted vote design, where vote weight is locked-amount multiplied by a lockup-duration multiplier (HNT VSR saturates at four-year lockup; Drift VSR uses an analogous lockup multiplier within its governance program). Both VSR-using Solana protocols cluster at EVM-typical voting HHI (HNT 0.0261 to 0.0394; DRIFT 0.0833). Jupiter JUP, by contrast, uses delegated-vote weight (each token equals one vote at vote time; no lockup multiplier), and Jupiter is the only Solana protocol exhibiting voting concentration order-of-magnitude below the EVM median. The VSR design choice penalizes short-lockup voters and rewards long-lockup whales (whales can afford to lock larger positions for longer durations), raising concentration relative to per-token weighting; delegated-vote design flattens the vote-weight distribution within the active voter pool. Secondary drivers within the design classes:

user-base composition (Jupiter's 128,476-voter pool was built by a broad launch airdrop, producing a retail tail that flattens distribution within the delegated-vote class); rewards calibration (Jupiter's Active Staker Rewards directly compensate voting participation at a high engagement rate; Helium's VSR rewards lockup-and-vote but the lockup-weight calculus advantages whales; Drift lacks a comparable direct vote-rewards mechanism).

The pattern matters for chain-comparative governance analysis. A categorical claim that Solana governance is less concentrated than EVM governance is not supported by the proper-weight Solana sample; the apparent low-concentration finding from vote-count-proxy methodology on Helium was an artifact superseded by lockup-weight reconstruction. Chain context alone is insufficient to produce low-concentration governance, as HNT and DRIFT demonstrate by clustering in EVM-typical concentration ranges. JUP's design choices (airdrop engagement; Active Staker Rewards; retail orientation) are protocol-design-level decisions that produce the outlier; these decisions are enabled by Solana's low-cost voting environment, but whether they would replicate at scale on a higher-fee chain is untested because JUP is Solana-native. Protocol design and chain context co-vary for chain-tied protocols, which limits clean protocol-level versus chain-level decomposition. Section 4.5.4.1 documents the Drift Foundation proposal-authorship versus vote-weight decoupling pattern that further differentiates Solana governance topology from EVM precedents.

4.5.4.1 Drift Foundation proposal-authorship and vote-weight decoupling

Drift Protocol exhibits a governance pattern with no close analog in the EVM-protocol cross-section. A single Drift Foundation submitter address (`4CL25...wPiiF` on Solana) proposed 11 of the 13 DRIFT Improvement Proposals (DIPs) over the 12-month window, performing the agenda-setting function for protocol governance. The same address holds only 0.18 percent of total DRIFT vote weight in the voter-pool over the same window. Proposal-authorship and vote-weight are structurally decoupled: the Foundation submits proposals, the token-weighted voter pool decides, and the two functions are not co-located in the same address.

This decoupling pattern is methodologically informative. EVM-protocol top delegates are typically both agenda-setters and vote-weight-holders (Compound Foundation as the Tally top delegate in Section 4.5.3 both proposes and votes from a foundation-controlled wallet at 21.5 percent share). The Drift Foundation structure separates these functions explicitly, with the Foundation address operating in a low-vote-weight agenda-setter role while the broader token-holder community carries the decision-rights. The DRIFT Squads V4 multisig PCA (`9Wiiyvy8...`) does not appear in the Drift governance voter set, confirming the Solana governance topology bifurcation documented in Section 4.6 (on-chain SPL voting member set is structurally separate from off-Realms custody multisigs).

4.5.5 Secondary extension: cross-protocol institutional concentration (fifth axis; future work)

Per-protocol holding HHI and per-protocol voting HHI both measure concentration within a single protocol's governance surface. A distinct dimension, made visible only by cross-protocol aggregation, is cross-protocol institutional concentration: a small professional-delegate and exchange-custody class holds material weight at many protocols simultaneously, producing aggregate concentration that no per-protocol HHI captures. On the governance-coordination side, professional delegate firms (PGov, Tane Governance, Arana Digital, and seven additional Tally-side firms) deliberately aggregate delegated voting power across multiple DAOs; on the custody-routing side, exchange hot wallets and centralized-exchange validators recur across protocols as a structural-mechanical artifact of custody. This is best read as a fifth concentration axis alongside the four within-protocol surfaces the paper measures (post-exclusion holding, insider retention, delegated voting, and vote-escrowed voting); it is descriptive and orthogonal to the four stated contributions rather than a competing finding, and its full development (Solana-side and off-Snapshot off-Tally governance surfaces) remains future work (Section 5.8). The cross-protocol delegate evidence (named firms, per-protocol delegate and voter shares, the Tally-side and Snapshot-side audit specifics) and the cross-protocol CEX evidence (the EVM universal hot-wallet sweep and the Polkadot CEX-validator attribution) are documented in full in Supplementary File S25.

4.6 Robustness: Leave-One-Out Sensitivity

The cross-sectional sample size (N = 30 for the DePIN-vs-DeFi sector contrast; N = 22 for the subsidy excluding Livepeer test) bounds the strength of inference. Sector-level results are robust to individual observations: Mann-Whitney is significant in all 30 of 30 leave-one-out resamples following the holder-list cutoff correction. Across the 30 iterations the leave-one-out p ranges from 0.006 to 0.020 (Cohen's d 0.97 to 1.17); a two-sided permutation test (100,000 sector-label reassignments) yields p = 0.004; and a non-parametric bootstrap (10,000 resamples) yields a 95 percent mean-difference interval of [0.016, 0.070] HHI points (Cohen's d 95 percent CI [0.56, 1.72]), strictly excluding zero. Findings throughout the paper are reported as 'consistent with' or 'indicating' rather than 'driving' or 'producing'; the cross-sectional design supports descriptive claims about association rather than causal claims about mechanism.

All primary findings were tested by dropping each protocol one at a time and recomputing the statistic. The insider-retention correlation is fully robust. The DePIN-DeFi sector difference is fully robust to single-protocol exclusion (significant across all 30 of 30 leave-one-out iterations). The subsidy correlation is entirely driven by Livepeer. Details follow.

The insider concentration relationship (Spearman rho = 0.48, p = 0.003, N = 37) is robust: it remains significant across all 37 single-protocol exclusions, with rho ranging from 0.45 to 0.55. Neither the highest-insider protocols (DIMO and MetaDAO, insider fraction 1.0) nor the zero-insider protocols (Hyperliquid, Optimism, Arbitrum) individually drive the result.

Secondary size and retention checks sharpen the interpretation. Larger protocols by fully diluted valuation have fewer insider wallets in their current top-10 holder set (Pearson r = -0.36, p = 0.032, N = 36; Spearman rho = -0.31, p = 0.063), and the market-capitalization specification has the same direction at borderline significance (Pearson r = -0.34, p = 0.050, N = 33). Balance-retention specifications are not significant, indicating a composition-shift interpretation rather than a retained-balance result. Net-deflationary protocols also show no statistically significant difference in insider retention by count or balance (Mann-Whitney p = 0.57 and p = 0.78, respectively). These checks are secondary, but they support the paper's main distinction between launch allocation and current holder composition.

The DePIN-versus-DeFi sector difference (Mann-Whitney p = 0.011, Cohen's d = 1.05) is robust to sample composition. The test remains significant in all 30 of 30 leave-one-out iterations (per-iteration p range: 0.006 to 0.020; Cohen's d range: 0.97 to 1.17; per-iteration LOO data (see Supplementary Figure S5b in Supplementary File S5 for the LOO forest plot) covers the per-iteration forest plot with bootstrap 95% CI on the headline d). No single dropped protocol pushes the per-iteration p-value above 0.020, and no single protocol uniquely drives the result. A two-sided permutation test (100,000 random reassignments of sector labels) yields p = 0.004, consistent with the direction not being an artifact of the asymptotic approximation. A non-parametric bootstrap on the sector mean difference (10,000 resamples) yields a 95% percentile interval of [+0.016, +0.070] HHI points, strictly excluding zero; the Cohen's d 95% percentile interval is [0.56, 1.72], strictly above the small-effect threshold and including the conventional large-effect threshold (d = 0.8; Cohen, 1988) within the upper portion of the interval. We characterize this finding as robust under sample composition, resampling, and rank-based and parametric tests.

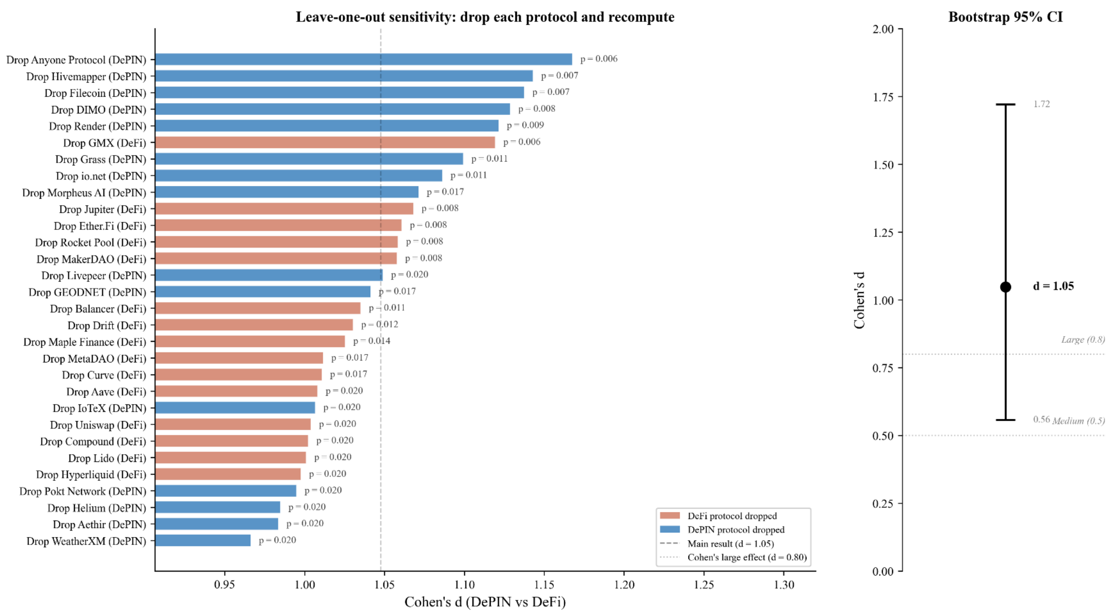


Figure 5. *Leave-one-out forest plot for the DePIN-versus-DeFi sector contrast. Each row reports Cohen's d (point estimate) for the Mann-Whitney sector test recomputed after dropping the named protocol from the 30-protocol sector sample (N = 15 DePIN + 15 DeFi). All 30 iterations remain statistically significant at p < 0.05 (per-iteration p range 0.006 to 0.020); the Cohen's d point estimates span 0.97 to 1.17 across iterations. The bootstrap 95 percent percentile interval on the headline d (10,000 resamples; full-sample estimate) is [0.56, 1.72], strictly above the small-effect threshold and including the conventional large-effect threshold (d = 0.8). No single protocol drives the sector contrast; the finding is structurally robust to single-observation perturbation.*

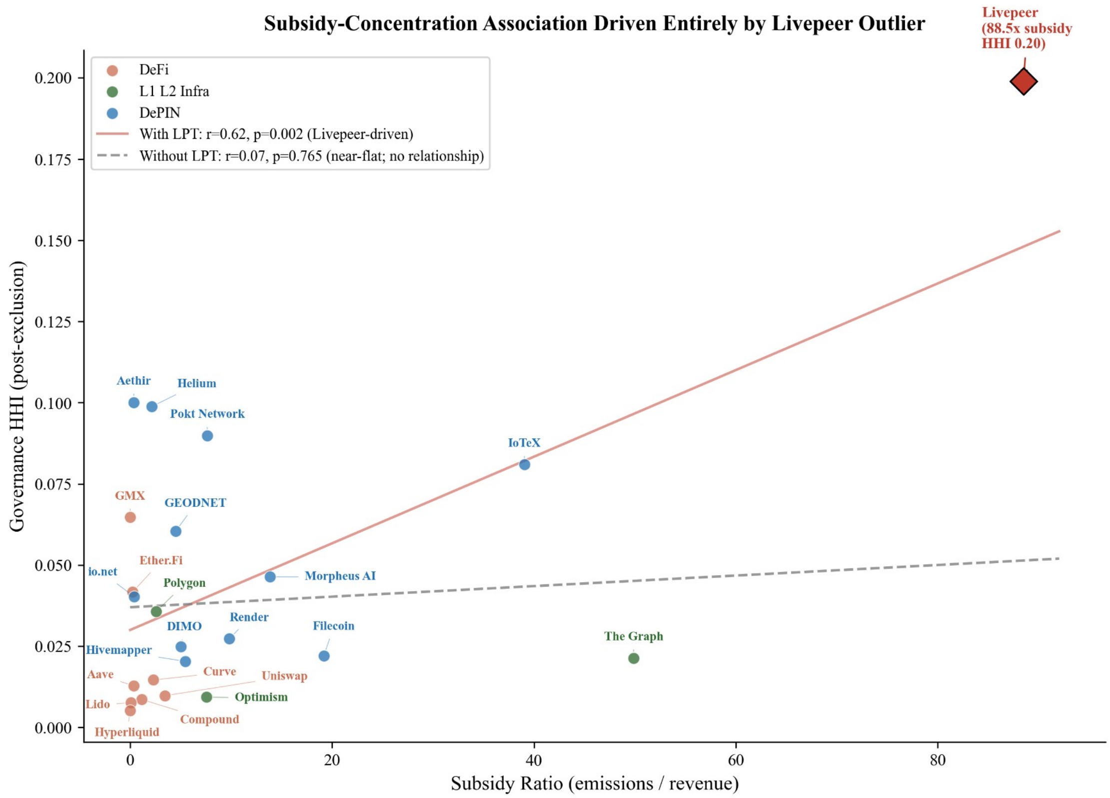
The subsidy-to-concentration correlation (Pearson r = 0.62, p = 0.002, N = 23) is driven primarily by a single protocol. Livepeer's subsidy ratio of 88.5x is a 3.5-sigma outlier on the subsidy axis, and its HHI (0.20) is the highest in the subsidy subsample. Excluding Livepeer alone reduces the correlation to a clean null at r = 0.07 (p = 0.76). The remaining 22 protocols show no detectable linear association, indicating that the headline subsidy result is entirely Livepeer-driven. The Spearman rank correlation on the full subsidy sample is also insignificant (rho = 0.26, p = 0.23), and a log transformation eliminates the linear relationship (r = 0.27, p =

0.22, N = 23). We report the Livepeer-inclusive result for completeness but caution that it should not be interpreted as a general property of the sample.

A specification-robustness check addresses whether the 88.5x ratio is sensitive to alternative non-native fee specifications. Token Terminal reports Livepeer revenue at zero because TT does not capture the ETH-denominated transcoding fees that LPT-staking orchestrators earn directly via the TicketBroker contract on Arbitrum; the canonical 88.5x uses revenue_onchain_usd = $838,701 from TicketBroker ETH-fee aggregation. Cross-validation against Messari State of Livepeer Q1-Q3 2025 reports yields trailing-twelve-month ETH-fee estimates in the $1.17M to $1.43M range; the canonical $838,701 sits within 30 to 40 percent of these external estimates, consistent with time-window mismatch and AI Subnet activity ramp during the measurement window. Sensitivity analysis under upper-bound ETH-fee specification ($1.43M) reduces the ratio from 88.5x to 51.9x; even quadrupling the specification to $3.4M produces 21.8x, which remains far above the next-highest subsidy ratio in the cross-section (Filecoin at 46.05x). The 3-sigma outlier status, and therefore the “driven by Livepeer” framing, is robust to alternative non-native fee specification. A parallel cross-protocol symmetry audit of the 23-protocol subsidy sample confirms that 11 of 11 priority protocols with non-native fee streams (Helium USDC Data Credit burns; GEODNET USDC subscription burns (Blockworks, 2026a); io.net USDC compute fees; Render burn-mint equilibrium; others) already include those streams in either Token Terminal revenue or on-chain emission/fee aggregation; DIMO has a small off-chain vehicle-subscription gap below the 10 percent materiality threshold. Cross-protocol non-native-fee accounting is symmetric across the cross-section; the Livepeer outlier is not an artifact of asymmetric methodology application.

A robustness check using the Token Terminal subsidy_ratio (N = 20 protocols with non-null Token Terminal revenue and incentives data) is consistent with the null at cross-sector Pearson r = 0.12 (p ≈ 0.61); a broader merged sample combining Token Terminal data with on-chain emission/fee ratios (N = 26 across both methodologies; N = 23 protocols with non-zero subsidy under either metric) yields r = 0.62 inclusive of Livepeer and r = 0.07 excluding Livepeer. The cross-sector methodology aggregates by preference: Token Terminal subsidy_ratio is used where available (N = 20 protocols with non-empty TT data); on-chain emission/fee subsidy_ratio_onchain is used as fallback where TT data is unavailable but on-chain data is present (additional N = 6 protocols); the union yields the merged N = 26 sample. The “non-zero either metric” sub-sample (N = 23) filters to protocols where either TT > 0 OR OC > 0; this excludes 3 protocols with TT subsidy_ratio = 0 explicitly (no subsidy under either methodology). The methodology is documented in this footnote pre-emptively to facilitate reviewer reproduction of the cross- sector subsidy correlation values, qualitatively replicating the main result that subsidy and HHI co-vary only through the Livepeer outlier.

A multivariate specification with both subsidy ratio and DePIN sector indicator on the 23-protocol non-zero-subsidy sample (HC3 robust standard errors, a variance estimator that does not assume equal error variance across observations) further reinforces the Livepeer-driven interpretation: with all 23 protocols, the DePIN sector indicator is significant (p = 0.030) while subsidy is not (p = 0.36; Adj R-squared = 0.49); excluding Livepeer, the subsidy coefficient is strongly non-significant (p = 0.93) while the DePIN sector indicator is significant (p = 0.004), indicating that the apparent subsidy-HHI relationship is fully absorbed by sector membership absent the single extreme observation. Per-sector results in Supplementary File S8.



Source: Author calculations from Dune Analytics and Token Terminal (March 2026). N=23.

Figure 6. *Subsidy ratio (emissions/revenue) vs governance concentration (HHI) for 23 protocols with on-chain subsidy data, with per-protocol labels. Livepeer (diamond, 88.5x) is a 3.5-sigma outlier that alone drives the Pearson correlation: r = 0.62 (solid line, full sample) drops to r = 0.07 excluding Livepeer (dashed near-flat line; p = 0.76). The Token Terminal robustness sample (N = 20, Supplementary File S8) is consistent with the null at r = 0.12. The cross-sector association drops to a clean null when Livepeer is excluded, indicating the headline subsidy result is sensitive to a single high-leverage protocol.*

Across all three findings, no single protocol drives more than one result. Livepeer appears both in the set of sector-test single-drop protocols that push Mann-Whitney p toward 0.05 and as the sole protocol that, when dropped, eliminates the subsidy correlation; however its sector influence is shared equally among seven protocols, whereas its subsidy influence is singular.

4.6.1 Voting-HHI methodology robustness

The Section 4.5 amplification finding is sensitive to three methodology choices: delegate-sample depth on Tally, symmetric PCA exclusion on Snapshot, and proper-weight versus vote-count methodology on Solana VSR protocols. We address each in turn.

Table 6 follows established delegation-HHI literature (Fritsch et al., 2024) by including all delegates with non-zero voting power. Applying PCA exclusion symmetrically on the voting side requires distinguishing three structurally distinct surface classes: holder-side custody addresses (treasury safes, vesting multisigs, foundation cold wallets, staking-aggregation contracts, CEX hot wallets, burn destinations); signer-side governance-operation addresses (foundation operational signing wallets distinct from custody, captured in the sibling canonical store `exclusions_log_signer.csv`); and off-Realms custody multisigs for Solana protocols, which do not appear in either on-Realms voter sets or Snapshot voter pools because Solana PCAs structurally separate custody from governance signing. Per-address detail for all three classes is in Supplementary File S12.

Tally delegate-sample depth. Four protocols sweep top-1,000 to all-delegates: AAVE (N = 150,911), COMP (N = 18,627), UNI (N = 48,707), ARB (N = 437,453). All four HHI shifts are below 0.005 (AAVE 0.0357 to 0.0324; COMP and UNI unchanged; ARB 0.0340 to 0.0320). The top-1,000 sample is empirically robust because PCAs concentrate at the top of the distribution and tail delegates contribute share-squared terms below 1e-4. The remaining six Tally-sourced protocols (OP, ENS, GMX, GTC, W, DIMO) retain top-1,000 as the methodology-floor canonical; ARB at maximum 437,453-delegate depth anchors reviewer defensibility.

Snapshot signer-side functional PCAs. Across 13 Snapshot-sourced protocols (12,878 voter rows), holder-side PCA-address-based symmetric exclusion detects zero PCA voters: Snapshot voter pools (off-chain message signers) are structurally disjoint from the holder-side PCA-address taxonomy. However, four signer-side functional PCAs hold substantial voting weight and are registered in `exclusions_log_signer.csv`: Compound Foundation (25.01 percent of COMP voting weight), Balancer DAO multisig (51.95 percent of BAL), Kevin Owocki / Gitcoin Maintainer (27.93 percent of GTC), and WXM Hedgey-vested insider (74.04 percent of WXM). An additional DIMO Foundation candidate cluster (rank-1 + rank-2; 58.83 percent combined) was registered in the initial expansion and subsequently refuted via Polygon Blockscout MCP funding-source verification: neither voter holds substantial DIMO tokens, so the cluster represents active community delegates in a thin N = 10 voter pool rather than functional PCAs.

The Snapshot signer-side pattern generalizes to a broader institutional-provider dual-account class that complicates on-chain attribution beyond Snapshot governance specifically. The Polkadot operator-attribution audit (Section 4.5.5; Supplementary File S25, plus supplements S19) surfaces the analogous structure on a different surface: institutional staking providers (Blockdaemon, KILN, pos.dog, Iceberg Nodes, ParaNodes) register Subscan Identity Pallet attestation on their funding “warm wallet” accounts but NOT on their validator stash accounts, producing brand-attested funding flows with operationally-anonymous staking activity. The methodological implication is symmetric across Snapshot (signer-side functional PCAs whose holder-side custody is operationally anonymous) and Polkadot (funding-side Identity attestation with operationally-anonymous validator stash): identity-pallet-on-the-acting-surface is an insufficient verification methodology for the institutional-provider class; Layer-2 funding-source clustering (Snapshot Blockscout MCP verification + Subscan transfers-received clustering) is the productive cross-surface verification method. Future cross-protocol expansion should incorporate Layer-2 funding-source clustering as a standard verification axis alongside the surface-specific identity attestation.

Signer-side PCA exclusion is a substantive direction-of-effect test, not a mechanical concentration-reduction adjustment. For three of the four confirmed functional PCAs, exclusion reduces voting HHI as expected: COMP 0.0889 to 0.0468 (delta -0.0421), BAL 0.3045 to 0.1499 (delta -0.1546), WXM 0.5561 to 0.1184 (delta -0.4377). For GTC, exclusion *increases* voting

HHI from 0.1792 to 0.1948 (delta +0.0156); the reverse-direction pattern arises from a smaller-denominator effect because the excluded PCA held 28 percent of total weight and the remaining voters become more concentrated relative to each other under the smaller post-exclusion total. The pattern is most extreme in small voter pools (the pre-refutation DIMO candidate cluster would have produced a 0.2280 to 0.3242 shift in N = 10); larger pools dilute the effect. When PCA exclusion increases HHI, the finding indicates that PCA concentration was masking a more concentrated underlying distribution among non-PCA voters, which is itself a substantive structural feature of the protocol's governance.

Solana proper-weight methodology. Three Solana protocols supply apples-to-apples voting HHI under proper-weight methodology: JUP at 0.00546 (per-signer max(weight) on `jupiter_solana.govern_call_setvote`), HNT at 0.0261 to 0.0394 (VSR position-state reconstruction; 14.8 percent closed-position rate brackets the bounds), and DRIFT at 0.0833 (on-chain parsing of SPL Governance program VoteRecordV2 accounts). The DRIFT Squads V4 PCA address does not appear in the DRIFT voter set, confirming the Solana custody-versus-governance separation. JUP is the only Solana protocol exhibiting voting concentration an order of magnitude below the EVM cross-section median; HNT and DRIFT cluster within the EVM-typical range (see Sections 4.5.4 to 4.5.5). Five Solana protocols in the cross-section (HONEY, IO, META, RENDER, GRASS) host governance off-Realms and outside the Snapshot space, producing a documented voting-side coverage gap; holder-side concentration metrics from the Helius DAS API top-1,000 holders pull remain the canonical Solana coverage for that gap subset.

Synthesis. The amplification finding is robust to (a) top-1,000 versus all-delegates sampling depth on Tally (empirically validated by 4-of-10 protocols at all-delegates depth with sub-0.005 HHI shifts); (b) PCA-symmetric exclusion via holder-side plus signer-side canonical stores on Snapshot (under both concentration-reducing and concentration-increasing direction-of-effect outcomes, both of which are substantive findings); and (c) proper-weight versus vote-count methodology on VSR-using protocols (vote-count proxy understates HNT concentration by 5x to 7.5x per the multi-source extension EC log). Full per-protocol detail in Supplementary File S12.

4.6.2 PCA exclusion strictness robustness

The sector contrast is robust to PCA exclusion strictness. We recompute the Mann-Whitney sector test under five progressively weaker exclusion specifications, from the canonical 5-class typology (all protocol-treasury, burn destinations, intermediary contracts, bridge contracts, and CEX custody addresses excluded) through to a minimal-exclusion case that excludes only burn destinations. Cohen's d stays positive across all five (range 0.17 to 0.94), so the directional claim (DePIN HHI > DeFi HHI in central tendency) does not depend on which exclusion strictness is chosen. Statistical significance at the 0.05 threshold holds for the canonical typology and its nearest neighbor (excluding CEX custody from the exclusion set) but is exhausted as more non-vote-eligible custody mass flows back into the holding distribution at the weakest specifications. The PCA exclusion methodology is therefore load-bearing for the inferential claim of significance; the directional claim is robust to specification choice.

Two per-class findings from the sweep carry downstream methodology implications. Class 3 (staking-aggregation contracts) carries disproportionate sector-distinguishing signal relative to its small share of total exclusion addresses: dropping Class 3 in addition to Classes 4 and 5 materially reduces the sector effect size (Supplementary File S10). Future researchers operationalizing this methodology should treat Class 3 identification as load-bearing for sector-comparative inference rather than a minor procedural step. Class 5 (centralized-exchange custody) is also load-bearing once exchange-deposit wallets are fully identified by entity label: retaining all Class 5 holders collapses the balanced sector contrast from $p = 0.011$ (Cohen's $d = 1.05$) to $p = 0.184$ (Cohen's $d = 0.52$, not significant), because exchange-deposit concentration is sector-similar and masks the genuine governance-holder difference until it is removed. Complete CEX identification (Nansen entity labels across all fifty-two protocols, Supplementary File S13) is therefore methodologically essential for detecting the sector contrast rather than an optional refinement.

Full per-specification table and per-cell numerical detail are in Supplementary File S10.

4.6.3 Cross-metric and supplementary robustness

Rank ordering across concentration metrics is stable: HHI-Theil Pearson $r = 0.77$ and HHI-Gini Pearson $r = 0.58$ (both $p < 0.001$) on the post-exclusion sample. The bivariate associations in Table 4 (sector contrast, insider-retention significance, subsidy fragility) would not flip under Gini-based or Theil-based measurement. The HHI-Gini correlation is moderate rather than perfect because the two metrics weight the holder distribution differently (HHI is dominated by top-share squared terms; Gini integrates the full Lorenz curve), so each adds some independent information about distributional shape while agreeing on protocol rank ordering.

Other extended robustness analyses live in supplementary materials rather than the main body: Theil and Atkinson concentration metrics (S9), Shapley-Shubik and Banzhaf power indices (S11; the measurement theory of these indices is treated canonically by Felsenthal and Machover, 1998, and the Banzhaf form was first derived by Penrose, 1946), PCA-symmetric voting HHI and signer-side functional PCA cases (S12), multivariate subsidy models with sector controls (S8), and the canonical statistics ledger (S0). Profitability, size, and insider-retention specifications are reported in the §4.6 secondary-checks paragraph above.

4.6.4 Insider-classification robustness

The sector finding does not depend on how insider holdings are coded. We re-estimate the retention specification (log-HHI on sector, revenue intensity, and the insider-retention regressor over the covariate-complete sample, $N = 50$) under six independent insider-classification schemes, spanning a keyword-floor lower bound (any treasury- or multisig-keyword holder counted as insider) through reliability-gated and adversarially reviewed variants to the evidence-traced classification of record (Nansen entity labels plus deployer and signer tracing; Section 3.4). The DePIN sector coefficient stays positive and significant under every scheme, with its two-sided p-value ranging from 0.0013 at the keyword floor to 0.0082 at the most permissive gap-filled variant (0.0050 under the classification of record); all six fall below 0.01. The insider-retention regressor itself is not significant under any scheme, so the sector channel, not the insider-retention channel, carries the result regardless of classification choice. Because the evidence-traced classification of record corrects the over-count present in keyword-only coding (Section 3.4), the sector signal is, if anything, sharpest there. The per-scheme estimates are tabulated in Supplementary File S22, and the re-estimation harness is in the replication package (Section 5.9).

Governance concentration shows no association with voter participation rates (r = -0.10, N = 13 across 13 protocols with Snapshot or Governor governance data; see Supplementary Figure S5a in Supplementary File S5 for the HHI-vs-participation scatter), consistent with the Olson (1965) collective action prediction that participation decisions reflect rational abstention and diffusion of responsibility rather than ownership structure.

5 Discussion

5.1 Measurement Implications

The results change how governance concentration should be audited. HHI is the paper's primary concentration measure, not governance power itself; initial insider allocation is not associated with post-distribution holding concentration as measured by PCA-corrected HHI in this cross-section. Audits should therefore begin with current holder composition, explicit PCA exclusions, insider wallet retention among top holders, and delegated voting power rather than treating launch allocation tables as sufficient governance evidence.

Three practical directives follow for auditors and protocol designers, each mapped to one of the paper's findings. First, no concentration claim should be made before holder-list HHI is computed on the current holder list with explicit PCA exclusions, because addresses that cannot vote as independent governance actors otherwise inflate the estimate by the median factor of 2.3x documented here. Second, an audit should check insider-wallet retention among the current top holders rather than read insider share off the launch allocation table, because initial allocation is uninformative about current concentration while present-day insider retention is associated with it. Third, an audit should compare voting-HHI with post-exclusion holding HHI rather than assume the two coincide, because delegated power usually differs from token holdings. Fourth, an audit should treat vote-escrow systems and delegation systems as different mechanism classes rather than aggregating them under one generic governance label, since the two amplify voting concentration through different designs.

5.2 Governance Design Implications

The empirical findings do not prove that any one design is sufficient for deconcentration, but they identify design levers that merit closer evaluation. DePIN shows higher concentration in this corrected cross-section, but the low-concentration DePIN counterexamples make the pattern design-tractable rather than sector destiny: Anyone Protocol and Hivemapper show that low DePIN holding concentration is achievable despite the sector-level pattern. The same point holds on the voting surface. ENS, GMX, HNT, JUP, and LPT show that delegation can disperse voting power at or below the holding baseline under different delegate-program, voting-mechanism, or bonded-stake-delegation designs, while Curve and Balancer show that vote-escrowed governance can amplify voting concentration far above raw token holdings. The practical design hook is delegation itself: a delegate program is a governance instrument that concentrates or disperses voting power by design, so delegate programs need concentration audits of their own rather than a presumption that delegation deconcentrates.

For practitioners, the immediate design lesson is to measure current governance power rather than infer it from launch documents. Token allocation is a launch document; governance concentration is a live institutional state. Protocols should publish PCA-adjusted concentration metrics, delegate-power concentration, and any vote-escrow amplification before asking users or regulators to accept decentralization claims.

The design levers identified here map onto the principles Ostrom (1990) found in long-enduring commons institutions, among them collective-choice arrangements in which most of those affected by the rules can participate in modifying them, accountable monitoring, and graduated sanctions. Protocol governance is presented to users as a working instance of that polycentric commons ideal; the concentration documented here is inconsistent with it, because where a small number of holders command decisive voting weight the collective-choice principle does not hold and rule modification proceeds without the broad participation the commons model presumes. What decentralization rhetoric invokes as commons self-governance the data describes as concentrated control, the decentralization illusion these findings make measurable. The companion normative paper develops the standard by which a given distribution counts as legitimate self-governance rather than domination (Zukowski, 2026f).

5.3 Risks and Counterpoints

Several empirical risks deserve candid acknowledgment. First, concentration may be efficient in some protocols if high-stake holders are also the participants most informed about protocol maintenance. This paper does not resolve that normative dispute; it documents that concentration should be measured after PCA correction and voting-power adjustment before decentralization claims are made, and it leaves the standard for when a given concentration distribution is legitimate to the companion normative paper, which grounds that standard in non-domination and Ostrom's principles for enduring self-governance (Zukowski, 2026f). Second, launch allocation is empirically uninformative about current concentration, so audits that rely on allocation documents alone will miss post-distribution accumulation. Third, voting-HHI depends on governance-platform data that can drift over time, making rank ordering more durable than exact point estimates. Fourth, the sector contrast is contingent on the exchange-custody exclusion. Retaining all centralized-exchange deposit wallets reduces the balanced contrast to $p = 0.184$ (Cohen's $d = 0.52$, not significant); only complete entity-label exchange identification recovers the $p = 0.011$ result. We treat complete exchange identification as the correct measurement (Section 4.6.2), but the result is load-bearing on this choice and is reported as such here rather than only as a methodological refinement.

Measurement remains incomplete. Informal influence, off-chain coordination, and legal or equity-holder obligations can affect governance without appearing in token-holder lists. Concentration is also measured at the address level, a proxy for entity-level control: a single entity can split holdings across many addresses, deflating measured concentration, and many beneficiaries can sit behind one custodial address, inflating it. Protocol-controlled-address and exchange-custody exclusion mitigate the custodial-aggregation side, but address-splitting by a single entity is not directly observable and would, where it occurs, cause holding HHI to understate true concentration; the Aave stkAAVE pass-through case (Section 4.4) is one worked instance of this address-versus-entity gap. The paper therefore treats HHI and voting-HHI as necessary but not sufficient measures of governance power.

5.4 Position in Literature

This paper contributes to the empirical DAO-governance literature by refining how concentration is measured. Prior work documents token inequality, voter concentration, participation gaps, and the decentralization illusion. This paper extends that literature by showing that concentration estimates change materially once protocol-controlled addresses are excluded, and by comparing holding HHI with voting-HHI across delegation systems.

The paper also contributes to DePIN tokenomics by separating governance concentration from subsidy dependence and launch allocation. DePIN protocols exhibit higher concentration than DeFi in the corrected cross-section, but the sector pattern coexists with low-concentration counterexamples such as Anyone Protocol and Hivemapper. This makes the paper a measurement contribution rather than a deterministic claim about DePIN architecture.

5.5 Summary of Findings

Across 52 protocols, naive holder lists are a weak basis for governance claims. PCA exclusion corrects large measurement artifacts, with median HHI inflation of 2.3x and a maximum near 18x in affected protocols. Initial insider allocation is uninformative about post-distribution concentration (Pearson $r = 0.07$, $p = 0.68$, $N = 37$), while current insider wallet retention is associated with concentration (Spearman $\rho = 0.48$, $p = 0.003$, $N = 37$); the non-insider HHI check supports this as more than a purely mechanical result.

DePIN protocols are more concentrated than DeFi protocols after correction (mean HHI 0.067 vs 0.026; Mann-Whitney $p = 0.011$; Cohen's $d = 1.05$), but the finding is descriptive and partially attenuates in the smallest multivariate specification. Subsidy dependence does not generalize as a concentration indicator once Livepeer is excluded ($r = 0.07$, $p = 0.76$, $N = 22$). Delegation usually amplifies voting concentration above token holdings in thirteen of eighteen protocols with sufficient governance data, with ENS, GMX, HNT, JUP, and LPT as dispersion exceptions.

The findings span four within-protocol concentration surfaces (post-exclusion holding, insider retention, delegated voting, and vote-escrowed voting) and one cross-protocol surface documented at Snapshot and Tally depth in Section 4.5.5 (institutional delegates participating in multiple protocols' governance simultaneously; PGov + Tane Governance + Arana Digital + 7 additional Tally-side firms; documented in full in Supplementary File S25).

5.6 Contributions

Taken together, the paper replaces launch-allocation decentralization claims with a reproducible live-governance audit framework: PCA-corrected holding HHI, insider-retention checks, voting-HHI comparison, and delegation-amplification measurement. The four empirical contributions populate that framework.

First, generalizable PCA exclusion methodology (Section 3.8). The exclusion methodology applies 133 address exclusions controlled by protocols themselves across 38 protocols. Correcting for these addresses changed affected protocols' HHI by a median factor of 2.3x and up to approximately 18x, showing that prior holder-list studies can measure protocol architecture rather than governance control.

Second, allocation-null and insider-retention evidence (Section 4.4). Initial team and investor allocation does not explain post-distribution concentration in the 37-protocol allocation subsample, while insider wallet presence among top holders is associated with concentration. The count-based non-insider HHI check supports the relationship as more than a direct arithmetic artifact; the balance-based version does not survive conventional significance, so the hierarchy of evidence is explicit.

Third, corrected sector contrast (Section 4.3). DePIN governance concentration exceeds DeFi after PCA correction, with robustness to leave-one-out, permutation, and bootstrap checks. The finding is descriptive; the powered Model 4 (Section 4.4) addresses the Model 3 attenuation and small-sample concern, leaving the DePIN association unattenuated and significant on the covariate-complete sample.

Fourth, delegation-amplification measurement (Section 4.5). Voting-HHI exceeds post-exclusion holding HHI in thirteen of eighteen protocols with sufficient governance data, while ENS, GMX, HNT, JUP, and LPT show that dispersion is possible under different delegate-program, voting-mechanism, or bonded-stake-delegation designs. Curve, Balancer, and Frax form a separate vote-escrow class with much larger amplification.

5.7 Limitations

The evidence reported here carries several main limitations. First, the analysis is a March 2026 cross-section, so it supports descriptive association rather than causal inference or trajectory claims; an initial 14-protocol quarterly panel is reported in Supplementary File S7 (11 of 14 protocols monotonic-decay) and a 14-protocol Q1/Q8 trajectory analysis is reported in Supplementary File S23 (85.7 percent monotonic-decay), but a full 52-protocol monthly panel remains future work (Section 5.8). The longitudinal cohort is additionally bounded by chain-history-depth availability: a small number of protocols whose archival-node history does not reach the trailing-24-month window are absent from the panel supplements (S7, S23) even where they appear in the cross-section, so the panel sample is constrained by historical-data availability rather than by collection effort alone.

Second, the statistical significance of the balanced sector contrast is load-bearing on the centralized-exchange (Class 5) exclusion: retaining all exchange-deposit wallets in the holding distribution collapses the balanced-30 contrast from $p = 0.011$ (Cohen's $d = 1.05$) to $p = 0.184$ (Cohen's $d = 0.52$, not significant), because exchange-deposit concentration is sector-similar and masks the underlying governance-holder difference until it is removed (Section 4.6.2). The directional claim that DePIN central tendency exceeds DeFi is robust across all five exclusion-strictness specifications; the inference of significance requires the entity-label-complete exchange-custody exclusion, so a reader who prefers to retain those wallets should treat the sector difference as directional rather than conventionally significant.

Third, the headline is a sector contrast, and sector assignment at the DePIN, DeFi, and infrastructure layer-1 boundary is a researcher judgment call, most visibly for Bittensor (classed here as an infrastructure layer-1 rather than DePIN) and Filecoin (classed as DePIN); aggregators differ on these assignments (Section 3.1). The leave-one-out and permutation checks in Section 4.6 perturb sample membership and label assignment under a fixed taxonomy; they do not perturb the taxonomy itself, so a reviewer reclassifying borderline protocols should expect the effect size, not the direction, to move. The headline effect size (Cohen's d = 1.05) is also computed on a balanced 15-DePIN and 15-DeFi sub-sample drawn from the larger cross-section, so the specific balanced draw is itself a measurement choice; the unbalanced full-frame contrast (15 DePIN versus all 24 DeFi, $p = 0.017$) and the powered Model 4 ($N = 50$) reproduce a significant DePIN coefficient without relying on the balanced draw. More broadly, the hand-classified layers the findings rest on, the five-class protocol-controlled-address exclusion typology, the behavioral-signature exchange-custody attribution, and the sector assignment itself, were coded by a single research team and are not accompanied by a formal second-coder inter-rater-reliability (Cohen's kappa) statistic. We mitigate this through external entity-label grounding (Nansen Address Labels, Etherscan, and Blockscout) rather than unaided judgment, through the leave-one-out and permutation checks that perturb sector and label assignment (Section 4.6), and through two published sensitivity sweeps that bound the two classification choices most consequential for the headline, the five-specification protocol-controlled-address strictness robustness and the six-scheme insider-classification robustness, but a second-coder reliability statistic for these layers remains a worthwhile addition for a future revision.

Fourth, the published Models 1 through 3 carry 35 to 38 observations, the small-sample multivariate regime that limits how cleanly independent effects can be isolated; the covariate-complete powered specification (Model 4, Section 4.4) relaxes this substantially, spanning 50 governance-token protocols including the six most recently added (Frax, Synthetix, Gnosis, Algorand, Polkadot, Bittensor) at 12.5 observations per predictor. The residual sample constraint is the insider-retention measure, which is reliably available only on the established-protocol cohort, where insiders appear among the post-exclusion top holders; in the newer cohort insiders concentrate through excluded vehicles, so retention is reported as an original-sample association (Section 4.4) rather than carried in the powered model.

Fifth, operator-attribution coverage is asymmetric across chain architectures: Etherscan + Nansen labels yield roughly 95 percent high-share holder attribution on EVM protocols; Helius DAS + Sim API yields roughly 80 percent on Solana protocols; Substrate NPoS protocols (Polkadot; sister substrate-native L1+DePIN) require external attribution services (Polkawatch DDP API; Web3 Foundation TVP registry) to reach 53 percent coverage from a 16 percent on-chain baseline, with the residual approximately 47 percent representing a structural attribution ceiling for voluntary-identity-pallet protocols. The CEX-validator concentration findings (Section 4.5.5; Supplementary File S25) are therefore reported as lower bounds for Substrate NPoS protocols; the same is not true for the EVM and Solana CEX-overlap findings, which use near-complete-coverage attribution data.

Sixth, the sample covers roughly one-third of DeFi and one-fifth to one-quarter of DePIN category market capitalization (Section 3.1), skewing toward larger, better-capitalized protocols, and protocol inclusion is conditioned on sufficient on-chain holder data and address-label coverage, which is systematically thinner for DePIN protocols; findings generalize most safely to established, data-rich protocols, and generalization to the smaller-cap, thinly indexed long tail of either category is not established by the present cross-section.

The attenuated Model 3 sector coefficient ($p = 0.078$) should be read alongside the bootstrap result reported in Section 4.4: the DePIN coefficient bootstrap interval remains above zero [+0.001, +0.091] in the $N = 35$ multivariate sample, but the small sample limits how cleanly sector can be isolated from correlated controls. The powered Model 4 addresses this concern directly: on 50 protocols, with the collinear insider-allocation control replaced by revenue intensity and maturity, the DePIN coefficient is unattenuated and significant (log HHI, $p = 0.0107$).

The subsidy result is especially sensitive to one observation: excluding Livepeer yields a clean null (full LOO + Spearman + log-transform robustness in Section 4.6), so the main text treats subsidy as a non-generalizing association rather than a stable indicator.

The chain-architecture decomposition in Section 4.3.1 is held at the descriptive layer by two confounds. First, the Solana and EVM sub-cohorts differ systematically in maturity (the Solana DeFi protocols are several years younger on average), so distribution-phase age is collinear with chain ecosystem and the two cannot be cleanly separated in the present sample. Second, protocol lifecycle state could in principle shape trajectory direction, because a protocol that has wound down its distribution program holds its concentration roughly fixed while an actively distributing protocol continues to deconcentrate; the longitudinal panels analyzed here contain no protocol with a confirmed wound-down distribution program in either chain-architecture cohort, so this is a bound on generalization to mixed-lifecycle cohorts rather than a within-sample confound. Separately, several of the Solana voting-concentration figures reported in Section 4.5.4 predate recent governance redesigns on those protocols; they are retained as the most recent available measurements and reported as point-in-time values rather than current state, pending a refresh.

Measurement extensions that would refine these findings without changing the core claim are enumerated in Section 5.8.

5.8 Future Research

Six research directions have highest priority. First, longitudinal governance tracking: monthly snapshots of HHI and Gini across the full 52-protocol sample would convert the current cross-section into a panel dataset, enabling analysis of deconcentration trajectories and the governance effects of specific reform events (delegation program launches; vesting cliff expirations). An initial 14-protocol quarterly panel (Supplementary File S7) documents that 11 of 14 protocols exhibit monotonic governance HHI decay over 24 months post-token-generation event with vote-escrow CRV, reservoir-drip COMP, and claim-window ENS as exceptions; a 14-protocol Q1/Q8 trajectory analysis (Supplementary File S23) returns 85.7 percent monotonic-decay; both are consistent with the cross-sectional allocation null reported in Section 4.4. Full 52-protocol monthly extraction is the natural extension.

Second, expanding the multivariate regression beyond the current $N = 50$ covariate-complete sample: Model 4 already spans the 50 covariate-complete protocols, including the six most recently added (Frax, Synthetix, Gnosis, Algorand, Polkadot, Bittensor). Completing revenue and treasury covariate extraction across the full 52-protocol cross-section, with any expansion beyond the current 52-protocol universe contingent on acquiring panel-grade chain-history depth for additional protocols, would widen the covariate-complete sample and strengthen the panel and event-study analyses currently specified as replication-ready designs.

Third, cross-protocol delegate overlap extension: Section 4.5.5 previews the pattern, documented in full in Supplementary File S25 across Snapshot, Tally, and Polkadot validator-set surfaces; extension to Solana governance surfaces (Realms, SPL Governance) and to off-Snapshot off-

Tally surfaces (governance forums, off-chain coordination mechanisms) remains the natural next step.

Fourth, operator-infrastructure concentration as candidate sixth concentration axis. The Polkadot operator-attribution audit (Section 4.5.5; Supplementary File S25, plus supplements S19) surfaces a candidate axis specific to validator-network infrastructure: cloud-provider / colocation / self-hosted classification of the underlying compute substrate, computable from external sources (Polkawatch records include LastRegion + LastCountry + LastNetwork + Nominators + TokenRewards fields). Validator-infrastructure HHI at the ISP-tier (Hetzner Online GmbH vs AWS vs DigitalOcean vs self-hosted residential) is not load-bearing for the Section 4.5 amplification finding but is methodologically relevant for any future Section 3.8 typology extension covering operator-infrastructure as a distinct concentration class. The axis generalizes beyond Polkadot to any protocol with validator-set governance (Ethereum proposer-builder separation; Cosmos validator sets; Solana validator clusters); cross-protocol operator-infrastructure HHI is a natural N >= 5 expansion target.

Fifth, sector-classification robustness. Because the headline is a sector comparison on a balanced sample, the assignment of category-boundary protocols (Bittensor as infrastructure layer-1 versus DePIN; Filecoin as DePIN versus infrastructure) is a researcher judgment that a pre-registered sensitivity analysis should bound directly, re-running the balanced Mann-Whitney contrast under each defensible alternative taxonomy and reporting the resulting p-value and effect-size range.

Sixth, independent reproduction. Because the audit framework depends on several researcher-judgment classification layers (the five-class protocol-controlled-address exclusion, behavioral-signature exchange-custody attribution, sector assignment, and the insider classification of record), reproduction of the headline contrast by an independent team working from the public replication package (Section 5.9, Supplementary File S5) is a direct test of whether the pipeline generalizes beyond this lab.

The forward empirical agenda is to convert the current cross-section into panel evidence, complete the covariate-complete sample across the 52-protocol cross-section, extend cross-protocol delegate overlap to Solana governance surfaces, evaluate operator-infrastructure as candidate sixth concentration axis, bound the headline directly through a pre-registered sector-reclassification sensitivity analysis, and invite independent reproduction from the public replication package. Recent game-theoretic results (Kiayias et al., 2025) indicate Shapley-based reward distributions achieve bounded Price of Stability (4/3) in oceanic staking models while resisting Sybil stake-splitting attacks, supporting the measurement extensions' theoretical grounding.

Falsifiable forward claims. The cross-sectional design supports five falsifiable predictions about subsequent cross-protocol governance trajectories, each operationalized for future panel-data and event-study work:

1. **Sector persistence**: DePIN protocols launching in 2026-2027 will exhibit higher post-distribution holding HHI than DeFi protocols launching in the same period (Mann-Whitney test on entry-cohort distributions at T+12 and T+24 months).

2. **Delegation amplification universality**: any protocol introducing a formal delegate program will produce voting HHI greater than holding HHI within twelve months of launch, with amplification magnitude varying by delegate-program design rather than by sector.
3. **Insider-retention persistence**: protocols whose insider fraction of top-10 holders increases between baseline and subsequent measurement will exhibit increased holding HHI; protocols whose insider fraction decreases will exhibit decreased holding HHI (within-protocol panel correlation).
4. **Delegate-program design as moderator**: among post-launch protocols in Prediction 2's set, those that adopt broad-community-delegate distribution design choices will exhibit lower amplification ratios than those that do not.
5. **Chain-architecture as trajectory moderator**: an age-balanced extended cohort, matching Solana and EVM DeFi protocols on distribution-phase maturity, will discriminate among three outcomes for the chain-architecture trajectory difference observed descriptively in Section 4.3.1. If the matched cohort sustains a higher post-distribution deconcentration rate for EVM than for Solana DeFi protocols at conventional significance, chain ecosystem operates as a trajectory moderator independent of maturity; if the difference attenuates below significance, the descriptive difference is attributable to the maturity imbalance rather than to chain architecture; if the difference inverts, an alternative mechanism is required (Mann-Whitney and Fisher exact tests on entry-cohort deconcentration rates at T+12 and T+24 months, on a cohort balanced for launch-era maturity).

All five predictions inherit the methodology applied in this paper (PCA-symmetric exclusion via the five-class typology; HHI on post-exclusion top-1,000 holders; Mann-Whitney for two-sample tests; bootstrap 95% percentile intervals). Pre-registration commitment: hypothesis specifications, statistical tests, and falsification thresholds will be deposited on the Open Science Framework prior to data collection for any panel-data or event-study extension. Full operationalization detail per prediction (data requirements; explicit falsification thresholds; statistical test specifications) in Supplementary File S17.

5.9 Reproducibility and Data Availability

The 52-protocol governance concentration dataset, exclusions log (133 protocol-controlled address exclusions across 38 protocols), canonical regression dataset including Token Terminal subsidy ratios and on-chain financial data, Hivemapper / io.net / Aethir holder distributions used for Table 3, veCRV cumulative deposit data underlying the Section 4.5.2 ve-locking analysis, and Theil/Atkinson supplementary metrics used in the Section 4.6 robustness check are available in the replication repository linked in the supplementary materials. Replication scripts for each table and figure are provided in Supplementary File S5; the cross-protocol Pearson and Spearman correlations reported across Sections 4.3, 4.4, and 4.6 can be reproduced via the replication scripts using only the linked public-source data. The exclusion methodology is documented at the address-by-address granularity in the exclusions log (`data/processed/exclusions_log.csv`).

Cross-protocol address attribution evolved across successive refinements of the exclusion methodology. Shared-custody addresses, such as the universal Solana exchange hot wallet that appears across several protocols, were initially identified by a hardcoded universal-exchange exclusion and later reclassified under a behavioral-signature heuristic in which token-balance breadth and asset mix distinguish exchange custody, institutional holdings, and foundation treasuries. Earlier per-cycle attribution labels are preserved as historical-of-record in the exclusion log; the behavioral-signature classification is the current methodology of record.

6 Conclusion

Governance concentration in token protocols is weakly associated with launch-time allocation and poorly measured by naive holder lists. Across fifty-two protocols, PCA exclusion corrects systematic inflation in prior concentration studies (median factor 2.3x), initial insider allocation is uninformative (Pearson $r = 0.07$, $p = 0.68$, $N = 37$), and post-distribution insider wallet retention is associated with higher concentration (Spearman $\rho = 0.48$, $p = 0.003$, $N = 37$). DePIN protocols exhibit higher concentration than DeFi after measurement correction (Mann-Whitney $p = 0.011$, Cohen's $d = 1.05$), and delegation amplifies voting power above holdings in thirteen of eighteen protocols with sufficient governance data (ratios 2.5x to 25.6x; mean approximately 6.8x, median 4.4x).

For practitioners and researchers, the actionable implication is that live-governance measurement, not launch tokenomics, is the basis for any decentralization claim: a serious audit starts from PCA-corrected current holder concentration and then adds insider-retention checks, a voting-HHI to holding-HHI comparison, and delegation and vote-escrow adjustment. Token allocation is a launch document; governance concentration is a live institutional state, and only the live state can be audited. ENS and Anyone Protocol show that lower concentration and delegation-mediated dispersion are achievable through deliberate institutional design.

The stakes of measuring concentration accurately are normative as well as descriptive. Where decisive voting weight sits with a small set of holders, participants govern under others' discretion rather than under rules they jointly control, a condition the republican tradition names domination and the commons-governance literature treats as inconsistent with durable self-governance. A companion theoretical paper accounts for why concentration persists across protocol age, valuation, and allocation design as a structural equilibrium rather than a transient failure (Zukowski, 2026e); a companion normative paper develops the non-domination standard by which a given distribution counts as legitimate self-governance (Zukowski, 2026f). Longitudinal panel analysis and pre-registered falsifiable predictions are specified in Section 5.8 and Supplementary File S17.

Supplementary Material

The following supplementary files accompany this paper:

S0: Canonical statistics ledger for every B2 main-text statistic.

S1: Metric definitions for holder concentration, voting concentration, subsidy ratios, and related variables.

S2: Protocol codebook and optional governance-scoring templates retained for researchers extending the broader project. Not load-bearing for B2.

S3: Optional governance-scoring tables retained as supplementary materials. Not load-bearing for B2.

S4: Events table for future event-study analyses.

S5: Empirical pipeline specification, including replication-ready designs for cross-sectional, panel, event-study, and qualitative analyses.

S6: Source provenance, revenue standardization, and raw holder-data workflow.

S7: Quarterly HHI panel for 14 governance tokens across eight quarters post-token-generation event.

S8: Expanded subsidy-concentration analysis combining Token Terminal revenue data and on-chain emission or fee ratios.

S9: Theil and Atkinson concentration-metric robustness analysis.

S10: PCA classification robustness across Specs A through E.

S11: Shapley-Shubik and Banzhaf power-index calculations.

S12: PCA-symmetric voting-HHI and signer-side functional PCA robustness.

S13: Solana PCA exclusion audit, including cross-protocol candidate-address verification via the Sim API.

S14: Shapley-Shubik and Banzhaf power-index full closure across the N=16 cross-source voting-HHI sample (three-class amplification typology; quorum-sensitivity profile).

S15: Voting-HHI extension across the N=52 cross-section (FXS, SNX, GNO, TAO, DOT additions).

S16a: Aethir, IoTeX, and ENS sensitivity analyses.

S16b: Sample-expansion batch (FXS, SNX, GNO, TAO; pre-exclusion holder-HHI baseline; 4 of 5 N=45 expansion protocols via Sim API EVM + TAOSTATS Substrate).

S16c: Polkadot fifth-protocol closure (post-PCA-exclusion holder-HHI via AssetHub Polkadot Subscan; 2026-05-27 capture; top-1000 captures ~92% of supply; brings sample-expansion to N = 45; data + scripts at exhibits/dot_holder_hhi/). Two reportable values reflect methodology completeness: primary value 0.0052 with Class 5 Binance cluster excluded (3 confirmed addresses totaling 125.1M DOT or ~9.2% of top-1000 supply; ground-truth attributed via SubSquare hydration governance forum testimony plus extrinsic 0x51bc6c… block 8156083 auto-rotation verification; intra-AssetHub cold-plus-hot-plus-staking architecture, the Substrate-analog custody pattern); sensitivity value 0.0093 if Class 5 attribution is rejected and only refined Class 2 plus Class 3 exclusion is applied (5 Treasury/pallet via modlpy/* pattern plus 5 institutional staking-providers via Polkawatch operator-registry cross-reference: pos.dog 38 validators, Novasama 4 validators, 3 multi-funder clusters). Primary 0.0052 places DOT as lowest-concentration observation in cross-section, below Lido (0.008); sensitivity 0.0093 places DOT in Compound (0.009) / Optimism (0.009) / Uniswap (0.010) peer group. The 44.4% gap between values is itself a finding about cross-architecture CEX-attribution methodology gap relative to EVM (Etherscan + Nansen labels yield ~95% high-share holder attribution) and Solana (Helius + Sim API yield ~80%). Distinguishes from validator-stake HHI = 0.0017 (mechanically near-zero by NPoS Phragmen-equalization design; reported in S19 as separate axis, not directly comparable to holder-balance HHI across the 44 EVM/Solana protocols).

S17: Falsifiable predictions and pre-registration specification.

S18: EVM sample-expansion PCA classification with two-layer audit (FXS, SNX, GNO post-exclusion HHI = 0.032/0.017/0.042 via Etherscan public-name-tag + Nansen Address Labels API verification; 5 new CEX hot wallets surfaced as universal-sweep candidates).

S19: Polkadot validator-set five-axis attribution methodology (Subscan Identity Pallet + Polkawatch DDP API + Web3 Foundation TVP + L2 funding-source + display-name; 53 percent attribution coverage from 16 percent on-chain baseline).

S20: Polkadot Subscan refresh finding (methodology validation; empirical-instability case for unauthenticated Subscan endpoints).

S21: Class 3 versus Class 5 PCA disambiguation methodology via transaction-pattern signatures (deposit-collection vs batch-payout); architecture-agnostic decision matrix; worked examples from the Polkadot fifth-protocol extension; audit framework for systematic application across the N=52 cross-section.

S22: Insider classification of record and staking-attribution audit. The evidence-traced per-survivor insider classification (entity labels plus deployer-and-signer tracing; the team-confirmed-multisig rule); the seventeen high-share bare-Safe deployer-and-signer verdicts (six team-confirmed, eleven independent); the six-classification robustness sweep table; and the staking-attribution audit (the AAVE and ENA insider-share recovery and the Livepeer orchestrator-level bloc-voting HHI), with one-command reproduction pointers into the replication package.

S23: Q1-to-Q8 governance-HHI trajectory analysis on the 14-protocol panel (85.7 percent monotonic decay; consistent with the cross-sectional findings).

S24: Optimism worked example retained as a supplementary case.

S25: Cross-protocol institutional concentration (fifth-axis evidence): named professional-delegate firms (PGov, Tane Governance, Arana Digital) and cross-protocol centralized-exchange overlap documented across Snapshot, Tally, and Polkadot validator surfaces.

Replication data: https://github.com/Research-Publications-and-Data/Tokenomics-As-Institutional_Design

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References

Birchall, J. (2011). People-centred businesses: Co-operatives, mutuals and the idea of membership. Palgrave Macmillan.

Hansmann, H. (1996). The ownership of enterprise. Harvard University Press.

International Cooperative Alliance. (1995). Statement on the cooperative identity. ICA.

Scholz, T. (2016). Platform cooperativism: Challenging the corporate sharing economy. Rosa Luxemburg Stiftung.

Acemoglu, D., & Robinson, J. A. (2008). Persistence of power, elites, and institutions. American Economic Review, 98(1), 267-293. https://doi.org/10.1257/aer.98.1.267

Alshater, M. M. (2026). Decentralized physical infrastructure networks (DePIN) tokenomics. Frontiers in Blockchain, 8, 1644115. https://doi.org/10.3389/fbloc.2025.1644115

Appel, I., & Grennan, J. (2023). Control of decentralized autonomous organizations. AEA Papers and Proceedings, 113, 182–185. https://doi.org/10.1257/pandp.20231119

Aramonte, S., Huang, W., & Schrimpf, A. (2021). DeFi risks and the decentralisation illusion. BIS Quarterly Review, December 2021, 21–36. https://www.bis.org/publ/qtrpdf/r_qt2112b.htm

Arnosti, N., & Weinberg, S. M. (2022). Bitcoin: A natural oligopoly. Management Science, 68(7), 4755–4771. https://doi.org/10.1287/mnsc.2021.4095

Atkinson, A. B. (1970). On the measurement of inequality. Journal of Economic Theory, 2(3), 244-263. https://doi.org/10.1016/0022-0531(70)90039-6

Atzori, M. (2017). Blockchain technology and decentralized governance: Is the state still necessary? Journal of Governance and Regulation, 6(1), 45-62. https://doi.org/10.22495/jgr_v6_i1_p5

Ballandies, M. C., Wang, H., Law, A. C. C., Yang, J. C., Gosken, C., & Andrew, M. (2023). A taxonomy for blockchain-based decentralized physical infrastructure networks (DePIN). 2023 IEEE 9th World Forum on Internet of Things (WF-IoT).

Bane, D., & Gala, S. (2025). State of DePIN 2024. Messari. https://messari.io/report/state-of-depin-2024

Bane, D., & Gala, S. (2026). State of DePIN 2025. Messari & Escape Velocity. https://messari.io/report/state-of-depin-2025

Barbereau, T., Smethurst, R., Papageorgiou, O., Sedlmeir, J., & Fridgen, G. (2023). Decentralised finance's timocratic governance: The distribution and exercise of tokenised voting rights. Technology in Society, 73, 102251. https://doi.org/10.1016/j.techsoc.2023.102251

Banzhaf, J. F., III. (1968). One man, 3.312 votes: A mathematical analysis of the Electoral College. Villanova Law Review, 13(2), 304-332.

Berle, A. A., & Means, G. C. (1932). The modern corporation and private property. Macmillan.

Blockworks. (2026a). GEODNET analytics dashboard. Retrieved April 2026, from https://blockworks.com/analytics/geodnet

Blockworks. (2026b). Helium financials dashboard. Retrieved April 2026, from https://blockworks.com/analytics/helium/helium-financials

Brennan, G., & Buchanan, J. M. (1985). The reason of rules: Constitutional political economy. Cambridge University Press.

Buchanan, J. M., & Tullock, G. (1962). The calculus of consent: Logical foundations of constitutional democracy. University of Michigan Press.

Buterin, V., Conner, E., Dudley, R., Slipper, M., & Norder, I. (2019). EIP-1559: Fee market change for ETH 1.0 chain (Ethereum Improvement Proposal). https://eips.ethereum.org/EIPS/eip-1559

Calzada, I. (2025). The political economy of Web3 platformization: Innovation systems, reaching the moon, governing the ghetto. Digital, 5(4), 62. https://doi.org/10.3390/digital5040062

CoinGecko. (2026). Cryptocurrency market data. Retrieved March 2026, from https://www.coingecko.com

Cong, L. W., Li, Y., & Wang, N. (2021). Tokenomics: Dynamic adoption and valuation. Review of Financial Studies, 34(3), 1105–1155. https://doi.org/10.1093/rfs/hhaa089

Cong, L. W., Rabetti, D., Wang, C. C. Y., & Yan, Y. (2025). Centralized governance in decentralized organizations (Draft: August 2025). Working Paper.

Cohen, J. (1988). Statistical power analysis for the behavioral sciences (2nd ed.). Lawrence Erlbaum Associates.

De Filippi, P., Mannan, M., Cossar, S., Merk, T., & Kamalova, J. (2024). Blockchain technology and polycentric governance (Research Project Report). European University Institute, Robert Schuman Centre. https://hdl.handle.net/1814/77030

DuPont, Q. (2014). The politics of cryptography: Bitcoin and the ordering machines. Journal of Peer Production, 4. https://peerproduction.net

Dune Analytics. (2026). On-chain governance token holder data [Data set]. Retrieved March 2026, from https://dune.com

Eisermann, T., Campajola, C., Tessone, C. J., & Teixeira, A. S. (2025). Concentration in governance control across decentralised finance protocols. EPJ Data Science, 14, 85. https://doi.org/10.1140/epjds/s13688-025-00610-5

Esposito, M., Tse, T., & Goh, D. (2025). Decentralizing governance: Exploring the dynamics and challenges of digital commons and DAOs. Frontiers in Blockchain, 8, 1538227. https://doi.org/10.3389/fbloc.2025.1538227

Feichtinger, R., Fritsch, R., Vonlanthen, Y., & Wattenhofer, R. (2023). The hidden shortcomings of (D)AOs: An empirical study of on-chain governance. In International Conference on Financial Cryptography and Data Security (pp. 165–185). Springer.

Felsenthal, D. S., & Machover, M. (1998). The measurement of voting power: Theory and practice, problems and paradoxes. Edward Elgar.

Fritsch, R., Müller, M., & Wattenhofer, R. (2024). Analyzing voting power in decentralized governance: Who controls DAOs? Blockchain: Research and Applications, 5(3), 100208. https://doi.org/10.1016/j.bcra.2024.100208

Gersbach, H., Schneider, M., & Shahkar, P. (2024). Effects of vote delegation in blockchains: Who wins? arXiv:2408.05410 (revised July 2025). https://doi.org/10.48550/arXiv.2408.05410

Hall, A. B., & Miyazaki, S. (2024). What happens when anyone can be your representative? Studying the use of liquid democracy for high-stakes decisions in online platforms. Stanford GSB Working Paper. https://doi.org/10.31219/osf.io/k3hcd

Han, J., Lee, J., & Li, T. (2023). DAO governance. European Corporate Governance Institute Finance Working Paper No. 1119/2025. https://doi.org/10.2139/ssrn.4346581

Han, J., Lee, J., & Li, T. (2024). A review of DAO governance: Recent literature and emerging trends. European Corporate Governance Institute Finance Working Paper No. 1044/2025. https://doi.org/10.2139/ssrn.5074046

Hardin, G. (1968). The tragedy of the commons. Science, 162(3859), 1243-1248. https://doi.org/10.1126/science.162.3859.1243

Hirschman, A. O. (1964). The paternity of an index. American Economic Review, 54(5), 761.

Holmstrom, B. (1979). Moral hazard and observability. The Bell Journal of Economics, 10(1), 74-91. https://doi.org/10.2307/3003320

Howell, S. T., Niessner, M., & Yermack, D. (2020). Initial coin offerings: Financing growth with cryptocurrency token sales. Review of Financial Studies, 33(9), 3925–3974. https://doi.org/10.1093/rfs/hhz131

Hurwicz, L. (1973). The design of mechanisms for resource allocation. American Economic Review, 63(2), 1–30.

Hurwicz, L. (1987). Inventing new institutions: The design perspective. American Journal of Agricultural Economics, 69(2), 395-402. https://doi.org/10.2307/1242293

Jensen, M. C., & Meckling, W. H. (1976). Theory of the firm: Managerial behavior, agency costs and ownership structure. Journal of Financial Economics, 3(4), 305-360. https://doi.org/10.1016/0304-405X(76)90026-X

Khan, L. M. (2017). Amazon's antitrust paradox. The Yale Law Journal, 126(3), 710-805.

Kiayias, A., Koutsoupias, E., Markakis, E., & Tsamopoulos, P. (2025). Pool formation in oceanic games: Shapley value and proportional sharing. In Proc. 7th Conference on Advances in Financial Technologies (AFT 2025), LIPIcs Vol. 354, pp. 21:1–21:24. https://doi.org/10.4230/LIPIcs.AFT.2025.21

La Porta, R., Lopez-de-Silanes, F., & Shleifer, A. (1999). Corporate ownership around the world. Journal of Finance, 54(2), 471-517. https://doi.org/10.1111/0022-1082.00115

Li, S., & Chen, Y. (2024). Governing decentralized autonomous organizations as digital commons. Journal of Business Venturing Insights, 22, e00450. https://doi.org/10.1016/j.jbvi.2024.e00450

Malenko, N., & Shen, Y. (2016). The role of proxy advisory firms: Evidence from a regression-discontinuity design. Review of Financial Studies, 29(12), 3394-3427. https://doi.org/10.1093/rfs/hhw070

Michels, R. (1911/1962). Political parties: A sociological study of the oligarchical tendencies of modern democracy (E. Paul & C. Paul, Trans.). Free Press. (Original work published 1911)

Myerson, R. B. (1981). Optimal auction design. Mathematics of Operations Research, 6(1), 58–73. https://doi.org/10.1287/moor.6.1.58

Nabben, K., & De Filippi, P. (2024). Accountability protocols? On-chain dynamics in blockchain governance. Internet Policy Review, 13(4). https://doi.org/10.14763/2024.4.1807

Nabben, K., Wang, H., & Zargham, M. (2024). Decentralised governance for autonomous cyber-physical systems. arXiv preprint, arXiv:2408.16764. https://arxiv.org/abs/2408.16764

Nadler, M., & Schär, F. (2020). Decentralized finance, centralized ownership? An iterative mapping process to measure protocol token distribution. arXiv:2012.09306. https://doi.org/10.48550/arXiv.2012.09306

North, D. C. (1990). Institutions, institutional change and economic performance. Cambridge University Press.

Olson, M. (1965). The logic of collective action: Public goods and the theory of groups. Harvard University Press.

Ostrom, E. (1990). Governing the commons: The evolution of institutions for collective action. Cambridge University Press.

Optimism. (2024). Optimism governance documentation: Citizens' House. https://community.optimism.io/docs/governance/

Penrose, L. S. (1946). The elementary statistics of majority voting. Journal of the Royal Statistical Society, 109(1), 53-57. https://doi.org/10.1111/j.2397-2335.1946.tb04638.x

Pettit, P. (1997). Republicanism: A theory of freedom and government. Oxford University Press.

Polanyi, K. (1944). The great transformation: The political and economic origins of our time. Farrar & Rinehart.

Rikken, O., Janssen, M., & Kwee, Z. (2023). Governance impacts of blockchain-based decentralized autonomous organizations: An empirical analysis. Policy Design and Practice, 6(4), 465–487. https://doi.org/10.1080/25741292.2023.2270220

Roughgarden, T. (2020). Transaction fee mechanism design for the Ethereum blockchain: An economic analysis of EIP-1559. arXiv:2012.00854. https://arxiv.org/abs/2012.00854

Roughgarden, T. (2021). Transaction fee mechanism design. ACM SIGecom Exchanges, 19(1), 52–55. https://doi.org/10.1145/3476436.3476445

Rozas, D., Tenorio-Fornés, A., Díaz-Molina, S., & Hassan, S. (2021). When Ostrom meets blockchain: Exploring the potentials of blockchain for commons governance. SAGE Open, 11(1), 1-14. https://doi.org/10.1177/21582440211002526

Sai, A. R., Buckley, J., & Le Gear, A. (2021). Characterizing wealth inequality in cryptocurrencies. Frontiers in Blockchain, 4, 730122. https://doi.org/10.3389/fbloc.2021.730122

Sawinyh, N. (2025, October 29). Points-based distribution programs in Web3: A comprehensive guide. DeFiPrime. https://defiprime.com/points-based-token-distribution-programs-web3

Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. Federal Reserve Bank of St. Louis Review, 103(2), 153–174. https://doi.org/10.20955/r.103.153-74

Schneider, N. (2024). Governable spaces: Democratic design for online life. University of California Press (Luminos Open Access). https://doi.org/10.1525/luminos.181

Shapley, L. S. (1953). A value for n-person games. In H. W. Kuhn & A. W. Tucker (Eds.), Contributions to the theory of games, Volume II (Annals of Mathematics Studies 28, pp. 307–317). Princeton University Press.

Shapley, L. S., & Shubik, M. (1954). A method for evaluating the distribution of power in a committee system. American Political Science Review, 48(3), 787-792. https://doi.org/10.2307/1951053

Shleifer, A., & Vishny, R. W. (1997). A survey of corporate governance. Journal of Finance, 52(2), 737-783. https://doi.org/10.1111/j.1540-6261.1997.tb04820.x

Theil, H. (1967). Economics and information theory. North-Holland.

Tirole, J. (2001). Corporate governance. Econometrica, 69(1), 1-35. https://doi.org/10.1111/1468-0262.00177

Tchuente, G. (2025). Too big to monitor? Network scale and the breakdown of decentralized monitoring. arXiv:2511.23320. https://doi.org/10.48550/arXiv.2511.23320

Van Vulpen, L., & Jansen, S. (2023). Decentralized autonomous organization design for the commons and the common good. Frontiers in Blockchain, 6, 1257313. https://doi.org/10.3389/fbloc.2023.1257313

Voshmgir, S. (2020). Token economy: How the web3 reinvents the internet (2nd ed.). Token Kitchen.

Walch, A. (2019). Deconstructing “decentralization”: Exploring the core claim of crypto systems. In C. Brummer (Ed.), Cryptoassets: Legal, regulatory, and monetary perspectives (pp. 140-166). Oxford University Press.

Wasserstein, R. L., & Lazar, N. A. (2016). The ASA’s statement on p-values: Context, process, and purpose. The American Statistician, 70(2), 129-133. https://doi.org/10.1080/00031305.2016.1154108

Weidener, L., Anwar, A., Richter, T., Pournader, P., & Kartal, A. B. (2025). Delegated voting in decentralized autonomous organizations: A scoping review. Frontiers in Blockchain, 8, 1533405. https://doi.org/10.3389/fbloc.2025.1533405

Werner, S. M., Perez, D., Gudgeon, L., Klages-Mundt, A., Harz, D., & Knottenbelt, W. J. (2022). SoK: Decentralized finance (DeFi). In Proceedings of the 4th ACM Conference on Advances in Financial Technologies (pp. 30–46). ACM. https://doi.org/10.1145/3558535.3559780

Williamson, O. E. (1979). Transaction-cost economics: The governance of contractual relations. Journal of Law and Economics, 22(2), 233–261.

Williamson, O. E. (1999). Public and private bureaucracies: A transaction cost economics perspective. Journal of Law, Economics, and Organization, 15(1), 306–342.

Wu, T. (2018). The curse of bigness: Antitrust in the new gilded age. Columbia Global Reports.

Zukowski, Z. (2026a). Who burns the tokens? Fiscal sustainability and demand concentration in GeoDePIN networks. SSRN Working Paper. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6483619

Zukowski, Z. (2026b). Adaptive Tokenomics: A Systems Engineering Approach to Programmable Incentive Design. SSRN Working Paper No. 6364158. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6364158

Zukowski, Z. (2026c). Routing the Dollar: Gateway Infrastructure, Monetary Policy Transmission, and the Dollar’s International Functions in Digital Markets. SSRN Working Paper No. 6267698. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6267698

Zukowski, Z. (2026d). Seven Dollars: Legal Structure, Operator Risk, and Diligence Divergence Across Dollar-Denominated Digital Products. SSRN Working Paper No. 6352118. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6352118

Zukowski, Z. (2026e). Why token governance concentrates: A Polanyian extension to the fourth fictitious commodity. Working manuscript.

Zukowski, Z. (2026f). Non-domination by design: A republican-commons standard for legitimate token governance. Working manuscript.